SURPRISES IN PROBABILITY

—Seventeen Short Stories—

Henk Tijms

CPD CERTIFIED
The CPD Certification Service

CRC Press
Taylor & Francis Group
A CHAPMAN & HALL BOOK

Surprises in Probability

Seventeen Short Stories

Taylor & Francis
Taylor & Francis Group
http://taylorandfrancis.com

Surprises in Probability

Seventeen Short Stories

By

Henk Tijms
Emeritus Professor of Applied Mathematics

CRC Press
Taylor & Francis Group
Boca Raton London New York

CRC Press is an imprint of the
Taylor & Francis Group, an **informa** business

CRC Press
Taylor & Francis Group
6000 Broken Sound Parkway NW, Suite 300
Boca Raton, FL 33487-2742

Printed on acid-free paper
Version Date: 20180507

International Standard Book Number-13: 978-0-367-00043-1 (Paperback)
978-0-367-00082-0 (Hardback)

Visit the Taylor & Francis Web site at
http://www.taylorandfrancis.com

and the CRC Press Web site at
http://www.crcpress.com

Contents

Foreword

This book is a unique collection of fascinating applications of probability. It grew out of a series of popular science columns I wrote for the magazine of the Netherlands Society for Statistics and Operations Research, and from contributions I made to the *New York Times' Numberplay* blog. Probability is one of the most fascinating branches of mathematics, with wide appeal, not only due to its usefulness in solving common problems in daily life and many areas of modern science, but also to its many surprises. Here, readers will find instructive and entertaining stories that cover a wide range of probability applications from gambling to optimal stopping.

The book consists of seventeen short chapters that can be read independently. Every attempt has been made to ensure that the book is appropriate for many different audiences, including:

• Students who want 'behind the scenes' examples of probability to complement what they are learning in class.
• Teachers who want an easy source of instructive fun material for their students.
• Anyone who wants to really understand how probability works and how it is applied.

It is my sincere hope that the material will generate enthusiasm among students and the general public for the field of probability, and will offer teachers a new way of teaching the subject.

I would like to thank Sherrill Rose for translating the original Dutch material and my colleague Lex Schrijver for providing financial support from his Spinoza fund in order to make the translation possible. Also, thanks to Ad Ridder for help with the computer simulations.

Introduction

Chapter 1. What is Casino Credit Worth? This chapter shows several surprising applications of the 17th century gambler's ruin formula of Christiaan Huygens and Blaise Pascal. It is used to analyze the facts of the notorious lawsuit of Zarin v. Commissioner, which revolves around a federal tax assessment claim brought by the IRS after gambling addict David Zarin's 3-million-dollar casino debt was discharged. Also, several interesting facts of the drunkard's walk – on a line, or in higher dimensions – are presented and used to estimate the average travel of a photon from the center of the sun to its surface.

Chapter 2. One hundred prisoners: Freedom or Death. Hundred prisoners must find their names in one of 100 hundred closed boxes, which are arranged in random order. Each prisoner may open only 50 boxes and cannot communicate with other prisoners. If they all succeed in finding their names, they are set free. At first glance, the situation appears hopeless. There is, however a strategy that provides a success probability of more than 30%. This strategy can also be used in a Monty Hall show 2.0 in which two finalists must find both the car and its key that are hidden behind three doors.

Chapter 3. Birthday surprises and 500 Oldsmobiles. The 2014 World Cup soccer championship with 32 teams of 23 players each convincingly shows the validity of the theoretical solution to the classic birthday problem. An insightful heuristic for the classic birthday problem and its variants is discussed. The birthday problem also appears in a Canadian lottery in which 500 Oldsmobiles are raffled off as a bonus prize among the 2.4 million lottery subscribers, and lottery officials are astonished when one subscriber wins two Oldsmobiles.

Chapter 4. Was the Champions League Rigged? In March 2013, a heated discussion raged in sport programs and social media about whether the UEFA Champions League quarter-final draw had been rigged using 'hot and cold balls'. The Reverend Bayes' 18th century formula is the right formula to shed light on the alleged manipulation in the Champions League. The formula of Bayes and its usefulness in a wide variety of situations are extensively discussed. In particular, examples in medicine and law are given. Also, the fundamental differences between Bayesian statistics and classical statistics are raised.

Chapter 5. **Benford goes to the Casino.** As a consequence of Benford's law, an attractive casino game turns out to be less favorable than it seemed. This amazing law says that the first nonzero digit in many types of data is not uniformly distributed but approximately follows a logarithmic distribution, where the number 1 appears as the most significant digit about 30% of the time, while the number 9 appears as the most significant digit less than 5% of the time. An intuitive explanation of the law is given. Applications of Benford's law that look at the investigation of financial data and the detection of possible fraud are discussed.

Chapter 6. **Surprising Card Games, or, It's All in the Cards.** A clever strategy for winning a card game with a standard 52-card deck of playing cards. Each of two players must choose a three-color sequence and players win one point each time their particular sequence appears in a series of three consecutive cards. The strategy is similar to the strategy in the Penney Ante coin-tossing game in which each of two players chooses a sequence of heads and tails of length three. The Kruskal count is a magic trick that enables a magician to correctly guess, with high probability, the playing card selected by a player according to a certain counting procedure.

Chapter 7. **The Lost Boarding Pass and the Seven Dwarves.** A surprising solution for the lost boarding pass puzzle. In this puzzle one hundred people line up to board a plane with 100 seats. The first person in line, however, has lost his boarding pass, so he randomly chooses a seat. After that, each person entering the plane either sits in their assigned seat, if it is available, or, if not, chooses an unoccupied seat randomly. What is the probability that the last passenger entering the plane will find his assigned seat to be free? A nice variant of this problem is the seven dwarfs dormitory problem.

Chapter 8. **Monte Carlo Simulation – the Interface.** Monte Carlo simulation, named after the famous gambling hot spot in the

Principality of Monaco, was initially used to solve neutron diffusion problems in atomic bomb research at Alamos Scientific Laboratory in 1944. Nowadays it is one of the most-utilized mathematical tools in scientific practice. The basic ideas of this tool are outlined and illustrated with geometric and combinatorial probability problems, which are not easy to solve analytically. Also, it is argued that Monte Carlo simulation is a perfect didactic tool for adding an extra dimension to the teaching and learning of probability. It may help students gain a better understanding of probabilistic ideas and to overcome common misconceptions about the nature of 'randomness'.

Chapter 9. **Lotto Nonsense: The World is Asking to be Deceived.** Loads of books on the market about lotto and roulette would have readers believing in the existence of systems that help you win. Two lotto systems making this claim are debunked using the normal probability distribution. Several roulette systems including the Labouchère system and the Big-Martingale system are analyzed in detail using Monte Carlo simulation. It is convincingly demonstrated that these systems only differ from one another in patterns of betting and in the way in which they reconfigure a player's losses and wins. Over the long run, the gambler cannot beat the casino's house edge.

Chapter 10. **March Madness Grips the USA.** One of the biggest sporting events in the USA is the yearly March Madness tournament. Sixty-four university basketball teams take part in a knockout competition of six rounds with a total of 63 games. In 2014, March Madness reached a sort of fever pitch. Warren Buffett wrote out a check for 1 billion dollars to be awarded to anyone who could correctly predict the winners of all 63 games. What is the probability that Buffett will actually have to pay out when 10 million Americans fill forms to predict the winners of the games? Using historical tournament data, an estimate of this probability can be given, showing that Buffett in fact, extended a riskless offer.

Chapter 11. **Coincidences and Impossibilities.** Many weird things just happen due to chance. The lottery principle says that so-called coincidences can nearly always be explained by probabilistic arguments. But are there not events that are so improbable that they will never occur? This question is discussed using Borel's law as it pertains to impossible events and is illustrated using the practical impossibility of four perfect hands occurring in the game of bridge. The New-Age Solitaire card game sheds light on the seven riffle shuffles recommended for bridge and on the subtleties of randomness.

Chapter 12. Gambler's Fallacy Misconceptions over the way that truly random sequences behave fall under the heading gambler's fallacy. This refers to the gambler who believes that, if a certain event occurs less often than average within a given period of time, it will occur more often than average during the next period. This misconception is illustrated with a discussion of the tangle of events that occurred on August 18, 1913, at the casino in Monte Carlo when the roulette ball fell on black 26 times in a row. A rule of thumb is offered to help readers get a feel for the length of the longest run in tosses of the coin and spins of the roulette wheel. The gambler's fallacy is also behind the Venice-53 hysteria in the national Italian lottery when the number 53 remained elusive for many months in the bi-weekly Venice lottery draw. Using the Markov chain model, it is shown that such a remarkable happening is less coincidental than it seems.

Chapter 13. Euler's Number e is Everywhere. The Euler number $e = 2.71828\ldots$ shows up in many probability applications. The frequent occurrence of the number e in probability can be explained using the Poisson distribution, which is the most useful discrete probability distribution. The Euler number appears in problems as diverse as the Santa Claus problem, the Las Vegas card game, the dating problem, and the Oberwolfach dinner problem. Using the Poisson heuristic, excellent approximations are derived for the complicated exact solutions of those problems.

Chapter 14. The 10 Most Beautiful Formulas in Probability. A list of the 10 most beautiful formulas in probability is always subjective. Any list, however, should contain the equation for the normal or Gaussian density function and the equation for Bayes' theorem. Both equations combine depth with simplicity. Where the beauty of the formula for the normal function is immediately evident from its intrinsic aesthetic appeal, the beauty of Bayes' formula arises because this formula underlies rational thinking and decision-making. Among the other beautiful probability formulas are De Moivre's equation for the standard deviation, the gambler's ruin formula of Huygens and Pascal, and the Pollaczek-Khintchine queueing formula.

Chapter 15. Beating the Odds on the Lottery. Every lottery player has fantasized about winning the jackpot. However, one's chances of actually winning the jackpot, alas, are inconceivably small. Some people, however, find or create loopholes to help them beat the odds of winning the lottery. In the Virgina lotto game, a syndicate won the jackpot by buying nearly all of the possible ticket numbers when the jackpot had grown such that the expected payoff for one ticket was greater than the cost of the ticket. In the Cash Winfall lottery in Masssachusetts, a jackpot of more than $2 million was split and distributed among lower prizes when the jackpot was not won. Syndicates used this feature to buy very large numbers of tickets when a roll-down of the jackpot was approaching.

Chapter 16. Investing and Gambling with Kelly. The Kelly criterion has come to be accepted as one of the most useful methods for bettors and investors. The Kelly formula calculates the proportion of your bankroll to bet or invest on an outcome whose odds are higher than expected, so that your bankroll grows exponentially over the long run. The Kelly bet also maximizes the expected logarithmic utility of your bankroll. The betting formula is derived using the law of large numbers. It is extended to situations in which you can place multiple bets at the same time on different outcomes, such as in horse races and soccer games.

Chapter 17. To Stop or Not to Stop? That is the Question. The most famous optimal stopping problem is the dating problem in which potential partners appear in random order and you don't know how your the current candidate will stack up against those that follow. When to stop browsing and make a choice if you want to maximize the chances of getting the best partner? Simple stopping rules are derived for this problem and several of its variants, including the dating problem in which the aim is to achieve a maximal probability of choosing one of the best two or three of the potential partners. The one-stage-look-ahead principle is used to find good stopping rules for the famous Chow-Robbins coin-tossing game and the devil's penny game in which you sequentially open boxes containing dollar amounts while trying to avoid the box containing the devil's penny.

Taylor & Francis
Taylor & Francis Group
http://taylorandfrancis.com

CHAPTER 1

What is Casino Credit Worth?

THE GAMBLER'S ruin formula plays a prominent role in the field of probability. This formula, which goes all the way back to Christiaan Huygens (1629–1695) and Blaise Pascal (1623–1662), relates to cases where two players, A and B play a game of chance. They stake one unit of money on each gamble until one of the players goes broke. The gambler's ruin formula gives the ruin probability for each of the two players. It can be applied in many situations, and, perhaps surprisingly, has turned out to be a useful tool for deciphering the facts in the famous lawsuit of Zarin v. Commissioner.

In 1980, an Atlantic City casino extended a more or less unlimited credit line to gambling addict David Zarin. They only cut him off when his gambling debt passed the 3-million-dollar mark. Partly due to New Jersey state laws that provide a shield for gambling addicts, the casino in question had no legal recourse to collect the full amount of the debt; in fact, it was required to discharge the lion's share of the debt. But the story doesn't end there. Shortly after the court's decision was rendered, Zarin received a federal tax assessment claim from the Internal Revenue Service demanding tax payment on the sum of 3 million dollars, which had been defined as income. Zarin returned to court to fight this assessment, and won. His most important argument was that he had received no cash money from the casino, only casino chips with which to gamble. In the end, the court determined that Zarin did not owe taxes on the portion of the debt that had been discharged by the casino. This lawsuit generated much interest and has since been taken up into the canon of required case studies for law students studying in the United States.

In coming to its decision, the court neglected to ask this simple question: what monetary value can be assigned to a credit line of 3 million dollars in casino chips, that allows a player to gamble at a casino? First of all, it must be said that the odds of the player beating the casino are small. Still, the player does have a chance of beating the casino, of claiming a profit, and, after repaying the 3-million-dollar advance, of going home with a sum of money that was gained on the loan. The gambler's ruin formula enables us to quantify the monetary value of this loan.

Getting back to our friend Zarin: his game at the casino was the tremendously popular game of craps. Craps is a dice game played with two dice. There are various betting options, but the most popular, by far, is the so-called 'pass-line' bet. We don't need to get into the intricacies of the game or of pass-line betting procedure; suffice it to say that, using the pass-line bet, the player has probability $243/3495 \approx 0.493$ of winning, and the casino has probability $251/495 \approx 0.507$ of winning. When the player wins, the player gets a return of two times the amount staked; otherwise, the player loses the amount staked. This is precisely the situation of the classical gambler's ruin problem. In this problem, the gambler starts with a units of money, stakes one unit on each gamble, and then sees his bankroll increase by one unit with probability p and sees it decrease by one unit with probability $q = 1 - p$. The gambler stops when his bankroll reaches a predetermined sum of $a+b$ units of money, or when he has gone broke. Letting $P(a,b)$ be the probability that the gambler reaches his target of $a+b$ units of money without first having gone broke, the classical gambler's ruin formula is

$$P(a,b) = \frac{1-(q/p)^a}{1-(q/p)^{a+b}},$$

where $P(a,b)$ must be read as $a/(a+b)$ when $p = q = 0.5$. This formula lets us show that, in David Zarin's case, it would not have been unreasonable to assign a value of 195 thousand dollars to his credit line of 3 million dollars. If a player wants to achieve the maximal probability of a predetermined winning sum in a casino game such as craps, then the best thing the player can do is to bet boldly, or rather, to stake the maximum allowable sum (or house limit) on each gamble. Intuition, alone, will tell us that betting the maximum exposes the player's bankroll to the casino's house edge for the shortest period of time. In Zarin's case, the casino had imposed a house limit of 15 thousand dollars for the pass-line bet in the game of craps. So, we may reasonably think that Zarin staked 15 thousand dollars on each gamble. In terms of the gambler's ruin formula then, 15 thousand dollars would be equal to one unit of money. We can further assume that Zarin's target goal was to increase his bankroll of $3{,}000{,}000/15{,}000 = 200$ units of money by b units of money, having assigned a value to b beforehand. What is a reasonable choice for b when the casino gives a credit line of a units of money as starting bankroll? Part of the agreement is that the player will owe nothing to the casino if he goes broke, and the player will go home with a profit of b units of money if he increases his bankroll to $a+b$ units of money. In order to demonstrate this, we must make use of the function $u(a,b)$, which is defined as the expected value of the sum with which the player will exit the casino. This utility function is given

$$u(a,b) = b \times P(a,b) + 0 \times (1 - P(a,b)).$$

For a given bankroll a, a rational choice for b is that value for which $u(a,b)$, as a function of b, is maximal. This value of $u(a,b)$ for the maximizing b could be considered, by the court, to be the value of the credit advance of a units of money extended by the casino to the player. An insightful approximation can be given to the maximizing value of b, which we denote with b^*, and the corresponding value of the credit line. For the case of a sufficiently large bankroll a, we have

$$b^* \approx \frac{1}{\ln(q/p)} \quad \text{and} \quad u(a, b^*) \approx \frac{e^{-1}}{\ln(q/p)}.$$

Here $\ln(x)$ is the natural logarithm with Euler number $e = 2.71828\ldots$ as basis. Surprisingly, the value of the bankroll a is not relevant. The approximations can be derived by rewriting the gambler's ruin formula as

$$P(a,b) = \frac{(q/p)^{-a} - 1}{(q/p)^{-a} - (q/p)^b}$$

and noting that, for large a, the term $(q/p)^{-a}$ can be neglected when $q/p > 1$. Thus, $u(a,b) \approx \frac{b}{(q/p)^b}$. Putting the derivative of $u(a,b)$ with respect to b equal to zero, the approximations follow after a little bit of algebra.[1] An interesting result is that, for a sufficiently large bankroll a and a target amount of $b = b^*$, the probability of reaching the target is approximately equal to

$$e^{-1} \approx 0.3679,$$

regardless of the precise value of the bankroll a. If we apply these results to David Zarin's case, using $a = 200$, $p = 244/495$ and $q = 251/495$, then we find that

$$b^* \approx 35 \quad \text{and} \quad u(a, b^*) \approx 13.$$

This means that the value of the credit line extended by the casino is about 13 units of money. Each unit of money represents 15 thousand dollars. So, we can conclude that the 3-million-dollar credit line extended to Zarin by the casino can be valued at about 195 thousand dollars. And that is the amount the American tax authorities would have been justified in taxing.

The gambler's ruin formula also has some unexpected applications. Let's say you are chairperson of a committee consisting of n persons. The committee meets and determines that it is time to choose

[1] Further details can be found in the original article of Michael Orkin and Richard Kakigi, "*What is the worth of free casino credit?*," The American Mathematical Monthly, Vol. 102 (1995), 3-8.

a new chairperson. The choice will be made by means of a lottery, the new chairperson being chosen from the remaining $n-1$ committee members, such that each member has the same chance $\frac{1}{n-1}$ of being chosen. How can this result be achieved if the group has only a fair coin at hand to work with? As chairperson, you suggest using the following procedure. You will toss the coin. If it turns up heads, you will pass the coin to the person on your right, and if it turns up tails, you will pass it to the person on your left. The one that gets the coin from you repeats the procedure, tossing the coin and passing it to right or left depending on whether it turns up heads or tails. This procedure is followed until there is just one person left who has not been the recipient of the coin. And that person is the new chairperson. How can we show that this procedure is a fair one, in which each of the $n-1$ candidates has the same chance of becoming chairperson? Choose a random person from the group of $n-1$ candidates for the chairmanship, and call that person Rose. We'll call the person to the right of Rose, Right, and we'll call the person on her left, Left. Now define the following probabilities. Let r represent the probability that the coin first goes to Right rather than to Left, and let s represent the probability that the coin will travel around the table from Right to Left without landing at Rose when it has first gone to Right. Next, note that s also represents the probability that the coin will go around from Left to Right without landing at Rose when it has first gone to Left, rather than Right. A conditioning argument, then, reveals that the probability of Rose being chosen chairperson is equal to

$$r \times s + (1-r) \times s = s.$$

If we can show that $s = \frac{1}{n-1}$, then we have demonstrated that the lottery procedure is a fair one. The probability s can be calculated with the classical gambling model of Huygens and Pascal. The probability s is nothing other than the probability that player A will ultimately win all of the money in a sequence of fair games $(p = q = 0.5)$ between two players A and B with starting bankrolls $a = 1$ and $b = n-2$. This win probability is given by

$$\frac{a}{a+b} = \frac{1}{1+n-2} = \frac{1}{n-1}.$$

So, the probability that Rose will be the new chairperson is $\frac{1}{n-1}$. Rose was chosen randomly as one of the $n-1$ candidates, which demonstrates that the lottery procedure must be fair. And that is quite a surprising result.

The classical gambling model of Huygens and Pascal can be seen as a drunkard's walk with absorbing barriers. The drunkard's walk or random walk is an the important probability model. An interesting case is the drunkard's walk with no barriers. Think of a drunkard who exits a pub and makes his way down an infinitely long, straight road by successively taking either a step to the left or to the right with respective probabilities p and $q = 1-p$. Each step is independent of the step before it, and each step covers the same unit of distance. An interesting question is this: after the drunkard has taken a large number of steps, what is the expected distance between the drunkard's position and his starting point (the pub)? This question seemingly falls into the category of pure entertainment, but, in actuality, nothing could be further from the truth. The drunkard's walk has many important applications in physics, chemistry, astronomy, and biology. These applications usually consider two- or three-dimensional representations of the drunkard's walk. The biologist looks at the transporting of molecules through cell walls. The physicist looks at the electrical resistance of a fixed particle. The chemist looks for explanations for the speed of chemical reactions. The climate specialist looks for evidence of global warming, etc. The model of the drunkard's walk is extremely useful for this type of research. The one-dimensional drunkard's walk is used by some financial analysts to model the path of the stock market. The drunkard's walk model, however, has never really been popular on Wall Street. That is because, if the model is correct, then blindfolded monkeys throwing darts at the financial pages would be just as capable of compiling a good investment portfolio as a group of seasoned financial experts.

For the one-dimensional, symmetric drunkard's walk ($p = q = 0.5$), the expected distance between the drunkard's position after n steps and his starting position is approximately proportional to the square root of n. It is given by

$$\sqrt{\frac{2n}{\pi}} \approx 0.798\sqrt{n}$$

when n is sufficiently large. Similar approximation formulas can be given for the drunkard's walk in the higher dimensions. For the two-dimensional drunkard's walk on an unbounded, flat surface, the expected distance between the drunkard's position after n steps and his starting position is approximately equal to

$$\frac{1}{2}\sqrt{\pi n} \approx 0.886\sqrt{n},$$

while for the three-dimensional drunkard's walk in an unbounded space, the approximation formula for the expected distance is given by

$$\sqrt{\frac{8n}{3\pi}} \approx 0.921\sqrt{n},$$

when we suppose that each step has the same unit length and is independent from each preceding step. The formulas used for 2- and 3-dimensional drunkard's walks can be applied both in the scenario of a symmetric drunkard's walk on a grid, and when the drunkard steps randomly in any direction on a flat surface or into space (in a symmetric drunkard's walk on a grid, each grid point bordering on the drunkard's current grid position has an equal probability of being chosen as the next stepping point). The drunkard's walk has some surprising characteristics. In the symmetric drunkard's walk, either on the line or on a flat grid, there is a probability of 1 that the drunkard will ever return to his starting point, although the expected value of the travel time needed to complete the walk is infinitely large. For the symmetric drunkard's walk on a grid in 3-dimensional space, the probability of the drunkard ever returning to his starting point is less than 1 and approximately equal to 0.65. These are deep results that can only be demonstrated by the use of advanced probability calculations. Markov chain theory plays an important role here.

A nice application of the model of the drunkard's walk in space is the following: what is the average travel time of a photon moving from the sun's core to its surface? En route to the surface, such a photon will have innumerable collisions with other particles in the sun's plasma. An estimate of the average distance between two collisions is 10^{-1} millimeters. The radius of the sun is 70 thousand kilometers, or rather 7×10^{10} millimeters. If we solve n from the equation

$$\sqrt{\frac{8n}{3\pi}} = \frac{7 \times 10^{10}}{10^{-1}},$$

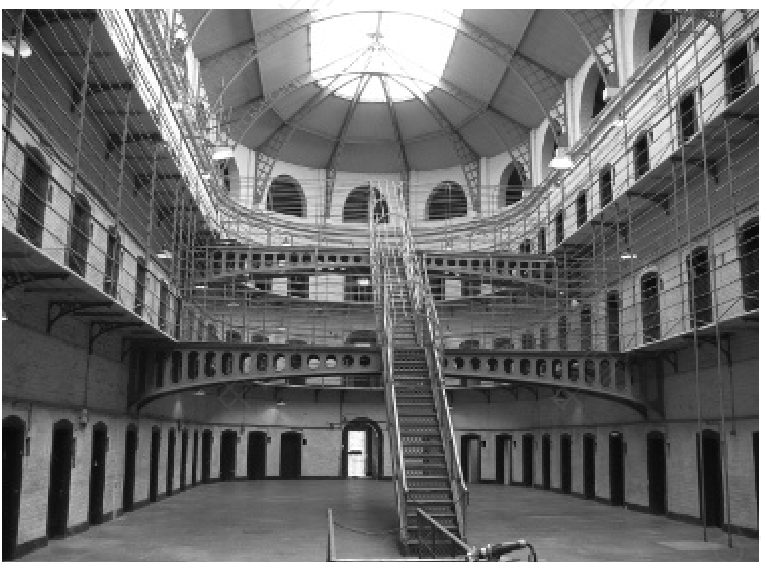
we learn that the average number of collisions a photon will undergo before reaching the sun's surface is about $n = 5.773 \times 10^{23}$. Photons travel at the speed of light, i.e., 300 thousand kilometers per second, such that the travel time between two collisions is equal to $10^{-1}/(3 \times 10^{11}) = 3.333 \times 10^{-13}$ seconds. The average travel time, then, of a photon from the sun's core to its surface is about

$$(5.773 \times 10^{23}) \times (3.333 \times 10^{-13}) = 1.924 \times 10^{11} \quad \text{seconds.}$$

If you divide the travel time by $365.25 \times 24 \times 3600$, you find that the average travel time of a photon from the sun's core to its surface can be estimated as 6,000 years. Which only goes to show that a random walk is not a very fast way to get somewhere! Once the photon finally reaches the sun's surface, it only takes about 8 minutes for it to travel the 149,600,000 kilometers from the sun to the planet Earth.

CHAPTER 2

One Hundred Prisoners: Freedom or Death



One hundred condemned prisoners are called together by a prison guard, who informs them that they are being transferred. Their new prison has a hard-as-nails reputation, so it is more or less assured that the 100 prisoners will be executed. The soft-hearted warden of their current prison has decided to give the prisoners a chance to be released rather than transferred. Towards this end, he has proposed a game thus. There are 100 closed boxes on a table. Each box contains the name of one prisoner, and each prisoner's name appears exactly one time, in one box. Each prisoner may open no more than 50 of the boxes, one box at a time, in the hope of finding his name. The prisoners may not confer with one another during game play, may not shift the order of the boxes, and may not extract and exchange any names within the boxes. If all prisoners find their own names, they will all be released. The prisoners are given one half hour to discuss the warden's proposal, after which they must inform the guard whether or not they intend to play. At first glance, this seems a futile undertaking. If each prisoner randomly inspects 50 boxes, then the chance of a collective release of all prisoners is equal to $(\frac{1}{2})^{100} \approx 7.9 \times 10^{-31}$, in other words, practically zero. Luckily, one of the prisoners was mathematician-cum-investment banker before embarking on a life of crime and landing in the pokey. Our derailed mathematician explains to the other prisoners that they have about a 30% chance of gaining their freedom. The others can hardly believe it. Nevertheless, it is true. The solution is simple and easy to understand. Each prisoner is assigned one particular box, and no two prisoners are assigned the same box. The prisoners all have phenomenal memories, and each knows not only which box is assigned to him, but also which box is assigned to each of the other prisoners. Each prisoner first opens the box assigned to him. If he encounters his own name, he stops. If he encounters a name other than his own, he opens the box assigned to that prisoner. He continues in this fashion until either finding the box with his own name, or until he has opened 50 boxes. This strategy gives a 31.2% chance of all prisoners finding their names. Yet we see again that math always comes in handy.

To understand this surprising result, we need to understand the concept of a cyclic of a permutation of the numbers $1, 2, \ldots, m$. A permutation is a rearrangement of the numbers 1 to m in an ordered list. We will elucidate the concept of cycle by using the permutation $(7, 3, 8, 6, 1, 4, 5, 2)$ of the list (1, 2, 3, 4, 5, 6, 7, 8). This permutation is made up of three disjoint cycles

$$1 \to 5 \to 7 \to 1, \;\; 2 \to 8 \to 3 \to 2, \;\; \text{and } 4 \to 6 \to 4$$

with lengths 3, 3 and 2, respectively. The chance of each prisoner finding his own name is no different from the chance that in a random permutation of the numbers $1, 2, \ldots, 100$, there will be no cycle with length greater than 50. This is most easily seen by imagining that the box in which the guard has placed the name of the ith prisoner has an invisible label i for all i and imagining that the ith prisoner will be assigned the box located in the ith position of the random order in which the guard has placed the boxes. What is the probability that a randomly generated permutation of the numbers $1, 2, \ldots, 100$ will include a cycle with length greater than 50? Stated more generally: what is the probability Q_n that a random permutation of the numbers $1, 2, \ldots, 2n$ will include a cycle with length greater than n? The answer is

$$Q_n = \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n}.$$

Though the derivation of this formula uses only elementary combinatorial mathematics, it would nonetheless require us to go too far into technical details to make the effort worthwhile.[1] The key to the formula for Q_n is the combinatorial result that the total number of permutations of the numbers 1 to $2n$ having the property that the permutation contains a cycle of length $k > n$ is equal to $\binom{2n}{k}(k-1)!(2n-k)!$. An insightful approximation for Q_n can be given by means of the celebrated approximation

$$1 + \frac{1}{2} + \cdots + \frac{1}{m} \approx \ln(m) + \gamma + \frac{1}{2m}$$

when m is sufficiently large. Here $\gamma = 0.57722\ldots$ is the Euler-Mascheroni constant. This leads us to the approximation

$$Q_n \approx \ln(2n) - \ln(n)$$

when n is sufficiently large. Noting that $\ln(a) - \ln(b) = \ln(a/b)$ for $a, b > 0$, we arrive at the handsome approximation

$$Q_n \approx \ln(2)$$

when n is sufficiently large. This shows that Q_n does not depend on n for all practical purposes when n is sufficiently large. A surprising result! The probability Q_n is the probability that not all of the

[1] A clear explanation is in Peter Taylor, "*The condemned prisoners and the boxes*," https://protect-us.mimecast.com/s/8PXsCBB8n5t7z667WfzIcgw?domain=mast.queensu.ca.

prisoners find their own name. Therefore, for the case of $2n$ prisoners, the probability of the prisoners being released is approximately equal to

$$1 - \ln(2) = 0.3069,$$

regardless of how large n is. In the case of 100 prisoners ($n = 50$) the exact value of the probability of release is equal to 0.3118, showing that $1 - \ln(2)$ is an excellent approximation.

The strategy used in the prisoner's problem can also be used in an amusing variant of the famous Monty Hall problem, which is covered in Chapter 8. In the final segment of a game show for couples, the last remaining couple is presented with three closed doors, behind which, in random order, they will find a car, the key to the car, and a goat. One of the two players is given the task of finding the car, the other must find the car key. If both are successful, they get to keep the car. Each of the two players may open two doors; the second player may not see what is behind the doors chosen by the first player. The couple may discuss a strategy before the game starts. What strategy will afford the best odds of winning the car, and what are the winning odds? The answer is simple. The player assigned with the task of finding the car opens door number 1 first. If that player finds the key behind door number 1, the player goes on to open door number 2. If the player finds the goat behind door number 1, the player goes on to open door number 3. Then it is the second player's turn. This player, assigned with the task of finding the key, first opens door number 2, and if the player finds the car there, the player goes on to open door number 1. If the player finds the goat behind door number 2, the player opens door number 3. The winning probability for this strategy is $\frac{4}{6}$, as we can immediately confirm by going through all of the six possible configurations of car, key and goat: the four configurations (car, key, goat), (car, goat, key), (key, car, goat) and (goat, key, car) are winning configurations, and the two configurations (key, goat, car) and (goat, car, key) are losers. The probability $\frac{4}{6}$ is 50% greater than the probability $\frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$, which is what the players would have if they didn't agree on this strategy beforehand, and if each of them simply opened two doors randomly.

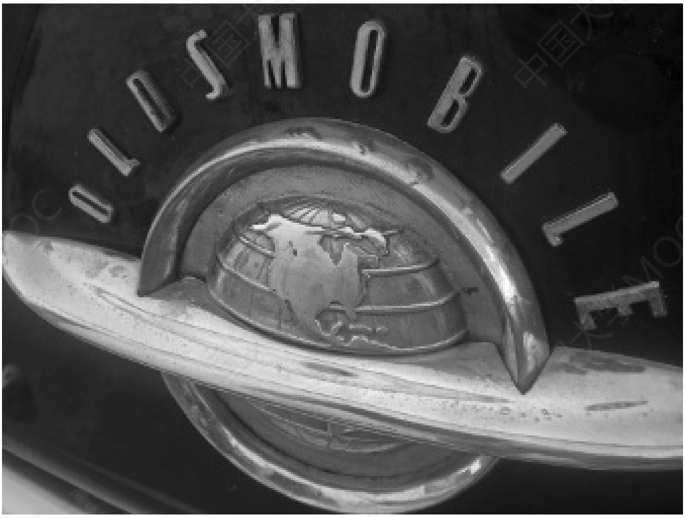
Lady Luck smiled on our 100 prisoners, and they were all released. But alas, they weren't footloose and fancy free for long. All of them soon landed back in the hoosegow. Each prisoner was placed into an isolated cell and had no contact with any of the others. One happy day, the warden calls all 100 of them together and tells them that they are about to get another chance to win their freedom. Each prisoner is given one, uniquely assigned, number from 1 through 100. Every day after that, the guard will randomly choose a number between 1 and 100. The prisoner assigned with that number will be brought to a table with a glass on top of it. The prisoner can leave the glass as it is, or turn it upside down. Then the prisoner is returned to his isolated cell. All prisoners are informed that the glass will be right-side up at the start of the game. Having visited the table, if one prisoner announces that "all prisoners have had a turn at the table," and this statement is indeed accurate, then all of the prisoners will be released. If the statement is uttered and is inaccurate, then the prisoners are doomed. The prisoners are given an opportunity to come up with a strategy before game play begins. What strategy should they use, and what are the odds of its leading to freedom? There is exactly one strategy that will result in freedom for the prisoners. What does that strategy look like? One prisoner is chosen as leader, and it is this one who, at a certain moment, will utter the words that will lead to their release. Each time the leader is summoned to the table and finds the glass upside down, he will turn it right-side up; otherwise, he will leave it as is. Each of the other prisoners will do the following:

• if the prisoner encounters the glass upside-down, or if the prisoner has already turned the glass around once, then the prisoner does nothing;
• if the prisoner approaches the table for the first time and encounters the glass right-side up, then he turns it upside-down.

If the leader approaches the table for the 99th time to find the glass upside-down, then he tells the warden that all prisoners have had a turn, and the warden rewards them all with freedom. Using this strategy, the prisoners will, indeed, achieve their freedom, but it will take them quite a few years to get there. Poor prisoners. They are on something of a wild goose chase. Rough calculations indicate that the expected value of the time until release is about 28 years, with a margin of error of about one year.

CHAPTER 3

Birthday Surprises and 500 Oldsmobiles



THE BIRTHDAY problem is a classic probability problem that goes like this: how many randomly picked people (no twins) must be assembled before there is an even chance or higher that two or more of them will share a birthday? The answer is 23, if we assume that each day of the year is equally likely as a birth date.

If you aren't familiar with this problem, this number might seem surprisingly low. You might be tempted to guess much higher, say, somewhere around 183. But if you really think about it, 23 is not so surprising after all: how often did two or more of your classmates in grade school have birthdays on the same day? BBC journalist James Fletcher conducted an interesting empirical study that reinforces these findings. During the 2014 World Cup soccer championship, 32 national teams of 23 players each took part. It turned out that exactly 16 of those teams had at least one double birthday, the Dutch national team (my pick!) included.[1] Another real-world example concerns the birth dates of American presidents. There have been 44 presidential births, and for a randomly formed group of 44 persons there is a probability of 93.3% that at least two persons will have been born on the same day. Among the 44 American presidents, there is one such coincidence: Warren G. Harding and James K. Polk were both born on November 2.

It should be pointed out here that, in fact, births are not evenly spread out over the calendar year. Interestingly, it can be mathematically shown that this uneven, or non-uniform, spread of birth dates only strengthens the probability that two or more people in a given group will have the same birthday. Without getting into the math, this idea makes sense if we imagine the existence of a country where births only occur in the spring, or the extreme example of a planet where everyone is born on the same day. A similar phenomenon occurs in the lottery: as the majority of tickets are handwritten tickets with popular numbers rather than random picks, the probability of a rollover of the jackpot will get larger (imagine the extreme case of all players choosing the same numbers). Our real-life circumstances, which reflect a deviation in the uniform distribution of birthdays over the year, give a probability not significantly higher, for common birthdays, than would be the case for a uniform distribution. And that means that, for a fifty-fifty chance of a common birthday, a group consisting of 23 people is sufficiently large.

That brings us to an interesting variant of the birthday problem that was a recent subject for discussion on a radio talk show: let's say there is someone who has several hundred or even thousands of Facebook friends. What are the chances of a birthday of one

[1] A similar study was done by Robert Matthews and Fiona Stones, "*Coincidences: the truth is out there*," Teaching Statistics, Vol. 20 (1998), 17-19.

or more of the Facebook friends on each day of the year? How many Facebook friends do you need to have before you get a fifty-fifty chance of this occurring? The answer is quite a few, 2287 to be exact, for a fifty-fifty chance. Calculating this exact value is not an easy task, but a good approximation can be determined using a simple and insightful method. Let's assume that someone has n Facebook friends, where $n \geq 365$. To keep it simple, we'll keep our year to 365 days, and leave the problem of February 29th aside. We'll also assume that all 365 days are equally likely to be a birth date. This is the procedure: we look at it as though we were conducting 365 trials. In the ith trial, we look to see whether the birthday of one or more of the friends falls on day i. The trial is deemed successful if this is not the case. A randomly chosen person does not have a birthday on day i with probability 364/365 and so the probability of success of each experiment is equal to $(364/365)^n$. Subsequently, the expected value of the number of successful trials is

$$\lambda = 365 \times \left(\frac{346}{365}\right)^n.$$

Before continuing, we digress for a moment and discuss the most important distribution from discrete probability, namely the Poisson distribution. This probability distribution has a single parameter $\lambda > 0$ and is given by

$$e^{-\lambda}\,\frac{\lambda^k}{k!} \qquad \text{for } k = 0, 1, \ldots .$$

The expected value of the Poisson distribution is equal to the parameter λ and the standard deviation is $\sqrt{\lambda}$. The Poisson distribution can be interpreted as the probability distribution of the total number of successes in a large number of independent trials each having the same small probability of success.[2] This physical background of the Poisson distribution explains why, in practice, the distribution is an appropriate model for many phenomena, such as the annual number of serious traffic accidents in a given area, the annual number of damage claims received by an insurance company, etc.

[2] Mathematically, it can be shown that the probability of getting exactly k successes in n independent chance experiments each having the same success probability p tends to $e^{-\lambda}\lambda^k/k!$ as $n \to \infty$ and $p \to 0$ such that $np \to \lambda$. This result can be generalized to chance experiments with non-identical probabilities of success.

The Poisson distribution is named after French mathematician Siméon-Denis Poisson (1781–1840), who discovered it in passing in the course of his research. Its great significance only became apparent years later, after which it came to be applied in countless situations. The Poisson distribution acquired its vast popularity on the heels of a study conducted by Russian-born statistician Ladislaus Bortkiewicz (1868–1931), the results of which were published in his book "The Law of Small Numbers" in 1898. In this book he analyzed the number of German soldiers that had been kicked to death by cavalry horses between 1875 and 1894 in each of 14 cavalry corps, showing that those numbers conformed remarkably well to a Poisson distribution.

The particular term $e^{-\lambda}$ in the Poisson distribution represents the probability of achieving not one single success. The appearance of the term $e^{-\lambda}$ for the probability of no success can be easily explained, using the simple but extremely useful approximation $e^{-x} \approx 1 - x$ for x close to 0. Imagine a sequence of m independent trials each having a very small success probability p. Then, the expected number of successes is mp, while the probability of no success occurring is $(1-p)^m \approx e^{-mp}$.

In our Facebook problem, the 365 experiments are not independent, but their dependence is weak enough to justify the use of the Poisson approximation with expected value $\lambda = 365 \times (364/365)^n$. If the trials end without a single occurrence of a success, that means that there is a birthday of one or more of the Facebook friends on every day of the year. The probability of this event is thus approximately equal to the Poisson probability $e^{-365\times(364/365)^n}$. Solving the equation

$$e^{-365\times(364/365)^n} = 0.5$$

gives the approximation $n = 2{,}285$ for the size of the friend group that would give a fifty-fifty chance of a birthday of one or more of the Facebook friends on every day of the year. The exact value can be calculated as 2,287, but this calculation is much more difficult to execute. Note that the approximate value is very close to the exact value. If the probability of a friend's birthday falling on each day of the year is to be at least 95%, then the approximate value for the size of the friend group is 3,234, and this is also the exact value. In actuality, however, a Facebook friend group consisting of several thousand members is not very common. The average size of the friend group on Facebook is about 130 people.

Coming back to the classic birthday problem, it is worth noting a couple of applications with real-life consequences. These applications show that coincidences are often less haphazard than we think, and can be explained by simple, probability based reasoning. In 1982 the organizers of the Quebec Super Lotto decided to use a fund of unclaimed winnings to purchase 500 Oldsmobiles, which would be raffled off as a bonus prize among the 2.4 million lottery subscribers in Canada.[3] They did this by having a computer randomly choose a number, 500 times, from the 2.4 million registration numbers assigned to subscribers. An unsorted list of the 500 winning numbers was published, whereupon, to the lottery officials' astonishment, they were contacted by one subscriber claiming to have won two Oldsmobiles. They had neglected to program the computer not to choose the same registration number twice. It isn't easy to detect double entries in an unsorted list of 500 winning numbers. The subscriber in question, on the other hand, a Mr. Antonio Gallardo, had no trouble parsing the list and finding his own number listed twice. The probability of a given subscriber winning the car two times is astronomically small, but not so the case of the probability that, out of 2.4 million subscribers, there will be someone whose number appears at least twice in the list of 500 winning numbers. The latter event has a probability of about 5%. That is quite a small probability, but not a negligible probability. How do we calculate this probability of 5%? To explain this, let's consider a birthday problem on a planet with d general days in the year and a randomly formed group of m aliens, where each day is equally likely as birthday for any of the m aliens. If d is large and m is much smaller than d, then the probability that two or more aliens from the group share a birthday can be approximated by

$$1 - e^{-\frac{1}{2}m(m-1)/d}.$$

How do we arrive at this formula? Several derivations are possible. We will use the same approach as above by placing the problem in the context of a sequence of trials. First, we must determine how many pairs we can assemble among the aliens. The answer to that is $m \times (m-1)/2$. After all, we have m choices for the first alien and $m-1$ for the second, and we divide by two because the order in which the pair is formed is irrelevant. Next, we note that the probability of two aliens in any given pair having the same

[3] This application is borrowed from the article by J.A. Hanley, "*Jumping to coincidences: defying odds in the realm of the preposterous*," The American Statistician, Vol. 46 (1992), 197-202.

birthday is equal to $1/d$. Considering that we will be comparing the birthdays of the two aliens in each pair, for every possible pair, it becomes evident that we are dealing with a very large number of experiments. The total number of experiments is $\frac{1}{2}m(m-1)$. An experiment is called successful if the two aliens making up the pair have the same birthday, and each experiment has the same small success probability of $1/d$. The expected value of the total number of successes is $\lambda = \frac{1}{2}m(m-1)/d$. This means that probability of achieving no successes is approximately equal to the Poisson probability $e^{-\lambda}$, or rather, the probability of at least one successful trial is given by the approximation $1 - e^{-\lambda} = 1 - e^{-\frac{1}{2}m(m-1)/d}$. This formula gives an approximate value that is very nearly equal to the exact value of the probability of at least one successful trial. The exact value is

$$1 - \frac{d \times (d-1) \times \cdots \times (d-m+1)}{d^m},$$

which rests on the fact that there are $d \times (d-1) \times \cdots \times (d-m+1)$ ways in which the birthdays of the m aliens can be different. The exact formula is less insightful than the approximation formula and might lead to technical calculation problems for large values of d and m.

The approximation formula gives a good idea of how large the value of m (= number of aliens) must be for a given value of d (= the number of days in the year), in order to get a probability of about 50% that two or more aliens will have birthdays that fall on the same day. Putting $e^{-\frac{1}{2}m(m-1)/d} = 0.5$ and taking logarithms for both sides of this equation, we get

$$m \approx 1.18\sqrt{d}.$$

If we are intent on finding a probability of about 95% of two or more birthdays falling on the same day, that will give us $m \approx 2.45\sqrt{d}$.

The problem of the Quebec Super Lotto can be seen as a birthday problem with $d = 2.4$ million days and $m = 500$ aliens. For our Oldsmobile problem, this gives a probability of about 5% that, among the 2.4 million subscribers, there will be some subscriber whose number appears at least twice in the list of 500 winning numbers.

American lottery officials were equally confounded when, as their lottery celebrated its second anniversary, they became aware of the fact that the same four-digit number had been drawn multiple times in the course of the 625 lottery drawings that had taken place. At each drawing, a four-digit number in the sequence 0000, 0001, ..., 9999 was drawn. In answer to the question of whether, in two years' time, the same four-digit number would come up multiple times, the lottery officials had declared that this was as good as impossible. They expected to execute some 5000 drawings before encountering this phenomenon and were taken aback when they saw that it had occurred not just once, but multiple times. Here, again, we see our birthday problem at work. If we take the formula shown earlier, $1 - e^{-\frac{1}{2}m(m-1)/d}$, and fill in the values $d = 10{,}000$ and $m = 625$, we get the value $1 - 3.4 \times 10^{-9}$ for the probability that the same four-digit number will come up two or more times in 625 lottery drawings. In other words, it is a sure thing. We can see just how likely it is even without the Poisson formula: the expected value of the number of combinations of two drawings with the same outcome is equal to $\frac{1}{2}m(m-1)/d = \frac{1}{2} \times 625 \times 624/10{,}000$, or rather 19.5. This makes clear that the probability is, practically speaking, equal to 1 that multiple drawings will have the same outcome when 625 drawings are executed. Seemingly astonishing results, in hindsight, often come with simple explanations!

In the American lottery above, the number of possible outcomes of a drawing is relatively small. This is different for the 6/45 lotto game in which six different numbers are picked from the numbers 1 through 45. The number of possible combinations of six different numbers is $45 \times 44 \times 43 \times 42 \times 41 \times 40$ divided by $6 \times 5 \times 4 \times 3 \times 2 \times 1$ (the order of the numbers picked is not relevant). That works out to the number of possible outcomes of a lotto drawing being 8,145,060. Interesting question: what is the probability that, in the coming 25 years, the same combination of six numbers will be chosen multiple times in a particular 6/45 lotto that conducts 64 drawings per year? If we fill in $m = 25 \times 64 = 1600$ and $d = 8{,}145{,}060$ in the formula $1 - e^{-\frac{1}{2}m(m-1)/d}$, we arrive at a probability of about 14.5%. Not huge, but far from negligible. The German national lotto 6/49 experienced once a repeat winning draw after 3,016 drawings. On Wednesday, 21 June 1995, the six numbers 15-25-27-30-42-48 were drawn – the same six numbers that had already been drawn on Saturday, 20 December 1986. So don't fall off your chair if you hear of a six number winning sequence that has already won before!

CHAPTER 4

Was the Champions League Rigged?

In March, 2013, the draw for quarter-finals of Champions League soccer resulted in the following four matches:

Málaga - Borussia Dortmund
Real Madrid - Galatasaray

Paris Saint Germain - Barcelona
Bayern München - Juventus.

THIS OUTCOME led to heated discussions on sport programs and in social media. Rumor had it that the draw had been rigged using 'hot and cold balls'. The 'big four' – the two Spanish teams, Real Madrid and Barcelona, and the two German teams, Bayern München and Borussia Dortmund – had not been slated to encounter one another in the quarter-finals. Such a match schedule is ideal for spectators following the tournament. How likely is it that the draw was rigged to produce such results? First, we'll calculate the probability that no two of the big four teams meet up when the teams are paired at random. This probability is equal to $\frac{8}{35}$. We don't need any fancy mathematics to arrive at this probability. If the draw is fair, the probability of an outcome in which no two of the big four teams meet up can be calculated as

$$\frac{4}{7} \times \frac{3}{5} \times \frac{2}{3} = \frac{8}{35}.$$

An $\frac{8}{35}$ probability of achieving a particular outcome given a fair draw does not, in fact, tell us much. What you really want to know is what the probability is of the lottery having been rigged given the outcome. To figure this out, we need look no further than the realm of Bayesian probability. In Bayesian probability, we work with subjective probabilities that differ for every individual, generally. Using the Bayesian approach, a sports journalist who suspects the fairness of the soccer draw must estimate, before that draw takes place, the probability of its turning out to be rigged in such a way as to prevent two of the four big teams from meeting up. This subjective probability is called the prior probability. If the journalist's subjective estimate is 20%, then after the outcome of the draw is made known, a revised value based on that estimate can be calculated to determine the probability of a rigged draw. In the situation of a prior probability of 20%, the revised value is equal to 52.2% if the laws of probability theory are consistently applied. The revised value is called the posterior probability. If the prior probability of a rigged draw is estimated as 50%, then the posterior probability of a rigged draw is equal to 81.4%.

How do we calculate the probabilities? For that we use Bayes' formula, one of the most important formulas in the field of probability and statistics. The basic principle of the Bayesian approach is simple. Let's say you have a certain hypothesis that may be true or not true. According to the Bayesian approach, you must first – before gaining access to relevant information (evidence) regarding the truth or falsity of the hypothesis – attach a value to the probability that the hypothesis is true. This prior probability is, in most cases, a subjective probability that will be estimated differently by each individual. After gaining relevant information regarding the hypothesis' truth/falsity, the prior probability of the hypothesis is revised using the formula of Bayes. There are several versions of this formula. The most insightful version is Bayes' rule in odds form. Let's first explain the concept of odds. The odds of an event occurring with probability p, and not occurring with probability $1-p$, is defined by

$$\text{odds} = \frac{p}{1-p}.$$

Conversely, odds O correspond to the probability $p = \frac{O}{1+O}$. Then, Bayes' rule in *odds form* states

$$\text{posterior odds} = \text{prior odds} \times \text{likelihood factor}.$$

The prior odds are equal to the prior probability of the hypothesis being true divided by the prior probability of the hypothesis being not true. The posterior odds are equal to the conditional probability that the hypothesis is true given the evidence, divided by the conditional probability that the hypothesis is not true given the evidence. The likelihood factor is defined as the conditional probability of the evidence given that the hypothesis is true, divided by the conditional probability of the evidence given that the hypothesis is not true.

Now, let's hypothesize that the Champions League draw was not fair, and was, in fact, rigged to ensure that the big four would avoid meeting up in the quarter-finals. Imagine that a sport journalist, *before* the draw takes place, has made a subjective probability estimate of $r\%$ that the draw will be rigged. After the outcome of the draw is made known, what is the revised value of our journalist's subjective probability estimate of a rigged draw? If we use Bayes' rule in odds form with the prior odds $\frac{r/100}{1-r/100}$ and likelihood factor $\frac{1}{8/35}$, then after some calculations we see that this posterior probability is equal to:

$$\frac{(35/8)r}{100-r+(35/8)r} \times 100\%.$$

This formula gives a posterior probability of 52.2% for a prior probability of 20%, and a posterior probability of 81.4% for a prior probability of 50%.

Incidentally, we can also arrive at these posterior probabilities without applying the Bayesian formula. This can be done using a heuristic method that works according to expected frequencies and is easily understood by the layman. Take our 8 quarter-final teams and imagine a large number of draws, say 1000, where for each draw there is a probability of r% that it is rigged to prevent the big four from meeting up. To keep things simple, let's say that $r = 20\%$. Then there will be on average $0.20 \times 1000 = 200$ rigged draws resulting in the big four not meeting up, and on average $0.80 \times 1000 \times \frac{8}{35} = 182.86$ non-rigged draws resulting in the big four not meeting up. Thus, among the draws resulting in the big four not meeting up in the quarter-finals, the proportion of rigged draws is $200/(200+182.86) = 0.522$. This is the same probability of 52.2% that we got earlier using Bayes, but this time we arrived at the result using a simple and insightful method. This method can be used by laymen in a host of situations. As another illustration, suppose that during a female patient's routine check-up, a doctor discovers a breast lump. It could be an indication of a malignant tumor. Without any further investigation, the chance of this is 1%. Mammograms will correctly diagnose the nature of the tumor (benign/malignant) for 90% of the cases tested. A positive test result means the possibility of a malignant tumor. If our patient gets a positive test result, what is the probability that she does, in fact, have a malignant tumor? To answer this question, we again opt to use expected frequencies rather than probabilities. Imagine a large number of women, say 1000. Of these 1000 women, 10 will have a malignant tumor on average, and 990 will not. Of the 10 women with tumors $0.90 \times 10 = 9$ will get a positive test result on average, and of the 990 women with no tumor $(1 - 0.90) \times 990 = 99$ will get a positive test result on average. This means that the probability of a malignant tumor given a positive test result is equal to $9/(9+99) = 0.083$, or rather a probability of 8.3%, and not one on the order of 80%, as so many people think.

The concept of conditional probability is at the heart of the Bayesian method. It is an intuitive concept. To illustrate this, most people reason as follows to find the probability of getting two aces when two cards are selected at random from an ordinary deck of 52 cards. The probability of getting an ace on the first card is $\frac{4}{52}$.

Given that one ace is gone from the deck, the probability of getting an ace on the second card is $\frac{3}{51}$. The probability of drawing two is therefore $\frac{4}{52} \times \frac{3}{51}$. Using this reasoning, you intuitively apply the fundamental formula:

$$P(A \text{ and } B) = P(A)P(B \mid A)$$

where $P(A \text{ and } B)$ is the unconditional probability that both event A ('first card is an ace') and event B ('second card is an ace') will occur, $P(A)$ is the unconditional probability that event A will occur, and $P(B \mid A)$ is the conditional probability that event B will occur given that event A has occurred. Interchanging the roles of A and B in $P(A \text{ and } B) = P(A)P(B \mid A)$ gives $P(B \text{ and } A) = P(B)P(A \mid B)$. This leads to the *basic form* of the celebrated formula of Bayes:

$$P(A \mid B) = \frac{P(A)P(B \mid A)}{P(B)}.$$

The formula of Bayes enables us to reason back from effect to cause in terms of probabilities. The conditional probabilities $P(A \mid B)$ and $P(B \mid A)$ are often confused for one another in practical situations. A classic example of this revolves around a crime scene stained with blood that can only have come from the perpetrator. The blood type is rare and only found among 0.001% of the population. Police records turn up the name of a person with that blood type, the same type as that of the perpetrator. The person is brought in for questioning even though police have no other incriminating evidence against him. The prosecutor argues: "the probability of the suspect having this blood type, if innocent, is 0.001%. The suspect has this blood type. Therefore, the probability that the suspect is innocent is 0.001%." If A is the event that the suspect is innocent, and B is the event that the suspect has the same blood type as the perpetrator, then the prosecutor has confused the probability $P(A \mid B)$ – the probability we're interested in here – with the probability $P(B \mid A)$.

In practice, the odds form of Bayes' rule is used rather than the basic form. The odds form follows directly from the basic form. Let H be the event that a particular hypothesis is true, $\overline{H}$ be the complementary event that the hypothesis is not true, and E be an event that provides information about whether the hypothesis is true or not true. Then, taking the ratio of the basic form of Bayes' rule with $(A = H, B = E)$ and the basic form with $(A = \overline{H}, B = E)$, we get Bayes' rule in odds form:

$$\frac{P(H \mid E)}{P(\overline{H} \mid E)} = \frac{P(H)}{P(\overline{H})} \times \frac{P(E \mid H)}{P(E \mid \overline{H})}.$$

In words, posterior odds = prior odds × likelihood ratio. The prior odds $\frac{P(H)}{P(\overline{H})}$ refer to the situation before the occurrence of event E and the posterior odds $\frac{P(H|E)}{P(\overline{H}|E)}$ refer to the situation after the occurrence of event E.

Conditional probabilities can also be confusing for other reasons. A poor or bad question may be asked, and a probability may be calculated on that basis, in which case the probability may be correct, but it will also be misleading. A textbook example of this is the notorious trial of O.J. Simpson – American actor, broadcaster and NFL football player. O.J. Simpson was accused of murdering ex-wife Nicole Brown and her friend, Ronald Goldman. During the trial, it was established that Simpson had abused his then-wife repeatedly. Simpson's chief counsel, Alan Dershowitz, reacted cleverly to this incontrovertible revelation by showing statistics to demonstrate that on average, only one out of 2500 cases of wife beating leads to the death of the wife. In other words, the probability is only about 0.04% that a woman will be murdered by her partner if the partner has a history of spousal abuse against her. In this way, Dershowitz threw the jury, who weren't mathematically savvy, off-kilter. The probability Dershowitz introduced was correct, but wholly irrelevant to the situation. The relevant probability is the conditional probability of a woman's murderer being her partner, given that the partner is known to have abused her in the past. This conditional probability is many times greater than the conditional probability that an abusive partner will go on to murder his wife. That can be argued to be about 90%, using violent crime statistics and Bayesian analysis.[1] O.J. Simpson's past as abusive partner was, indeed, relevant to the case.

Bayesian thinking should be part and parcel of many a course of study, especially in the fields of law and medicine. Awareness of Bayesian thinking could prevent a lot of juridical and medical blunders, as in the British case against Sally Clark, who unjustly received a life sentence after the crib deaths of two babies born

[1] See Steven Strogatz, "*Chances Are*," New York Times, April 25, 2010.

successively[2], and in the Dutch case of Lucia de Berk, who, through a miscarriage of justice, was sentenced to life for the deaths of seven hospital patients who all passed away during a period of time when she was working the night shift. In both cases, a colossal statistical error was made: that of not taking into consideration the minutely small prior probability of the parties involved actually committing murder. The prior odds of a mother or nurse being a murderer are extremely small and the likelihood ratio should be balanced with these odds.

Bayes' rule in odds form is also the foundation of Bayesian statistics. This statistical approach differs fundamentally from classical statistics. This is best explained by imagining that promising test results have been obtained for a new medical treatment. Classical statistics departs from the null hypothesis that mere fluke was the cause of the test results and calculates the probability of getting data that are *at least* as impressive as the observed data, given that the null hypothesis is true. Bayesian statistics, however, calculates the probability of that which you want to know, namely, the probability that mere fluke were the true cause given the observed data. In classical statistics, the conditional probability of getting data at least as impressive as the observed data if it is assumed that the observed data are just fluke results is called the p-value. If the p-value is below some threshold value, the null hypothesis is rejected and the findings are considered to be 'statistically significant'. The concept of p-value is often misunderstood. As p-values are calculated assuming fluke is the real cause, they cannot simply be flipped around to give a measure that the null hypothesis is correct. The p-value cannot tell you whether the results are true or whether they're due to random chance. In practice, however, the p-value is often incorrectly interpreted as the probability that the null hypothesis (the effect is just a fluke) is true given the observed data. Worse, p-values typically prove to be radical underestimates of this probability, by which the drug could be erroneously regarded as effective.[3] As a result, the statistical conclusions of thousands of research papers that use the $p < 0.05$ test for 'statistical significance' are flawed or non-reproducible. The p-value is not a litmus

[2] See also the notes of Philip Dawid, "*Probability and proof*," to be downloaded from https://protect-us.mimecast.com/s/BkBiCDkY05i5Dvv5Yf5UXK2?domain=cambridge.org, P.

[3] See also Robert Matthews, "*The Great Health Hoax*," The Sunday Telegraph, September 13, 1998.

test with $p = 0.05$ as borderline. Findings that meet the 0.05 standard can actually arise when the data provide very little or no evidence in favor of an effect.

Nevertheless, the p-value is a useful diagnostic tool when the experiment is carefully set up, and sufficiently small p-values, say smaller than 0.001, are required. This can be illustrated using the famous Salk experiment from 1954, in which the effectiveness of a polio vaccine was tested. A large group of 400 thousand children were equally divided into a treatment group and a control group. In the treatment group, 57 children came down with polio, and in the control group 142 children came down with polio. This made 199 total cases of polio among the children, collectively. Let's assume that these 199 children would have come down with polio regardless of the group in which they were placed. Under this hypothesis, the number of children in the treatment group that became infected with polio is binomially distributed with expected value $199 \times 0.5 = 99.5$ and standard deviation $\sqrt{199 \times 0.5 \times 0.5} = 7.1$. The observed value of 57 is approximately 6 standard deviations under the expected value. This means we have a p-value of about 10^{-9}. In combination with a carefully executed experiment, this p-value overwhelmingly indicates that the vaccine works. The p-value was also used by physicists searching for proof of the existence of the Higgs boson. Physicists reported the 'discovery' of the Higgs boson in 2012 after a statistical analysis of the data showed that they had attained a confidence level at the five-sigma level − about a one-in-3.5 million probability that the experimental results would have been obtained if there were no Higgs particle. Again, in all clarity, this is not the probability that the Higgs boson doesn't exist.

CHAPTER 5

Benford Goes to the Casino

You're strolling past a casino and you notice an eye-grabbing sign:

The Multiplication Game, Carpe Diem!

YOU'RE CURIOUS, so you go into the casino to find out what the rules of the game are. The game is played at a table between a player and a casino employee. The employee presses a button to generate a slip of paper printed with a four-digit integer on its reverse side. The player may not see this integer until he chooses one for himself. The player must choose a positive integer having as many digits as the player wishes. The player's integer is then multiplied together with the four-digit casino integer. The player wins if the product of the two integers begins with a 4, 5, 6, 7, 8 or 9; otherwise, the casino wins. If the player wins, he gets 2.45 dollars for every dollar staked. This sounds tempting, seems too good to be true. Before deciding whether or not to play this game, you go off to calculate for yourself the possible products of two four-digit integers, from 1000 to 9999, in case you should play using a four-digit integer. Your computer program alerts you to the fact that 43.0% of the products begin with 4, 5, 6, 7, 8 or 9. This would mean that, in the long run, you will win on average $0.43 \times 2.45 - 1 = 0.0535$ dollar for every dollar staked, giving you on average a winning margin of slightly more than 5% for any four-digit integer chosen. It certainly looks as though the casino has blundered, but hey, that is not your problem. You hightail it to the nearest casino to claim your winnings. After playing the game a great number of times, however, you are baffled to find yourself on the losing end of things. How could this happen? Your calculations were correct, and indicated a comfortable winning margin, but there you are, losing. Unfortunately, this is, in fact, the expected outcome. No matter what strategy the player uses, in the long run the casino wins at least 60.1% of the games. The trick is that the casino uses randomization in generating its four-digit integer. Randomization is a technique that has many applications in mathematics and computer science. For every interaction, i.e., every time the game is played, the random-number generator on the casino's computer picks a randomly chosen number u between 0 and 1. It uses this number to calculate a number $a = 10^u$ and prints out the largest four-digit integer below $10^3 \times a$ on the casino's slip of paper, generated at the start of the game. In this way, the casino guarantees itself winning odds of at least $\log_{10}(4) - 10^{-3} = 0.60106$, no matter what strategy the player applies. This truly surprising outcome was discovered by American mathematician Kent Morrison.

Elementary probability is all you need to shed light on the result. Define the random variable A as $A = 10^U$, where U is a randomly chosen number between 0 and 1. For any two numbers a and u with $1 < a < 10$ and $0 < u < 1$, the inequality $10^u < a$

applies if and only if $u < \log_{10}(a)$. This implies that the probability $P(A < a)$ is equal to the probability $P(U < \log_{10}(a))$. The probability $P(U < u)$ is by definition equal to u for any $0 < u < 1$. This gives $P(A < a) = \log_{10}(a)$ for $1 < a < 10$. Therefore

$$\begin{aligned} P(k \leq A < k+1) &= P(A < k+1) - P(A < k) \\ &= \log_{10}(k+1) - \log_{10}(k) = \log_{10}(1 + 1/k). \end{aligned}$$

That's exactly what we're looking for. The probability distribution

$$\log_{10}\left(1 + \frac{1}{k}\right) \quad \text{for } k = 1, \ldots, 9$$

is the famous Benford distribution. This is precisely the probability distribution of the discrete random variable B that is defined as the value of random variable A, rounded down. If we take the product of a Benford-distributed random variable B and a positive, integer-valued random variable S, we can prove that the first digit of the product will also exhibit the Benford distribution when B and S are independent of each other. This describes what is happening in the casino game, in which the random variable B corresponds to the casino's four-digit integer and the random variable S corresponds to the integer chosen by the player. Keeping in mind that the Benford distribution assigns a total probability mass

$$\sum_{k=1}^{3} \log_{10}\left(1 + \frac{1}{k}\right) = 0.60206$$

to the numbers 1, 2, and 3, it is no longer surprising that the player is at a disadvantage.

The Benford distribution is named after Frank Benford, an American physicist who published an article in 1938 in which he demonstrated empirically that the first nonzero digit in many types of data (lengths of rivers, metropolitan populations, universal constants in the fields of physics and chemistry, numbers appearing in front page newspaper articles, etc.) approximately follows a logarithmic distribution. In fact, a similar empirical result had already been noted by renowned astronomer Simon Newcomb. In 1881, Newcomb published a short article in which he observed that the initial pages of reference books containing logarithmic tables were far more worn and dog-eared than the later pages. He found that integers beginning with a 1 were looked up more often than integers beginning with 2, integers beginning with 2 were looked up more often than integers beginning with 3, etc. For digits 1 through 9, Newcomb found the relative frequencies to be

$$30.10\%, 17.61\%, 12.49\%, 9.69\%, 7.92\%, 6.69\%, 5.80\%, 5.12\%, 4.58\%$$

which is consistent with the mathematical formula $\log_{10}(1+1/k)$ for $k = 1, \ldots, 9$. This result was more or less forgotten until Benford published his article in 1938, in which he presented a mountain of empirical evidence supporting the logarithmic law. Benford's article received more attention than it otherwise might have done, partly due to a lucky placement in the journal that published it – right after the much-cited article of a famous physicist. This led to the re-discovered result being dubbed Benford's law rather than Newcomb's law.[1] Benford's law has the remarkable characteristic of being scale invariant: if a data set conforms to Benford's law, then it does so regardless of the physical unit in which the data are expressed. Whether river lengths are measured in kilometres or miles, or stock options are expressed in dollars or euros, it makes no difference for Benford's law. That said, there are some types of data sets that do not conform to Benford's law. Take, for example, Olympic 400–meter time trials. This data set will not include many qualifying times that begin with a 1! But if you collect the numbers appearing in front-page newspaper articles, you will find that Benford's law does more or less apply. This goes against intuition. You would expect that, in a randomly formed data set, each of the digits 1 through 9 would appear as the first digit with the same frequency, but this is, surprisingly, not the case. A satisfying mathematical explanation for Benford's law was a long time coming. In 1996, the American mathematician Ted Hill proved that if numbers are picked randomly from various randomly occurring number sets, the numbers from the combined sample approximately conform to Benford's law.[2] This is a perfect description of what happens with numbers appearing in front-page newspaper articles.

It is surprising that some specific number sequences such as the sequence of Fibonnaci numbers $(1, 1, 2, 3, 5, 8, 13, \ldots)$, the sequence of powers of two $(2, 4, 8, 16, 32, \ldots)$, and the sequence of factorials $(1, 2, 6, 24, 120, \ldots)$ also conform to Benford's law. Many other se-

[1] Benford's law is an example of Stigler's law of eponymy: No scientific discovery is named after its original discoverer. Another example of this phenomenon is the Pythagorean theorem, which was known in Babylonia long before Pythagoras.

[2] See also Ted Hill, "*The first digit phenomenon*," American Scientist, Vol. 86 (1998), 358-363.

quences, such as the sequence of prime numbers $(2, 3, 5, 7, \ldots)$, the sequence of square roots of positive integers $(1, \sqrt{2}, \sqrt{3}, \ldots)$ and the sequence of squares $(1, 4, 9, \ldots)$ do not conform to Benford's law. Table 5.1 shows the relative frequencies (in percentages) of the first significant digit of the powers 2^n, the Fibonacci numbers F_n and the prime numbers p_n for the first ten thousand terms.

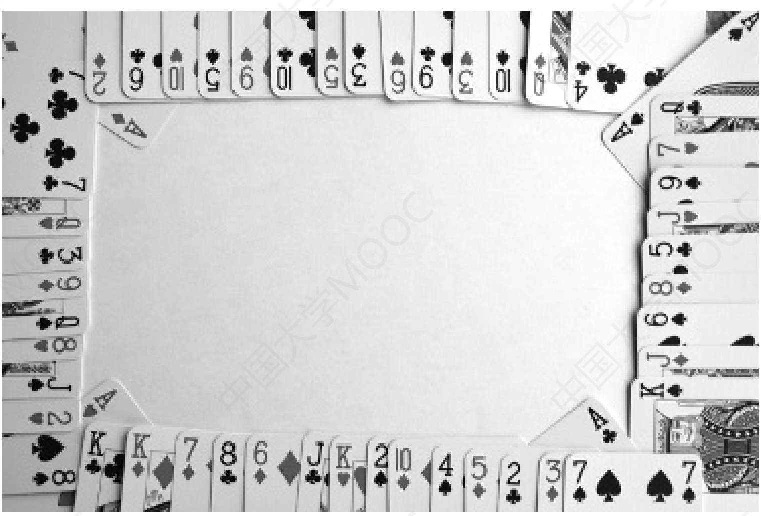
Table 5.1 Benford and some number sequences

	1	2	3	4	5	6	7	8	9
2^n	30.10	17.61	12.49	9.70	7.91	6.70	5.79	5.12	4.58
F_n	30.11	17.62	12.50	9.68	7.92	6.68	5.80	5.13	4.56
p_n	16.01	11.29	10.97	10.55	10.13	10.13	10.27	10.03	10.06

Although it may seem bizarre at first glance, the Benford's law phenomenon has important practical applications. In particular, Benford's law can be used for investigating financial data – income tax data, corporate expense data, corporate financial statements. Forensic accountants and taxing authorities use Benford's law to identify possible fraud in financial transactions. Many crucial bookkeeping items, from sales numbers to tax allowances, conform to Benford's law, and deviations from the law can be quickly identified using simple statistical controls. A deviation does not necessarily indicate fraud, but it does send up a red flag that will spur further research to determine whether or not there is a case of fraud. This application of Benford's law was successfully applied for the first time by a District Attorney in Brooklyn, New York. He was able to identify and obtain convictions in cases against seven fraudulent companies. In more recent years, the fraudulent Ponzi scheme of Bernard Madoff – the man behind the largest case of financial fraud in U.S. history – could have been stopped earlier if the tool of Benford's law had been used. Benford's law can also be used to identify fraud in macroeconomic data. Economists at the IMF have applied it to gather evidence in support of a hypothesis that countries sometimes manipulate their economic data to create strategic advantages, as Greece did in the time of the European debt crisis. This is a different kettle of fish altogether from the quaint application regarding dog-eared pages in old-fashioned books of logarithm tables. Nowadays, Benford's law has multiple statistical applications on a great many fronts.

CHAPTER 6

Surprising Card Games, or, It's All in the Cards



British mathematician Steve Humble and Japanese mathematician Yutaka Nishiyama came up with an exciting card game. You play against an opponent using an ordinary deck of 52 cards consisting of 26 black (B) cards and 26 red (R) cards, thoroughly shuffled. Before play starts, each player chooses a three-card-code sequence of red and black. For example, your opponent chooses BBR and you choose RBR. The cards are laid on the table, face up, one at a time. Each time that one of the two chosen sequences of red and black appears, the player of this sequence gets one point. The cards that were laid down are removed and the game continues with the remaining cards. The player who collects the most points is the winner, with a tie being declared if both players have the same number of points. Your opponent is first to choose a sequence. The 64,000 dollar question is this: how can you choose, in response to the sequence chosen by your opponent, in such a way as to give you the maximum probability of winning the game? The counter-strategy is simple and renders you a surprisingly high win probability. But let's defer that revelation momentarily to take a look at a precursor to that game: the Penney Ante coin toss game introduced in 1969 by Walter Penney in a magazine for recreational mathematics. In this game, a fair coin is tossed for which the probability of an outcome of heads (H) and the probability of an outcome of tails (T) is equal to $\frac{1}{2}$. The game is played by two players, 1 and 2, each of whom must choose, beforehand, a sequence of H's and T's of length three. The coin is then repeatedly tossed until one of the two chosen sequences appears for the first time. And that is the end of the game. The winner is the player whose sequence appears first. Let's say that player 1 chooses a sequence first, and shows it to player 2. Player 2 will always have an edge, because depending on which sequence player 1 chooses, a simple method of choosing an appropriate counter sequence will give player 2 a higher win probability. Player 1 chooses from among these combinations:

$$HHH,\ HHT,\ HTH,\ HTT,\ TTT,\ TTH,\ THT,\ THH,$$

which leaves player 2 to parry with the following countermoves

$$THH,\ THH,\ HHT,\ HHT,\ HTT,\ HTT,\ TTH,\ TTH.$$

That is, HHH is parried with THH, HHT with THH, and so on. Then, the probability of player 2 winning the game has the respective values:

$$\tfrac{7}{8},\ \tfrac{3}{4},\ \tfrac{2}{3},\ \tfrac{2}{3},\ \tfrac{7}{8},\ \tfrac{3}{4},\ \tfrac{2}{3},\ \tfrac{2}{3}.$$

These probabilities can be calculated using the Markov chain model. This model was developed by Russian mathematician A.A.

Markov (1856–1922) at the beginning of the 20th century. It is a very powerful probability model that is used today in countless applications in many different areas, such as voice recognition, DNA analyses, stock control, telecommunications and a host of others.[1] Google's search algorithm is also based on Markov chains. A Markov chain can be seen as a dynamic stochastic process that randomly moves from state to state with the property that only the current state is relevant for the next state. In other words, the memory of the process goes back only to the most recent state. A picturesque illustration of this would show the image of a frog jumping from lily pad to lily pad with appropriate transition probabilities that depend only on the position of the last lily pad visited. In order to plug a specific problem into a Markov chain model, the state variable(s) should be appropriately chosen in order to ensure the characteristic memoryless property of the process. The basic steps of the modeling approach are:

- Choosing the state variable(s) such that the current state summarizes everything about the past that is relevant to the future states.

- The specification of the one-step transition probabilities of going from one state to another in a single step.

Using the concept of state and choosing the state in an appropriate way, many probability problems can be solved within the framework of a Markov chain. Putting yourself in the shoes of someone who must write a simulation program for the problem in question may be helpful in choosing the state variable(s). Let's formulate a Markov chain model for the Penney Ante game, given a situation in which player 1 chooses the sequence HHT, and player 2 parries by choosing THH. For this situation, seven states suffice in order to describe a Markov chain. The states are 0, H, T, HH, TH, HHT, and THH. The auxiliary state 0 indicates the beginning of the game. State HH means that the last two tosses are heads, and state TH means that the last two tosses are tails and heads. The meaning attached to state T is a bit subtler. Whereas state H means that the first toss comes up heads, state T can mean either that the first toss is tails, or that the last toss is tails (use of the two extra states TT and HT, when the players' chosen sequences are HHT and THH, is not necessary). When one of the two states,

[1] A nice introductory article on Markov chains is Brian Hayes, "*First links in the Markov chain*," American Scientist, Vol. 101 (2013), 92-97.

HHT or THH, appears, the game is over. Each of these two states is a so-called *absorbing state*. Once the process enters an absorbing state, it always remains there. The seven states contain all necessary information in order to specify the one-step transition probabilities. It is convenient to use a matrix $\mathbf{P}$ for the one-step transition probabilities:

$$\begin{array}{l} \text{from}\backslash\text{to} \\ 0 \\ H \\ T \\ HH \\ TH \\ HHT \\ THH \end{array} \begin{array}{c} \begin{array}{ccccccc} 0 & H & T & HH & TH & HHT & THH \end{array} \\ \begin{pmatrix} 0 & 0.5 & 0.5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0.5 & 0 & 0 & 0 & 0.5 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{array}.$$

From state 0 there is a one-step transition probability 0.5 to each of the states H and T, from state H there is a one-step transition probability 0.5 to each of the states H and HH, from state T there is a one-step transition probability 0.5 to each of the states T and TH, from state HH there is a one-step transition probability 0.5 to each of the states HH and HHT, and from state TH there is a one-step transition probability 0.5 to each of the states T and THH. The states HHT and THH are absorbing and this is expressed by a one-step transition probability of 1 from the absorbing state to itself.

The matrix $\mathbf{P}$ of one-step transition probabilities describes how the Markov chain moves from state to state. The probability that player 2 will win is the same as the probability that the Markov chain will reach the absorbing state THH during the tosses of the coin. In general, the (i, j)th element of the n-fold matrix product of the matrix of the one-step transition probabilities of a Markov chain gives the probability of the process being in the jth state after n steps given that the process starts in the ith state. In our particular Markov chain with absorbing states, the process will ultimately settle in either state HHT or state THH and remain there forever. As n gets larger, the n-step transition probability of going from state 0 to state THH in n steps will tend to the sought probability of getting absorbed in state THH, being the winning state for player 2. Trying several values of n, we found that $n = 50$ matrix multiplications are enough to have convergence of all of the n-step transition probabilities in four or more decimals. For any

$k \geq 50$, the matrix product $\mathbf{P}^k$ is equal to

$$\mathbf{P}^{50} = \begin{pmatrix} 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.2500 & 0.7500 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.5000 & 0.5000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 1.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 1.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 1.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 1.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 1.0000 \end{pmatrix}$$

From the first row of this matrix we read off that the ultimate probability of absorption in state THH is 0.75 when starting in state 0. In other words, the win probability of player 2 is 0.75 and that of player 1 is 0.25. For any other sequence choice made by player 1, the probability of player 2 winning the game can be calculated by means of an appropriate Markov chain.

Back, now, to the Humble-Nishyama card game. You play this game against an opponent who is the first to choose a three-card-code sequence of red and black. The key to analyzing this game lies in carefully choosing your countermove, as made in response to the sequence choice of your opponent. The first element in your countermove should be the opposite of the second element in the sequence chosen by your opponent. The last two elements in your countermove should be the same as the first two elements in the sequence of your opponent. The third element of the sequence of your opponent is not relevant to your strategy. Now we can give an answer to the question of how you can best react to any choice made by your opponent. Your opponent chooses first a three-card-code sequence of red and black. The possible choices for your opponent are:

$$BBB,\ BBR,\ BRB,\ BRR,\ RRR,\ RRB,\ RBR,\ RBB,$$

which leaves you to parry with the following countermoves:

$$RBB,\ RBB,\ BBR,\ BBR,\ BRR,\ BRR,\ RRB,\ RRB.$$

Then, your probability of winning the card game has the respective values:

$$0.995,\ 0.935,\ \ 0.801,\ 0.883,\ \ 0.995,\ 0.935,\ 0.801,\ 0.883,$$

whereas the probability of the card game ending in a tie has the respective values 0.004, 0.038, 0.083, 0.065, 0.004, 0.038, 0.083, and 0.065. Your probabilities of winning the card game are surprisingly large, even greater than the probabilities of player 2 winning in the Penney Ante game! The reason behind the bigger win probabilities is that, in the card game, play does not end when one of the chosen sequences appears; the game continues with cards remaining in the deck. In this way, your probability of being the final winner can only increase. The win probabilities can be calculated most easily by computer simulation. They can also be calculated using Markov chain theory, but this demands a Markov chain with a 4-dimensional state space, and that leads to rather cumbersome calculations.

Another fascinating card game, or magic trick if you like, goes by the name of Kruskal's count, and it originates from the work done by American physicist Martin Kruskal in the 1970's. Kruskal's work was popularized by renowned American pop-science writer Martin Gardner, whose articles often tackled the complexities of one or another mathematical puzzle and game. The card trick goes like this: a magician invites a spectator to thoroughly shuffle a deck of cards. Then the magician lays out the cards, face up, in one long row (or in a grid). Each card has a number value: aces and face cards (king, queen, jack) have a value of 1, and the other cards take the value of the number on the card. The spectator is asked to think of a number from one to 10. The magician explains that the spot corresponding to that secret number, in the row of cards, is the spectator's first 'key card', and that the value of this key card determines the distance, in steps, to the next key card. If the secret number chosen by the player is 7, then the 7th card in the row of cards will be the spectator's first key card. If the 7th card is a 4, then the 11th card in the row is the new key card. If the 11th card is a jack, the 12th card in the row is the new key card, etc. The spectator counts in silence until reaching a key card with a value that doesn't permit continuing on because the deck has been exhausted and there aren't enough cards left. This ultimate key card is called the spectator's 'final card'. The magician then predicts which card is the final card. And more often than not, the magician will be right! So, what's the trick? It is astonishingly simple. Just as the spectator does, the magician also chooses a secret number between 1 and 10, and starting with this initial key card, silently counts through the row of cards just as the spectator does. Even though the initial key cards of the spectator and the magician are not necessarily one and the same card, there is a high probability that the two will land on the same card in their 'walk' through the row of cards, and from that point on, their paths will be the same. If we suppose that the spectator and the magician each, blindly and independently of one another, choose a number between 1 and 10, then the probability of the magician correctly 'predicting' which card is the spectator's final card is about 93.1%. If the magician uses a double-deck of cards, i.e., two times 52 cards, then this probability of success increases to about 99.5%. Skeptical? Try it out on a willing victim or write a simulation program. An exact formula for the magician's success probability in Kruskal's Count is not known, but here we have the approximation formula:

$$1 - \left(1 - \frac{1}{a^2}\right)^N,$$

in which N is the number of cards and a is the average value of a card. In the example above where not only the aces, but also the king, queen and jack count as 1,

$$a = \frac{1}{13}(1 + 1 + 1 + 1 + 2 + 3 + \cdots + 10) = \frac{58}{13}.$$

Simulation shows that the approximation formula gives amazingly good results. The approximation formula looks simple, but explaining it is anything but that. Markov chain theory underlies the theoretical analysis of this card game.

Another excellent approximation formula can be derived, without using Markov chains.[2] It goes along with the following heuristic argument. The probability is $\frac{9}{10}$ that the magician will choose a different starting number than the spectator. For both magician and spectator, the expected value of the position of the first key card in a row of N cards is equal to $\frac{1}{10}(1+2+\cdots+10) = 5.5$. The first key card has the expected value a for both magician and spectator. The number a also gives the average length of the steps in the 'walk' through the row of cards. This means that, in the row of N cards, the density of the spectator's key cards can be seen as $\frac{1}{a}$. The magician will use, on average, $\frac{N-5.5}{a}$ key cards after the first key card. Each of these key cards has a probability of $\frac{1}{a}$ of coming out at the same spot in the row of cards as the spectator's key card

[2] This approximation is taken from Colm Mulcacy, "*Predictability Outranks Luck*," MAA blogs, December 24, 2012.

has done, or rather, there is a probability of $1 - \frac{1}{a}$ that this will not happen. Some reflections now show that the probability that neither the magician's first key card, nor any key card thereafter, will end up coming out on the same spot in the row of cards as the spectator has done can be approximated by $\frac{9}{10}\left(1-\frac{1}{a}\right)^{(N-5.5)/a}$. So, an alternative approximation formula for the success probability of the magician is:

$$1 - \frac{9}{10}\left(1 - \frac{1}{a}\right)^{(N-5.5)/a}.$$

For one deck of cards ($N = 52$) and for two decks of cards ($N = 104$) respectively, with the card values 1 through 10, this approximation formula gives the success probabilities 93.6% and 99.7%, nearly the same approximate values given by the first approximation formula. In Kruskal's original version of the game, the king, queen and jack were all assigned a value of 5, instead of 1. In that case, $a = \frac{70}{13}$, which gives our magician a success probability of about 85% with 52 cards, and about 98% with 104 cards.

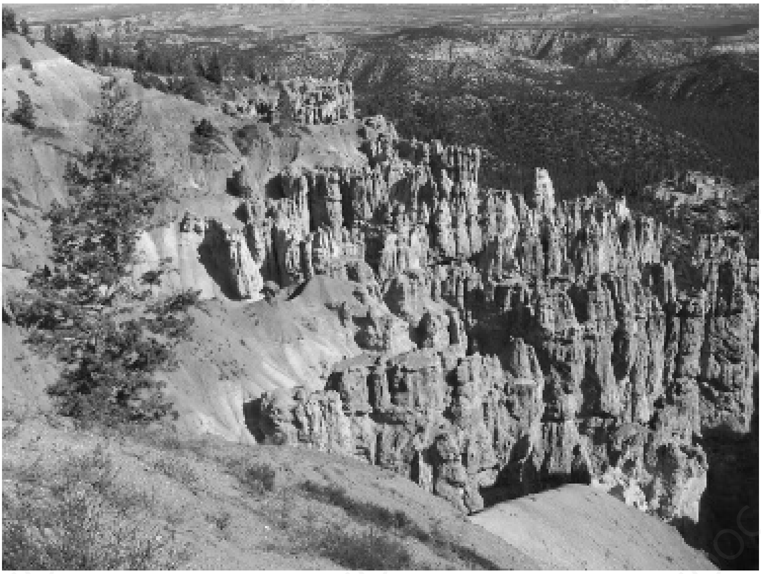
Kruskal's count can also be applied to textual passages. One illustration of this, considered to be fairly mystical by some, is this passage from the King James Version of the Book of Genesis, 1:1-3.

1 *In the beginning God created the heaven and the earth.*
2 *And the earth was without form, and void; and darkness was upon the face of the deep. And the Spirit of God moved upon the face of the waters.*
3 *And God said, Let there be light: and there was light.*

Pick a word in the first verse. Count the number of letters in that word. If the word has L letters, walk forward L words. Continue this procedure until you arrive at a word in the third verse. Stop then. Regardless of which word you choose as your first 'key word', you will always come out on the same word in verse three. This trick works not only for this text, but for many others, as well. The longer the text, the greater the chance it will work. Try it out on a passage from your favorite book and astonish your friends!

CHAPTER 7

The Lost Boarding Pass and the Seven Dwarves



One hundred passengers line up to board a fully booked flight from Reykjavìc to Mallorca. The passengers are Icelandic pub-keepers and they board the airplane one at a time. The first person who ascends the stairs to the plane lets his boarding pass fall from his hands and sees it twirling down through the stairs and under the plane. He decides not to go down and to pick up his boarding pass in order to see what his seat number is. He enters the plane and chooses a seat at random. Icelanders, and especially Icelandic publicans, are relaxed people. So, each person, in turn, finding his assigned seat occupied, reacts amicably by simply choosing a free seat at random in which to sit. What are the chances that the last person entering the plane finds his seat free? This is the famous lost boarding pass puzzle introduced by Peter Winkler.

The surprising answer to the question posed is that the probability of the last passenger finding his seat free is 50%, regardless of the number of passengers. How to get this answer? A useful problem-solving strategy in mathematics is to start first with a reduced version of the problem that is easier to solve. Taking this problem-solving strategy, let us first consider the trivial case of two passengers boarding a two-passenger plane. The last passenger gets his own seat only if the first passenger happens to choose his own seat. Therefore the probability of the last passenger finding his seat free is 50%. Thus, for $n = 2$, we have $P_n = \frac{1}{2}$, where P_n is defined as the probability that the last passenger finds his seat free when there are n passengers boarding an n-passenger plane. Next consider the case that three passengers are boarding a three-passenger plane. Then, by conditioning on the choice of the first passenger, we have

$$P_3 = \frac{1}{3} \times 1 + \frac{1}{3} \times P_2 + \frac{1}{3} \times 0.$$

The argument is simple. If the first passenger takes his own seat, the second passenger can take his assigned seat and then the last passenger will find his seat free. However, if the first passenger chooses the seat of the second passenger, then we can reduce the problem to the previous case of two passengers for a two-passenger plane by imagining that the seat of the first passenger becomes the seat of the second passenger with the second passenger playing the role of the first passenger. The equation for P_3 gives

$$P_3 = \frac{1}{3} + \frac{1}{3} \times \frac{1}{2} = \frac{1}{2}.$$

Similarly, in the case of four passengers boarding on a four-passenger plane, the problem can be reduced to the $n = 3$ problem if the first passenger chooses the seat of the second passenger, and it can be reduced to the $n = 2$ problem if the first passenger chooses the seat of the third passenger. Thus

$$\begin{aligned} P_4 &= \frac{1}{4} \times 1 + \frac{1}{4} \times P_3 + \frac{1}{4} \times P_2 + \frac{1}{4} \times 0 \\ &= \frac{1}{4}\Big(1 + \frac{1}{2} + \frac{1}{2}\Big) = \frac{1}{2}. \end{aligned}$$

Continuing in this way, we get the recursion

$$P_n = \frac{1}{n} + \frac{1}{n}\sum_{j=1}^{n-1} P_{n-j},$$

where $P_1 = 0$. Then, we find by induction that

$$P_n = \frac{1}{2} \quad \text{for any } n \geq 2.$$

A remarkable result, which cries out for an intuitive explanation without mathematical formulas. The key observation is that the last free seat is either the seat of the first passenger or the seat of the last passenger. This is an immediate consequence of the fact that any of the other passengers always chooses his own seat when it is free. Each time a passenger finds his seat occupied, the passenger chooses a free seat at random and then the probability of the first passenger's seat being chosen is equal to the probability of the last passenger's seat being chosen. It is also true that the first passenger chooses his own seat with the same probability that he chooses the seat of the last passenger. Thus the last free seat is either the seat of the first passenger or the seat of the last passenger, each with a 50% chance.

An interesting question is: what is the probability that the person who enters as kth the plane will find his seat occupied? Let's denote this probability by $p_k(n)$ for $k = 1, 2, \ldots, n$. Obviously,

$$p_{n-1}(n) = \frac{n-1}{n}.$$

Using a similar recursion as for the P_n, we get

$$p_k(n) = \frac{1}{n-k+2} \quad \text{for } k = 2, \ldots, n.$$

The details of the derivation are omitted. This result enables us to give a nice formula for the expected value of the number of people who will not be seated on their own seats. This expected value is given by

$$\sum_{k=1}^{n} p_k(n) = \frac{n-1}{n} + \frac{1}{n} + \frac{1}{n-1} + \cdots + \frac{1}{2}$$
$$= 1 + \frac{1}{2} + \cdots + \frac{1}{n-1}.$$

Alternatively, the expected value of the number of people who will not be seated on their own seats can be calculated as

$$\frac{n-1}{n} + \frac{1}{n}\,\mu_0 + \frac{1}{n}\,\mu_1 + \cdots + \frac{1}{n}\,\mu_{n-2},$$

where μ_k is defined as the expected value of the number of additional passengers who will not be seated on their own seats if the first passenger chooses the seat of the $(n-k)$th passenger. Using the recursion $\mu_k = \frac{1}{k+1} \times (1+0) + \frac{1}{k+1}\sum_{j=0}^{k-1}(1+\mu_j)$ with $\mu_0 = 1$, you may easily verify that $\mu_k = \sum_{l=1}^{k+1}\frac{1}{l}$. Recursive thinking is very rewarding! It is known from calculus that the partial sum of the harmonic series can be very accurately approximated as

$$1 + \frac{1}{2} + \cdots + \frac{1}{n-1} \approx \ln(n-1) + \gamma + \frac{1}{2(n-1)},$$

where $\gamma = 0.57722\ldots$ is the Euler-Mascheroni constant. This gives an excellent and insightful approximation for the expected value of the number of people who will not be seated on their own seats. The expected value is about 5.17 when $n = 100$.

The problem of the birthday candles has much ground in common with the problem of the lost boarding pass. Suppose it's your birthday and you have become n years old. There is a birthday cake with n burning candles You are asked to blow out the n candles. The expected number of times you must blow until all of the n candles are blown out is then equal to $\sum_{j=1}^{n}\frac{1}{j}$, as is readily verified from the recursion $E_k = 1 + \frac{1}{k}\sum_{j=0}^{k-1} E_j$ for the expectation as function of the number of burning candles.

A funny variant of the lost boarding pass problem is the seven dwarves dormitory problem. Each of the seven dwarves sleeps in his own bed in a shared dormitory. Every night, they retire to bed one at a time, always in the same sequential order, with the youngest dwarf retiring first and the oldest retiring last. On a particular evening, the youngest dwarf has had too much to drink.

He randomly chooses one of the seven beds to fall asleep on. As each of the other dwarves retires, he chooses his own bed if it is not occupied, and otherwise chooses another unoccupied bed at random. What is the probability that the oldest dwarf sleeps in his own bed? This probability is $\frac{1}{2}$, and the expected number of dwarves who do not sleep in their own beds is 2.45.

To conclude, let us consider the following modification of the seven dwarves dormitory problem. The youngest dwarf is in a jolly mood and decides not to go to his own bed but rather to choose one at random from among the other six beds. Then, as you may easily verify, the probability that the oldest dwarf can sleep in his own bed is $\frac{5}{6} \times \frac{1}{2} = \frac{5}{12}$. The expected number of dwarves who will not sleep in their own beds can be calculated as $\frac{7}{6}(1 + \frac{1}{2} + \cdots + \frac{1}{6}) = 2.858$. The seven dwarves remain a source of inspiration!

CHAPTER 8

Monte Carlo Simulation and Probability — the Interface

```
# Python simulation Monty Hall
from random import choice
def monty(games=100000):
    switch_wins = 0
    stay_wins = 0
    doors = [1, 2, 3]
    for game in range(games):
        prize_door = choice(doors) # randomly choose prize_door
        chosen_door = choice(doors) # randomly choose initial door guessed
        opened_door = choice(list(set(doors) - set([prize_door, chosen_door])))
            # door opened by monty
        switch_door = (set(doors) - set([chosen_door, opened_door])).pop()
            # door chosen if switching
        if chosen_door == prize_door:
            stay_wins += 1
        if switch_door == prize_door:
            switch_wins += 1
    print('Win rate with a "dont́ switch" strategy:', stay_wins / games)
    print('Win rate with a "switch" strategy:', switch_wins / games)
if __name__ == '__main__':
    monty()
```

MONTE CARLO simulation is a powerful probabilistic analysis tool, widely used in both engineering fields and non-engineering fields. It is named after the famous gambling hot spot, Monte Carlo, in the Principality of Monaco. Chance and random outcomes are central to the modeling technique, much as they are to games like roulette, dice, and slot machines. Monte Carlo simulation was initially used to solve neutron diffusion problems in atomic bomb research at Los Alamos National Laboratory in 1944. From the time of its introduction during World War II, Monte Carlo simulation has remained one of the most-utilized mathematical tools in scientific practice. And in addition to that, it has also functioned as a very useful tool for adding an extra dimension to the teaching and learning of probability. It may help students gain a better understanding of probabilistic ideas and to overcome common misconceptions about the nature of 'randomness'. Using computer simulation, a concrete probabilistic situation can be imitated on the computer. A key concept such as the law of large numbers can be made to come alive when students can observe the results of many simulation trials. The nature of this law is best illustrated through the coin-toss experiment. The law of large numbers says that the percentage of tosses to come out heads will be as close to 50% as you can imagine, provided that the number of coin tosses is large enough. But how large is large enough? Experiments have shown that the relative frequency of heads may continue to deviate significantly from 0.5 after many tosses, though it tends to get closer and closer to 0.5 as the number of tosses gets larger and larger. The convergence to the value 0.5 typically occurs in a rather erratic way. The course of a game of chance, although eventually converging in an average sense, is a whimsical process. To illustrate this, a simulation run of 100,000 coin tosses was made. Table 8.1 summarizes the results of this particular simulation study; any other simulation experiment will produce different numbers. The statistic $H_n - \frac{1}{2}n$ gives the observed number of heads minus the expected number after n tosses and the statistic f_n gives the observed relative frequency of heads after n tosses. It is worthwhile to take a close look at the results in the table. You see that the realization of the relative frequency, f_n, indeed approaches the true value of the probability in a rather irregular manner and converges more slowly than most of us would expect intuitively. The law of large numbers does not imply that the absolute difference between the numbers of heads and tails should oscillate close to zero. It is even typical for the coin-toss experiment that the absolute difference between the numbers of heads and tails has a tendency to become larger and larger and to grow proportionally with the square root of the number of tosses. The mathematical explanation of this phenomenon is that the number of heads minus the number of tails after n tosses is symmetrically distributed around 0 with standard deviation $\sqrt{n}$ (the square-root law). Figure 8.1 displays a simulated path of the actual number of heads minus the actual number of tails for 2,000 tosses of the coin. This process is called a *random walk* (or *drunkard's walk*), based on the analogy of an object that moves one step higher if heads is thrown and one step lower, otherwise. Results such as those shown in Figure 8.1 are not exceptional. On the contrary, in the coin-toss experiment, it is typical to find that, as the number of tosses increases, the fluctuations in the random walk become larger and larger and a return to the zero-level becomes less and less likely. This result is otherwise not in conflict with the law of large numbers, since $\sqrt{n}/n$ goes to zero if n gets large.

Table 8.1 Simulation results for 100,000 coin tosses

n	$H_n - \frac{1}{2}n$	f_n	n	$H_n - \frac{1}{2}n$	f_n
10	1	0.6000	5 000	−9.0	0.4982
25	1.5	0.5600	7 500	11	0.5015
50	2	0.5400	10 000	24	0.5024
100	2	0.5200	15 000	40	0.5027
250	1	0.5040	20 000	91	0.5045
500	−2	0.4960	25 000	64	0.5026
1 000	10	0.5100	30 000	78	0.5026
2 500	12	0.5048	100 000	129	0.5013

The coin-toss experiment is full of surprises that clash with intuitive thinking. Unexpectedly long sequences of either heads or tails can occur ('local clusters' of heads or tails are absorbed in the average). We come back to this matter in Chapter 12. One of the most surprising results in the coin-toss experiment is the so-called arcsinus-law. Suppose that you are involved in a coin-toss experiment. You have committed yourself to a pre-agreed number of tosses and you keep to the agreement. The arcsinus-law tells us that the random walk describing the difference between the actual number of heads and tails tends, most of the time, to occur on one side of the axis line. Intuitively, one would expect that the most likely value of the percentage of total time the random walk occurs on the positive side of the axis would be somewhere near 50%. But

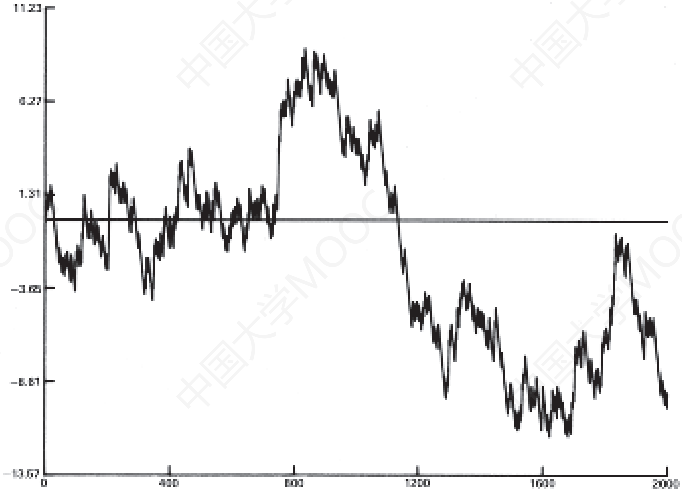


Figure 8.1 Realisation of a random walk in coin-tossing

quite the opposite is true, as may be verified by simulation. If the pre-agreed number of tosses is large, then more than 80% of the time, the random walk will stay on the same side of the axis with a probability of about 0.590. More than 95% of the time, it will stay on the same side with a probability of about 0.287. This is a counter-intuitive result that analysts do well to keep in mind when analyzing financial markets.

How do we simulate the coin-toss experiment on the computer? We make use of the computer's *random-number generator*. This generator produces a sequence of numbers that are picked at random from between 0 and 1 (excluding 0 and 1). The result is something like the fickle finger of fate pointing to a number between 0 and 1. The outcome of the coin toss is simulated by a random number u between 0 and 1: if $u \leq 0.5$, the outcome is heads; otherwise, it is tails. Of course, the numbers generated by the computer are not, strictly speaking, truly random numbers; a computer generates pseudo-random numbers. This is achieved through a deterministic procedure that is iterative by nature. The older random-number generators generate a sequence of positive integers z_0 (seed), $z_1, z_2, \ldots$ according to the procedure $z_n = a \times z_{n-1}$ (modulo m) for suitably chosen positive integers a and m. The best choice is $a = 16{,}807$ $(= 7^5)$ and $m = 2^{31} - 1$. This scheme produces each of the numbers

1 to $m-1$ before it repeats itself ($u_k = z_k/m$ is a random number between 0 and 1). The cycle length is a little over two billion numbers. But today, this is not enough for more advanced applications, such as those used in physics and financial engineering. The newest random-number generators do not involve multiplications or divisions at all. They are the Christopher Columbus generator with a cycle length of about 2^{1492} and the Mersenne twister generator with a cycle length of $2^{19\,937}-1$. These generators are very fast, have incredibly long periods and provide high-quality pseudo-random numbers that have passed all statistical tests for randomness.

Simulation is not only good for purposes of clarifying basic concepts in probability theory by means of experimentation. It is also very useful for quickly finding answers to probability problems when it is not immediately clear whether such problems are analytically solvable. One small alteration to an easily solvable probability problem can quickly lead to a probability problem that is very difficult to solve analytically. Let's use the classic birthday problem to illustrate. In this problem, we want to know the probability that two or more persons in a randomly formed group of m persons will have birthdays on the same day. This question is an easy one to answer analytically, but that is not the case when the question is altered to ask for the probability of two or more persons having birthdays within one day of one another. The simulation program, however, does remain simple: in every simulation run, you generate m times a random integer from the integers 1 to 365, and then you go through these m integers to see whether there are two integers that correspond with birthdays occurring just one day apart. A random integer from $1, 2, \ldots, 365$ is found by having the computer generate a random number u from $(0, 1)$ and rounding up the number $365 \times u$. If we take a group of $m = 10$ persons and we execute a large number of simulation runs, say 100,000, then we find that the probability of two or more persons having birthdays one day apart is about 31.5%. Also, by simulation, we find that the probability of two or more persons having birthdays within a week of one another is about 86.4% for a group of 10 persons.

Many geometric probability problems can be quickly and easily solved using simulation. Take the problem of the expected value of the distance between two randomly chosen points in a unit square (side length 1) or a unit circle (radius 1). This problem is difficult to solve analytically, and advanced integral calculus is needed. Simulation, on the other hand, offers a simple and fast approach to the problem of finding the average distance. For this approach, all you need to know is how to generate a random point within a square or a circle. In a square with vertices (a, b), (c, b), (a, d) and (c, d), you generate a random point (x, y) by having the computer generate two random numbers u_1 and u_2 from $(0, 1)$ and taking $x = a + (c - a)u_1$ and $y = b + (d - b)u_2$. In the unit circle with $(0, 0)$ as center, you generate a random point by enclosing the circle in a square with vertices $(-1, -1)$, $(1, -1)$, $(-1, 1)$ and $(1, 1)$ and continuing to generate random points in this square until you arrive at a random point (x, y) satisfying $x^2 + y^2 \leq 1$. This is an application of the powerful *hit-and-miss method.*[1] You generate two random points a great many number of times within a given area, and each time you use Pythagoras to calculate the distance between the two points. Finally, you take the average of all of the distances. It is as simple as that! One million simulation runs show that the average distance between two randomly chosen points is about 0.521 for the unit square and about 0.905 for the unit circle. For the unit cube and the unit sphere, simulation is just as simple and results in values of about 0.662 and 1.029 for the average distance between two randomly chosen points. The hit-or-miss method used to generate random points inside the circle can also be used to generate random points in any given bounded region in the plane or in other spaces in higher dimensions. Enclosing the region inside a box enables you to estimate the area of the region by generating a large number of random points inside the box. The ratio of the number of points falling inside the region and the total number of generated points estimates the ratio of the area of the region and the area of the box. In this way high-dimensional integrals can be calculated using simulation.

There are countless probability problems that can be quickly solved using simulation, whereas finding analytical solutions requires no small effort. This is often the case with tricky combinatorial problems. Take the following entertaining problem. What is the probability that no two adjacent letters will be the same in a random permutation of the eleven letters of the word Mississippi? This is a very tricky problem to solve analytically, but simulation is easily done by using a simple procedure for generating a random permu-

[1] The idea of the hit-or-miss method was introduced to the statistics community by physicists N. Metropolis and S. Ulam in their classical paper "*The Monte Carlo method*," *Journal of the American Statistical Association,* 1949, 44, 335-341.

tation of the integers 1 to n with $n = 11$. The sought-after probability is about 5.8%. Another interesting combinatorial problem has to do with the lotto. Anyone who keeps an eye on lotto results will have noticed that consecutive numbers, say 32 and 33 for example, come up quite frequently. How large is this probability? This is difficult to determine analytically, but not so using simulation. Simulation tells us the probability of two or more consecutive numbers coming up in a lotto 6/45 draw is about 52.9%.

How many simulation runs should be made in order to get a desired level of accuracy in the estimate? It is never possible to achieve perfect accuracy through simulation. All you can measure is how likely the estimate is to be correct. When doing a simulation, it is important to have a probabilistic judgment about the accuracy of the point estimate. Such a probabilistic judgment is provided by the concept of 'frequentist confidence interval'. Suppose you want to estimate the unknown probability p of a particular event E. If n simulation runs are performed and the event E occurs in s of these n runs, then $\hat{p} = s/n$ is the estimate for the sought-after probability p. The accuracy of this estimate is expressed by the 95% confidence interval

$$\Big(\hat{p} - 1.96\frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}},\, \hat{p} + 1.96\frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}}\Big).$$

The 95% confidence interval should be interpreted as follows: with a probability of about 95% the interval will cover the true value of p if n is large enough. The effect of n on the term $\sqrt{\hat{p}(1-\hat{p})}$ fades away quickly if n gets larger. This means that the width of the confidence interval is nearly proportional to $1/\sqrt{n}$ for n sufficiently large. This conclusion leads to a practically important rule of thumb: to reduce the width of a confidence interval by a factor of two, about four times as many observations are needed. Let's illustrate by finding the probability that, in a house where 7 students reside, two or more students will have birthdays within a week of one another. If we make 25,000 simulation runs for the birthdays of 7 students, we get the probability estimate 0.6072 with $(0.6011, 0.6132)$ as 95% confidence interval, whereas 100,000 simulation runs result in an estimate of 0.6056 with $(0.6026, 0.6086)$ as 95% confidence interval. The confidence interval has indeed been narrowed by a factor of two.

How can we construct a confidence interval if the simulation study is set up to estimate an unknown expected value rather than an unknown probability? The expected value is estimated by $\overline{X}(n) = \frac{1}{n}\sum_{k=1}^{n} X_k$ when $X_1, \ldots, X_n$ represent the independent observations from the n simulation runs. The corresponding 95% confidence interval is

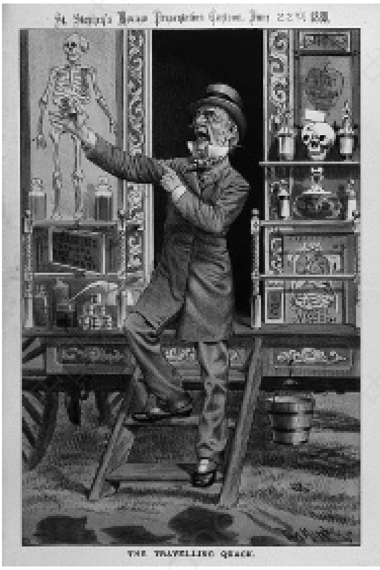
$$\Big(\overline{X}(n) - 1.96\frac{\sqrt{\overline{S}^2(n)}}{\sqrt{n}}, \overline{X}(n) + 1.96\frac{\sqrt{\overline{S}^2(n)}}{\sqrt{n}}\Big),$$

where $\overline{S}^2(n) = \frac{1}{n}\sum_{k=1}^{n}[X_k - \overline{X}(n)]^2$ is an estimator for the variance of the X_i. The statistic $\overline{S}^2(n)$ reduces to $\overline{X}(n)\big(1 - \overline{X}(n)\big)$ if the X_i can take on only the values 0 and 1.

The following statement, often attributed to Martin Gardner, contains a lot of truth: “In no other branch of mathematics is it so easy for experts to blunder as in probability theory.” This is why simulation is so very useful as a tool for verifying whether the calculation of a particular probability is correct, or for showing the correctness of a surprising solution to a probability problem. Two engaging examples of probability problems with surprising solutions are the hundred prisoners problem, described in Chapter 4, and the lost boarding pass problem, described in Chapter 7. But when we’re talking about the effectiveness of simulation to convince an audience of the correctness of a particular solution, then perhaps the most engaging problem of all is the three-doors problem, also known as the *Monty Hall dilemma*. This problem is named after the first presenter of the popular American quiz show, *Let’s Make a Deal*, who became a global celebrity thanks to American columnist Marilyn vos Savant, who took on the three-doors problem in a 1990 column for *Parade Magazine*. The problem goes like this: the quiz show finalist is presented with a choice of three doors. Behind one door is an expensive car, and behind the other two are gag prizes. The finalist, with no more than luck for a guide, chooses one of the doors. The quiz master knows exactly what is behind each door. He has pledged to open one of the ‘gag’ doors after the finalist chooses a door (an essential component of the problem!), and this, in fact, is what happens. Next, he asks the candidate whether he would like to shift his choice of door to the other, remaining door? The candidate is then faced with a dilemma. What to do? The advice given by Marilyn vos Savant in her column was to switch, and go with the remaining door. This way, the candidate increases the probability of winning the car to $\frac{2}{3}$. This reasoning is correct. Nevertheless, vos Savant was flooded with thousands of letters, mainly from readers who disagreed with her solution. Some registered their disapproval in no uncertain terms. Ninety percent of the letter-writers were of the opinion that switching doors would have no effect on the outcome of the 'deal'. Their argument was that the two unopened doors left at the final phase of the game each had a probability of $\frac{1}{2}$ of hiding the car. Some of the letter writers were mathematicians. One of them wrote in: "As a professional mathematician it concerns me to see a growing lack of mathematical proficiency among the general public. The probability in question must be $\frac{1}{2}$. I caution you in future to steer clear of issues of which you have no understanding." The most well-known of the mathematicians to write in a challenge to vos Savant was world-famous mathematician Paul Erdös. He was convinced of the inaccuracy of vos Savant's answer until a colleague showed him a simple simulation program displaying crystal clear evidence that vos Savant was right (such a simulation program is shown at the beginning of the chapter).

CHAPTER 9

Lotto Nonsense: The World is Asking to be Deceived



LOTTO AND ROULETTE are games of pure chance; nonetheless, there are loads of books on the market that would have readers believing in the existence of systems that help you win. These systems and their inventors always remind me of snake oil peddlers depicted in the old westerns. Usually a bit long-in-the-tooth and down-at-the-heels, these glib, fast-talking swindlers palmed off utterly useless health elixirs and salves on unsuspecting farmers and homesteaders. Of the many strategic systems claiming to increase one's chances of winning the lottery, let's focus on debunking two: the 'Balanced Numbers' system and the 'Rainbow' system. The former was conceived of by an American who has been promoting herself as America's most trustworthy lottery expert for more than 30 years, and who has a huge following of gullible believers. Her books are all best sellers and have earned her much more money than she ever would have gained by playing the lottery. The Balanced Numbers system is one of the jewels in her crown. She lures faithful followers by claiming that the system is based on the principle of the 'bell curve', expropriating, in this way, the name of one of the greatest mathematicians of all time, Carl Friedrich Gauss. How is the system used? It is brilliantly simple. To illustrate, let's take the lottery game 6/49, in which six different numbers are randomly chosen from the series of numbers 1 through 49. For this lottery, the 'balanced numbers' method advises players to choose six different numbers such that their sum will fall between 117 and 183. This advice is based on the fact that the probability distribution of the sum of the six drawn numbers can be accurately approximated by a normal distribution with its characteristic bell-shaped curve. Figure 9.1 shows the probability distribution of the sum of six randomly drawn numbers in the game of 6/49, for which the simulated probability distribution is based on 1 million simulation runs. The expected value and the standard deviation of the approximating normal distribution are 150 and 32.8. More generally, for the r/s lottery in which r distinct numbers are randomly drawn from the numbers 1 to s, the sum of the drawn numbers has an approximate normal distribution whose expected value and standard deviation are

$$\mu = \frac{1}{2}r(s+1) \quad \text{and} \quad \sigma = \sqrt{\frac{1}{12}r(s+1)(s-r)}.$$

For the normal distribution with expected value μ and standard deviation σ about 68% of the probability mass lies between $\mu - \sigma$ and $\mu + \sigma$. And that is why players of the 6/49 lottery are advised to choose six numbers that add up to a sum between 117 and 183. This strategy purportedly increases the player's odds of winning a lottery prize. Pure poppycock! This advice completely neglects the fact that there are many more combinations of six numbers

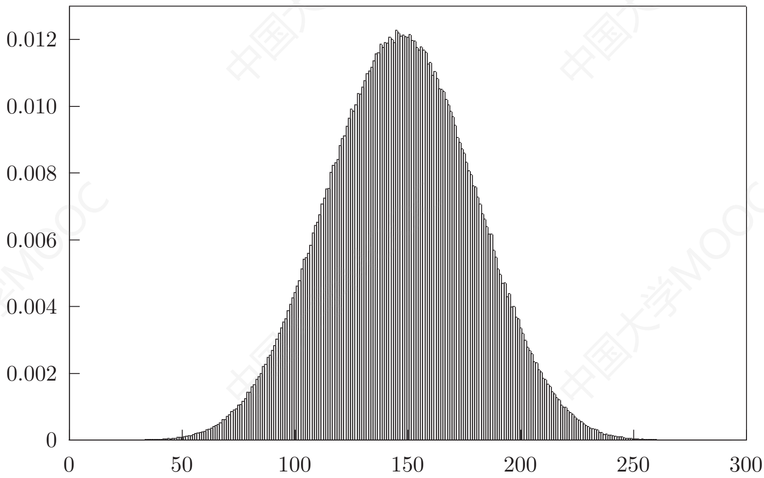


Figure 9.1 Distribution of the sum of the winning numbers.

whose sum falls between 117 and 183 in the middle of the sum's distribution, than there are combinations of six numbers whose sum falls between, say, 21 and 87, on the far end of the distribution. In any case, if you play this lottery, you are betting on the six individual numbers that are going to be drawn, and not on what their sum will be.

The 'Rainbow' method was developed by a Brazilian mathematician who reckoned he could earn more money outside the hallowed walls of academe than within them, and hastened to put this theory to the test. He unveiled the principle behind the Rainbow method in an article published in 2013 in a Brazilian scientific journal. This article made a big splash in the media, after which our mathematician decided to build a website geared towards playing not only the Brazilian 6/48 lottery, but that also advised on the best way to fill in lottery tickets, worldwide. As far as I can tell, the article makes no sense at all. It claims that the Rainbow Numbers system is based on the law of large numbers, but it appears to employ statistical intimidation, rather than real evidence, as a means to bamboozle unsuspecting lottery players. The workings of this so-called miracle method are described using the Brazilian lottery 6/48 Super Sena, in which six different numbers are randomly drawn from the numbers 1 to 48. If I understand the author's hocus-pocus theory correctly, the method goes as follows. The numbers 1 through 48 are divided into five groups, and each group is assigned a color. The numbers 1 through 9 get the color yellow, numbers 10 through 19 get the color blue, numbers 20 through 29 get the color red, numbers 30 through 39 get the color violet, and the numbers 40 through 48 get the color purple. Then, each sequence of six numbers from the numbers 1 to 48 takes on a color code. For example, the sequence $2-5-8-17-25-38$ takes on color code YYYBRV, and the sequence $7-15-23-32-43-44$ takes on color code YBRVPP, the order of the symbols in the color code being not relevant. The central argument of the Rainbow method is that some color codes are more likely to come up than others, among the six numbers in the lottery draw. For example, there is a probability of about 1 in 150 that a six-number sequence from the YYYBRV color group will be drawn, and a probability of about 1 in 38 that a sequence from the color group YBRVPP will be drawn. The probability of the latter group turning up, then, is four times higher than that of the former group. The website tries to sell players on the idea that they can increase their chances of winning the jackpot by filling in six-number sequences with the appropriately chosen color codes; these codes are offered for sale for a modest sum. And although the level of flim-flammery contained in this color-scheme advice is even greater than that of the 'Balanced Numbers' method, there are apparently enough takers (there's a sucker born every minute, according to the old adage) to keep the enterprise up and running. They take the Rainbow method at face-value, not realizing that there are about 4 times as many ticket numbers with a YBRVPP color code as there are ticket numbers with a YYYBRV color code. Lottery win-schemes are no different from any other form of quackery that offers miraculous results.

Where lotteries are concerned, there is only one advice 'system' based on solid evidence, and that is this: steer clear of the popular numbers used by players in lottery games. Never choose sequences such as $1-2-3-4-5-6$ or $7-14-21-28-35-42$ for your six numbers if you are playing the lottery 6/45. Not because of anything having to do with your probability of winning the jackpot, but rather because, in the extremely unlikely case of the jackpot falling on one of these sequences, you can be sure that you will have to share the jackpot with a huge number of other players. People often use their birth dates, arithmetical sequences, lucky numbers, etc., in choosing their lottery numbers. In lotteries where the majority of the tickets are filled in by hand, it appears that the number of winners of the jackpot is largest when most of the six numbers drawn fall in the lower range.

Lottery win-systems claiming to increase your chances of getting rich are utterly worthless. If you want to be sure of doubling your money, my advice is this: fold it in half! Now that we know our onions, as it were, when it comes to the lottery, we can move on to the subject of win-systems for roulette. Let me be very clear: they do not exist! No matter how you play the game of roulette, you cannot escape the fact that, in the long run, the casino has a house edge. To illustrate, let's look at European roulette, which uses a wheel containing 37 numbers from 0 through 36.[1] Of the remaining numbers not equal to 0, there are 18 red and 18 black numbers. The croupier releases the ball into the game by rolling it onto the wheel in the direction opposite to the one in which the wheel is traveling. The number assigned to the pocket where the ball comes to a stop is the winning number. Players may bet on either a single number or on a combination of k numbers, for which k is equal to 1, 2, 3, 4, 6, 12 or 18. If the ball stops on a number in the player's chosen combination of k numbers – there is a probability of $\frac{k}{37}$ that this will happen – then the player wins $\frac{36}{k}$ times the amount staked; otherwise, the player loses the stake. In particular, if the player bets on red or black ($k = 18$), then the player wins two times the amount staked if the ball comes to rest on that color. The number 0 is always a win for the house in European roulette, and this gives the casino an advantage over the player. Indeed, the expected value of a casino win on every euro staked on a combination of k numbers is positive and is equal to $1 - \frac{k}{37} \times \frac{36}{k} = \frac{1}{37}$. In other words, the casino's house edge is $\frac{1}{37} \times 100\% = 2.70\%$ regardless of which combination of numbers the player bets on. In the long-run, you simply cannot beat this house edge, no matter what system you use. Nevertheless, loads of betting systems claim to be winners. But there is no winning combination to be made from individual stakes that are individual losers. This can be shown mathematically, but for most people, the results are more readily convincing when achieved by means of computer simulation.

[1] Blaise Pascal – the French philosopher and mathematician – is the father of the roulette wheel. He was not trying to invent a casino game, rather he was looking to invent a perpetual motion machine. Roulette emerged in the early 19th century as a glorious attraction in the casinos of Europe after legalization of the game by Napoleon in 1806.

A popular betting system for roulette is the Labouchère system. Using this system, you must decide beforehand how much you want to win, and after that you make a list of positive numbers whose sum add up to this amount. The list is updated after each bet. You bet on red each time. For each bet, you stake an amount equal to the sum of the first and last numbers on your list (if the list consists of just one number, then that number is the amount of your stake). If you win the amount staked, you cross off the amounts you used from your list. If you lose, you add the amount lost to the bottom of your list. You continue in this manner until your list is used up (target amount has been achieved) or until you have lost all of your money. The Labouchère system is an exciting system, and there is no reason not to enjoy using it for its entertainment value, but here also, with regard to long-term play using this system to play European roulette, your average loss per staked euro inevitably comes out at, or near the 2.70 cent mark. Let's show this for a specific situation. Say your win-target is 250 euro and that you have a bankroll of 2500 euro. Off you go to the casino. You configure a starting list made up of the numbers: 50, 25, 75, 50, 25, 25. In the first round, then, you stake $50 + 25 = 75$ euro. If you win, your list is narrowed down to 25, 75, 50, 25, and if you lose, the list is extended to 50, 25, 75, 50, 25, 25, 75. It is worth noting that at a given moment the sum of the first and last numbers on the list might be larger than the amount of money still in your possession at that time. For example, let's say that at a particular moment your first and last numbers are 50 and 125, respectively, whereas you only have 150 euro left to bet. In that case you would bet the entire sum of 150 euro. In each simulation run, you begin with a bankroll of 2500 euro, and play according to the Labouchère system until either your bankroll has increased by 250 euro, or you have gone broke. Executing 100,000 simulation runs produces the result that the player, on average, loses money with an average loss of about 0.0274 euro per euro staked (the average loss per round was about 62 euro and the average amount staked per round was about 2265 euro). This simulated value of 0.0274 confirms that, for the long run, you cannot beat the house edge of 2.7% for the casino. We can conclude, then, that there simply is no system for playing roulette that leads to a player's win over the long run. If we are talking short run play, it is possible for a player to win at roulette. To illustrate this: we get a simulated value of 0.8865 for the probability P_w that a one-time execution of the Labouchère system will end in a win for the player of 250

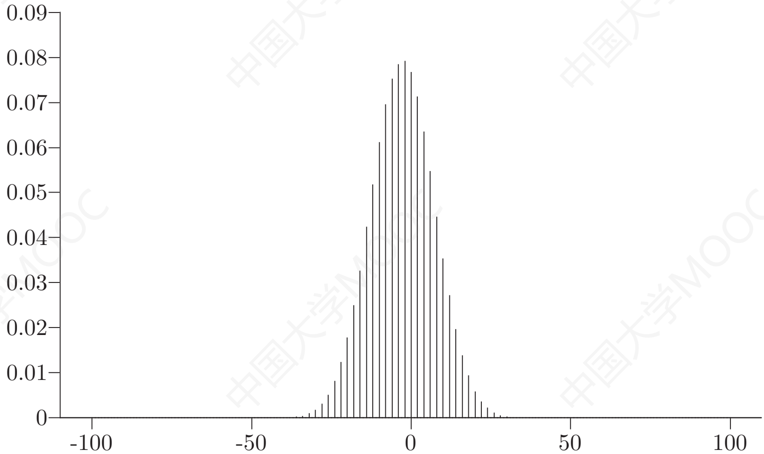


Figure 9.2 Gain/loss distribution for the flat system.

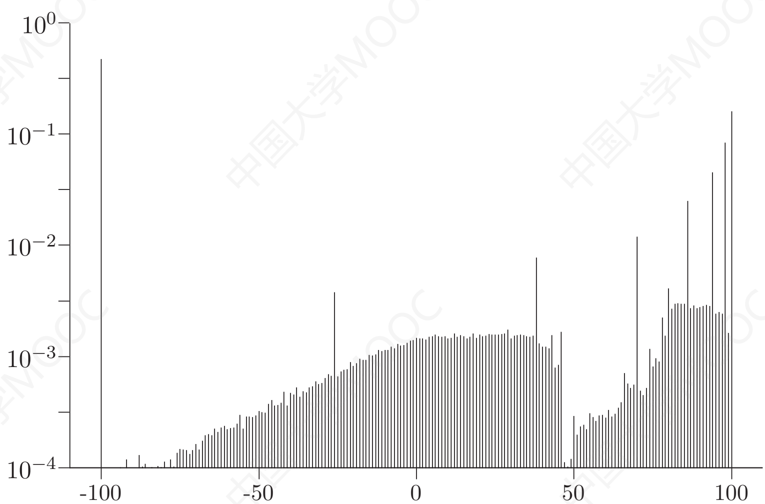


Figure 9.3 Gain/loss distribution for the Big-Martingale system.

euro. This large win-probability for a one-time execution does not contradict the fact that the system is a losing one when applied over a large number of executions.

No matter which roulette system you try, Labouchère, Big-Martingale or the Fibonacci system, for play extended over the long run you lose on average 2.7 cents for every euro staked when playing European roulette. The roulette ball has no memory, and this means that it is impossible to make winning combinations of bets when each individual bet is a losing proposition. It is true that the one system is more exciting than the other, but when it comes right down to it, over the long run, each system has an average loss of 2.7 cents for each euro staked. These systems only differ from one another in patterns of betting and in the way in which they reconfigure a player's losses and wins. This is illustrated in Figures 2 and 3, where the flat betting system is compared with the Big-Martingale betting system. In a flat betting system, the player stakes 1 chip on each bet. The Big-Martingale system works like this: your first stake is 1 chip. After a loss, your next stake will be two times the former stake plus 1 chip (if you don't have enough chips to allow this, then you simply stake whatever amount you do have); after a win, your next stake will, again, be 1 chip. For example, if, using this system, your first win comes after 4 stakes, the first four will have consisted of 1, 3, 7 and 15 chips. Assuming you have bet on red each time, after the 4th stake, you receive 30 chips from the house; your profit after 4 stakes, then, is equal to $30-(1+3+7+15) = 4$ chips. For both systems, we have simulated a game consisting of one million rounds. In each round, you begin with 100 chips and may place a maximum of 100 bets, always on red. Using the flat betting system, you will stake precisely 100 bets, whereas using Big-Martingale, you will stake fewer than 100 bets if you have lost all of your chips before the game gets to the hundredth bet. Figures 9.1 and 9.2 clearly show that the probability distribution of the number of chips lost at the end of a play round is entirely different for the 'duller' flat betting system than for the more 'exciting' Big-Martingale system (a logarithm scale is used in the figure for the Big-Martingale system). You won't find it surprising to hear that for the Big-Martingale system the probability mass of the gain/loss in a single round is much more strongly concentrated in the tails than is the case for the flat system. For both systems, however, it holds that, over the long run, you cannot beat the casino's house edge of 2.7%. The simulation of one million play rounds gives an average loss of 2.71 out of every 100 chips per play round in the flat betting system, and an average loss of 2.69 out of every 100 chips per play round in the Big-Martingale system. Attempts to influence your average loss over the long run by using a betting system for roulette are just as fruitless as the bizarre attempt of a long-ago despot to influence the ratio of newborn boys to girls by ordering that, as soon as a woman bore a son, she was prohibited from bearing any more children. In the long run the gambler's chances in roulette are the same as those of a lamb in the slaughterhouse.

A non-mathematical but nonetheless convincing proof of the fact that a winning betting system does not exist for the game of roulette is evident from the fact that casinos have never shown any resistance to the use of any such a system at the roulette table. Going back in time, there are gamblers who have broken the bank by detecting biased roulette wheels or by using electronic equipment to predict the path of the ball. The first and most famous biased wheel attack was made in 1873 by the British mechanical engineer Joseph Jagger. He recruited a team of six clerks to record the outcomes of the six roulette wheels at the Monte Carlo casino in Monaco. Jagger detected a biased roulette wheel showing up the nine numbers 7, 8, 9, 17, 18, 19, 22, 28, and 29 far more often than a random wheel would suggest. In a cat-and-mouse game with the casino, Jagger and his team ultimately won two million francs, which was a fortune for 1873! More recently, gamblers smuggled a laser scanner linked to a microcomputer in a mobile phone into the casino at the Ritz hotel in London, and using these high-tech devices they won 1.3 million British pounds on the evening of March 16, 2004. The scanner noted where the ball had dropped and measured the declining speed of the wheel. These factors were beamed to the microcomputer, which calculated the most likely section of six numbers upon which the ball would finally settle. This information was flashed onto the mobile phone just before the wheel made its third spin, by which time all bets must be placed. Then, the gamblers placed their bets in the area the computer had pinpointed as the ball's most likely resting place. You only have to increase your odds by 3% to go from losing on average to winning on average. The gamblers were arrested by Scotland Yard afterwards, but there were no legal grounds for charging them with 'cheating'. They did not break the law by using a laser scanner – the scanner has, after all, no effect on the outcome of the roulette wheel – and they were permitted to keep their winnings. Of course, they were barred from ever again entering a casino. Casinos have the right to bar you – without cause – but especially if they notice you consistently beating the odds. At the end of the day, the only sure way to get rich by means of roulette is to open your own casino!

CHAPTER 10

March Madness Grips the USA

THE YEARLY 'March Madness' tournament organized by the American basketball authority, NCAA, is one of the biggest sporting events in the USA. It goes on for about a month and is covered by all of the major television stations. Sixty-four university basketball teams take part in the tournament, and a total of 63 games are played. It is a knockout competition comprising six rounds. According to a preset playing roster, two teams compete, and the losing team is eliminated from the tournament. In advance of tournament play, millions of Americans fill in forms known as 'brackets'. These forms represent a diagram for the sequence of games to occur; participants fill in the diagram with their predictions of the winners. A perfect bracket is one that has correctly predicted the winners of all 63 games. In the thirty years since the 64-game tournament was established, a perfect bracket has never surfaced. The best attempt to date is that of a 17-year-old schoolboy from Chicago, who, in 2010, correctly predicted the results of all 48 games in the first two rounds. Quite a feat, considering that even predicting the outcome of all 32 games in the first round is next to impossible. President Obama, a big fan of the hoops, could hardly contain himself when, in 2011, he correctly predicted nearly all of the first-round wins, thereby gaining admittance into the group of best 'predictors' up to that point. Feeling rather well-pleased at such an accomplishment, he couldn't prevent himself from crowing just a bit, in front of the television cameras. This brought on a wave of criticism. Did the president have nothing better to do? At that time, an earthquake and tsunami were wreaking havoc in Japan and elsewhere, and tensions were high in North Africa and the Middle East.

In 2014, March Madness reached a sort of fever pitch. Warren Buffett wrote out a check for 1 billion dollars , to be awarded to anyone who could deliver a perfect bracket for that year, i.e., one that correctly predicted the winners of all 63 games. Participants were required to provide some personal information via a form submitted to Berkshire Hathaway Inc., of which Buffet is the number one boss. Participants were required to be at least 18-years-old, and only one form per participant was permitted. Further, only the first 15 million forms submitted would be considered for the 1-billion-dollar prize money. Buffett knew well enough that he ran very little risk of having to cough up the dough. In announcing the billion-dollar contest, he explained that if you picked your winners by tossing a fair coin, you would have a probability of 1 out of 9.2 quintillion (that's a 1 followed by 18 zeroes) of having all 63 winners correct. At these odds, Buffett could well afford to tempt a population of 7.3 billion souls to take the gamble. Then, the probability that he would actually have to pay up can be calculated as

$$1 - e^{-7.3\times 10^9 \times (1/2)^{63}} \approx 7.9 \times 10^{-10},$$

being the Poisson probability of at least one success in 7.3×10^9 independent trials each having a success probability of $(1/2)^{63}$. Of course, the reality of the situation in that year, as in any year, was that not all participating teams are evenly matched, and quite a few basketball fans know their subject. Statisticians quickly got to work and came up with a probability of about 1 in 128 billion for an expert correctly predicting the outcome of all 63 games. How did they arrive at this probability? They calculated an estimate for the probability that each of the 63 games would be won by the higher-ranked team (the teams are ranked before the tournament begins), and they took this probability as an estimate for the probability that a basketball expert would achieve a perfect bracket. Of course, they needed good estimates for the win-probabilities of the higher-ranked teams for every round in the tournament. First, let's look at how the tournament is organized: the first four rounds are played in four different regions of the USA, with 16 teams per region. In the first round, the team ranked #1 plays against the team ranked #16, the team ranked #2 plays against the team ranked #15, and so on. For the first round, then, the probabilities of wins for the higher-ranked teams are estimated at:

$$\#1-\#16: 1, \quad \#2-\#15: 0.93, \quad \#3-\#14: 0.86, \quad \#4-\#13: 0.79,$$
$$\#5-\#12: 0.72, \quad \#6-\#11: 0.65, \quad \#7-\#10: 0.58, \quad \#8-\#9: 0.51.$$

The first and last of these probabilities are estimated on the basis of historical data. The remaining probabilities are estimated by starting with a probability of 1, and decreasing each successive probability by 0.07; this gives a picture consistent with historical tournament data. Assuming that the outcomes of the various games are independent from one another, we get the estimate

$$p_1 = (1 \times 0.93 \times \cdots \times 0.51)^4 \approx 5.85 \times 10^{-5}$$

for the probability that, in the first round, each game taking place in the four regions ends in victory for the higher-ranked team. This probability is about 1 in 17 thousand. In the second round, 8 teams compete in four regions: the team ranked #1 plays against the team ranked #8, the team ranked #2 plays against the team ranked #7, and so forth. For each second-round game, the value 0.65 (estimated according to historical data) represents the probability that the higher-ranked team wins. This means there is a probability of $p_2 = 0.65^{16}$ that each of the 16 games in the second round will be won by the higher-ranked team. This brings us to the estimate

$$p_1 \times p_2 \approx 5.94 \times 10^{-8}$$

for the probability that, in the first two rounds, each game will be won by the more highly ranked team. This probability is about 1 in 16.8 million. In the third round, we find the team ranked #1 versus the team ranked #4, and the team ranked #2 versus the team ranked #3, in each of the four regions. The probability of a win for the higher-ranked team in round three is estimated at 0.60, which gives the estimate $p_3 = 0.60^8$ for the probability that each of the 8 games in the third round will be won by the higher-ranked team. Next, the probability that , in the first three rounds, each game will be won by the higher-ranked team is estimated by

$$p_1 \times p_2 \times p_3 \approx 9.98 \times 10^{-10}.$$

This probability is about 1 in one billion. After the first three rounds, there are still 7 games to be played in the tournament. For each of these 7 games, it is assumed that the win probability for the higher-ranked team is equal to 50%, so the probability that the outcome of all 7 of these games will be correctly predicted is equal to $p_4 = 0.5^7$. This leads to the estimate

$$p_1 \times p_2 \times p_3 \times p_4 \approx 7.80 \times 10^{-12}$$

for the probability that, in all 63 games, the higher-ranked team will win. This probability is about 1 in 128 billion. About ten million Americans took Buffett up on his 1-billion-dollar challenge. If we can agree that all ten million of these Americans were basketball experts, then the probability that not one of them will achieve a perfect bracket (one that correctly identifies the winners of all 63 games) can be estimated by the Poisson probability

$$e^{-10^7 \times (7.80 \times 10^{-12})} = 0.999922.$$

Thus, the probability that Buffett would actually have to pay out the 1-billion-dollar prize is on the order of 7.8×10^{-5}. In reality, the probability of a pay-out is even lower than this. Not all participants are experts on the subject of basketball. Right from the start it was clear that Buffett would be keeping his billion right in his own pocket. Not that he would have had trouble coming up with the cash. It is not too bold to say that his estimated net worth of 74.4 billion dollars would have kept him out of the poorhouse if he had had to deliver the goods. In any case, he did pay out twenty times 100-thousand dollars to the participants who submitted the best brackets. But even this is a drop-in-the-proverbial-bucket when you consider how much valuable information he was able to gather from his participants. Not only did he get 10 million email addresses, he also got information about income and property ownership. This is valuable information for Buffett's Berkshire Hathaway organization, information purchased for 20 cents on the dollar per participant, as opposed to the stiff price-tag normally attached to efforts of acquiring such a wealth of information. It is not for nothing that Buffett is one of the richest men in the world! A risk-free investment, that is what you could call Warren Buffett's 1-billion-dollar challenge.

James Randi displayed more daring, and less self-interest, when he put up his own money to combat and unmask the multitudes of 'psychics' with so-called paranormal powers. In the TV-show "Exploring Psychic Powers Live," broadcast live in the USA on the 7th of June, 1989, James Randi put a group of psychic mediums through a series of tests. The show offered \$100,000 to any participating psychic who could demonstrate genuine paranormal powers. James Randi, himself a well-known magician, pledged \$1,000 of his own money towards each eventual \$100,000 pay-out. One participating psychic attempted a 'sorting' of 250 concealed Zener cards. For each concealed card, she would divine (guess!) which of the five Zener symbols was displayed. She would win the \$100,000 if she could correctly identify 82 or more of the 250 cards. Zener cards were developed in 1930 by Karl Zener together with celebrated parapsychologist Joseph Rhine in order to conduct experiments in ESP (extrasensory perception). Randi's Zener-card psychic fell far short of the 82 cards she had to correctly identify in order to win the cash. She clearly had no understanding of probability theory. Otherwise, she would have known that the number of correct guesses for a deck of 250 cards has a binomial distribution with expected value $250 \times \frac{1}{5} = 50$, and standard deviation $\sqrt{250 \times \frac{1}{5} \times \frac{4}{5}} \approx 6.325$. Correctly guessing 82 or more of the 250 cards comes in at

$$\frac{82 - 50}{6.325} \approx 5.06$$

standard deviations above the expected value. Even without calculating further, a disparity such as this indicates that the probability of correctly guessing 82 or more cards is very small indeed (the exact value of the probability is 1.36×10^{-6}).[1] In 1996, professional skeptic James Randi set up the James Randi Educational

[1] As a rule of thumb, a binomial distribution with parameters n and p has nearly all probability mass within three standard deviations from the expected

Foundation. The goal of the Foundation was to inform the general public and the media about the dangers of lending scientific credence to the exploits of psychics. The Foundation offered a prize of one million dollars to anyone who could demonstrate extrasensory or paranormal powers under scientific test criteria, which would be agreed-on beforehand. Various psychics took the challenge, including 'baby whisperer' Derek Ogilvie, but none came through it cash-in-hand. More about this in Chapter 13.

value when $np(1-p) \geq 25$. The same is true for the Poisson distribution with parameter $\lambda \geq 25$.

CHAPTER 11

Coincidences and Impossibilities

NOT TOO LONG AGO, British tabloid *The Sun* published a remarkable news item. They reported that the newly born third child of an English couple had arrived into the world at 7:43 a.m., that is, at exactly the same time of day as the couple's two other children. Based on the curious premise of a 12-hour window during which births may occur, the paper informed its readers that the probability of three children born successively to the same family at precisely the same time (either a.m. or p.m.) is equal to 1 in $720 \times 720 \times 720$, or rather a probability of about 1 in 370 million. Observant readers will immediately see a red flag; the newspaper made the same classic error reflected in the wrong answer $\frac{1}{49}$, commonly made in response to the question: what is the probability that two randomly chosen people will have birthdays on the same day of the week? Naturally, the newspaper should have declared that the probability is 1 in $720 \times 720 = 518\,400$. This is still quite a small probability, but considering there are approximately 3 million families in the UK with three or more children, the event described by *The Sun*, reported with much ballyhoo and hullabaloo, was perhaps less remarkable than it seems at first glance. There are a great many so-called coincidental occurrences in our daily lives that, put into perspective, turn out not to be so very coincidental. In 1986, within a span of four months, Evelyn Mary Adams won the jackpot in the New Jersey State Lottery, twice. An event such as this one seems less surprising when you consider the fact that many regular lottery participants try their luck every week with multiple lottery tickets. Using realistic presuppositions, simple probability calculation will show that there is a probability close to 1 that somewhere among the millions and millions of lottery players, there will be one who, within a short span of time, wins the jackpot two or more times. Coincidences can nearly always be explained by probabilistic arguments! Another remarkable event occurred on Wednesday, June 21, 1995, in the German national lottery. On that day, the six numbers drawn, 15-25-27-30-42-48, were the same six numbers that had been drawn on Saturday, December 20, 1986. This was the first time that a repeat had occurred in the 3016 drawings of the German lotto 6/49. But looking more closely, this phenomenon is not so very unusual. Just think of the birthday problem. An event occurring in the Bulgarian national lottery, by contrast, is unusually coincidental. On the 6th and the 10th of September, 2009, in two consecutive draws, the same six numbers were drawn from the numbers between 1 and 42. This really is a surprising occurrence, but if you consider how many lotteries there are in the world that have a weekly or bi-weekly draw of six numbers, year after year, the occurrence is not so inconceivable as to raise the specter of fraud. Many weird things happen just due to chance. What happened in the Bulgarian lottery got a lot of publicity. We take no particular notice of likely events, rather casting our spotlight on unlikely events that grab the attention. Lottery jackpot winners always make the news, but the unflagging multitudes that soldier on for 20 years without winning are relegated to the annals of obscurity.

There are many such examples of so-called coincidences that turn out to be something less than coincidental after proper analysis. The lottery principle (statisticians Persi Diaconis and Frederick Mosteller call it the law of very large numbers) says that, however small the probability of a given event, when conducive circumstances present themselves often enough, the event will always occur just due to chance. But is it really so? Are there not events that are so improbable that they will never occur? Where do we draw the line? Mathematician Emil Borel applied himself to find answers to these questions. This very famous French mathematician – who made fundamental contributions to the field of probability theory – published a book for a broad reading public called *Les Probabilités et La Vie*, which was published in English in 1962 by Dover under the title *Probabilities and Life*. In this book, Borel made a distinction between three different categories of probabilities that are so small they can practically be said to be negligible: an event of probability of 10^{-6} is negligible on a ‘human scale’. an event of probability of 10^{-15} is negligible on a ‘terrestrial scale‘, and an event of probability of 10^{-50} is negligible on a ‘ cosmic scale’. This principle is known as Borel’s law. This law is a rule of thumb that exists on a sliding scale, depending on the phenomenon in question. It is not a mathematical theorem, nor is there any hard number that draws a line in the statistical sand saying that all events of a given probability and smaller are impossible for all types of events. The suggestive title of his book, *Probabilities and Life*, notwithstanding, Borel was not entering into a discussion of evolutionary topics (this came in a later publication, *Probabilité et Certitude*). Nevertheless, Borel’s law is freely bandied about in discussions between creationists and evolutionists.

Let’s now stick to things more down-to-earth. Like the game of bridge. Every bridge player has fantasized about getting the perfect hand, or about a perfect game during which each of the four players gets a perfect hand, i.e., 13 cards of the same suit (clubs, diamonds, hearts or spades). What is the probability of four perfect hands being dealt in a game of bridge when the 52-card deck in play is arranged in a completely random order? This probability is negligible and is given by

$$\frac{4!}{\binom{52}{13}\binom{39}{13}\binom{26}{13}\binom{26}{13}\binom{13}{13}} \approx 4.47 \times 10^{-28}.$$

The probability that a specific player will be dealt a perfect hand is

$$\frac{4 \times \binom{13}{13}}{\binom{52}{13}} \approx 6.30 \times 10^{-12},$$

and the probability that at least one of the four players will be dealt a perfect hand is about four times as large and is thus about 2.52×10^{-11}. In order to get an idea of just how small the probability of four perfect hands is, imagine that 7.3 billion people across the world play 8 hands of bridge per hour, every day 12 hours a day. This means about 6.4×10^{13} deals per year. Then, four perfect hands would happen about once every 35 trillion years. Such a coincidence is impossible on a terrestrial scale. Nevertheless, the British tabloid *Daily Mail* made much of an item it published on November 25, 2011, namely, that during a bridge party in a county Warwickshire village, each of the four players was dealt 13 cards of the same suit. The news item was accompanied by a photograph of four respectable looking pensioners, faithful churchgoers, to reassure readers that this was not a case of someone taking the mickey. The article pointed out that readers could safely assume that this was the first time in the history of the game that four perfect hands had been dealt. The editor of the article apparently hadn't made the effort to do some googling on the subject. It would have been immediately clear that stories of perfect hands are a dime-a-dozen. Funny thing is, the prefect hands always surfaced during private bridge parties, never at a professional tournament. The world is filled with jokers. On April 1, 1959, at a bridge game in the estimable St. James' Club in London, a claim of four perfect hands among the club's bridge-playing Lords surfaced. Quite possibly, the players informed the newspapers of this incredible event in all honesty and good conscience. But the only explanation, other than that it was an April fool's gag, is that the cards were not shuffled well. Let's say you open a new deck of cards that, straight from the pack, is ordered according to suit. You shuffle the cards by performing a perfect riffle shuffle, two times. After the second shuffle, you'll see that the cards will have returned to their original, perfect order. In this case, you can only achieve four perfect hands if each player is dealt 13 cards from the top of the deck, in one go. A perfect shuffle is a riffle in which the deck is split into two equal-sized packets, and in which the interweaving between these two packets strictly alternates between the two.

Shuffling in such a way as to achieve a sufficiently random order in the deck, without placing the cards into a tumble-dryer, is much more difficult than most people realize. The ordinary, imperfect riffle shuffle should be done seven times to achieve an ordering of the cards that is close enough to randomness for the game of bridge, as was mathematically shown by Dave Bayer and Persi Diaconis in 1992 (the imperfect riffle shuffle is modeled by cutting the deck binomially and dropping cards one-by-one from either half of the deck with probability proportional to the current sizes of the deck halves).[1] Shuffling more than this does not significantly increase the 'randomness'; shuffle less than this and the deck is 'far' from random. In the game of bridge, few players are willing to shuffle the deck seven times. They usually shuffle about four times. After the findings of Bayer and Diaconis were published, automatic shuffling machines began to crop up more and more in bridge circles. When using automatic shuffling machines, some bridge players had to revise their intuitive chance calculations that were based on nonrandom orderings of the cards! No manual shuffle process is fully random. Achieving randomness is not what we have in mind when we puzzle over how many times to shuffle, rather we seek the number of shuffles that will get us reasonably close to randomness. Seven random riffle shuffles create enough randomness for the game of bridge.

However, the degree of randomness created by seven riffle shuffles may not be enough for other purposes. This is nicely demonstrated by the game of New-Age Solitaire that was invented by Peter Doyle. To play this game, begin with a brand-new deck of cards. In a brand-new deck of cards with the deck laying face-down, the suits are in the following order, from top to bottom: ace, two,. . ., king of hearts, ace, two, . . . , king of clubs, king, . . . , two, ace of diamonds, and king, . . ., two, ace of spades. Hearts and clubs are Yin suits, and diamonds and spades are Yang suits. Let us number the cards in the starting deck, turned face down, in top to bottom order $1-2-\cdots-26-52-51-\cdots-27$. A yin pile and a yang pile are now made as follows. Riffle shuffle the deck seven times. Then deal the cards one at a time from the top of the deck and place them face up on the table. The Yin pile is started as soon as card 1 appears and the Yang pile starts as soon as card 27 appears. The cards must be added to the Yin pile in the order $1-2-\cdots-26$ and to the Yang pile in the order $27-28-\cdots-52$. If a card comes up that

[1] An article worth reading on this subject is Brad Mann, "*How many times should you shuffle a deck of cards*," UMAP Journal, Vol. 15 (1994), 303-332.

is not an immediate successor of the top card in either the Yin pile or the Yang pile, it is placed in a third pile. A single pass through the deck is normally not enough to complete the Yin pile or the Yang pile. When finished going through the whole deck, take the third pile, turn it face down and repeat the procedure until either the Yin pile or the Yang pile is completed. Yin wins if the Yin pile is completed first. If the deck was thoroughly shuffled, the yins and yangs will be equally likely to be completed first. But it turns out that, after seven riffle shuffles, it is significantly more likely that the Yins will be completed before the Yangs. Using simulation, it can be verified that the probability of Yin winning is about 81% for seven riffle shuffles. This is 31% more than we would get if the deck were in truly random order. The probability of Yin winning is about 67%, 59%, and 54%, respectively, after eight, nine, and ten riffle shuffles. Only after fifteen riffle shuffles can we speak of a nearly 50% probability of Yin winning. This shows, once again, the difficulty of getting a fully random mix of the cards by hand.

CHAPTER 12

Gambler's Fallacy

DURING THE FIRST meeting of his yearly course in probability theory, American mathematician Ted Hill would always give his students the home assignment of performing a chance experiment, and bringing the results along to the next class meeting. The students were asked to do one of two things: either to toss a fair coin 200 times, taking note of the order of the outcome, or to write up a 'fake' outcome list of 200 tosses reflecting the outcome they anticipated getting if they did toss a coin 200 times. Astonishment would always run high among the students when, the following week, Prof. Hill easily identified just about all of the 'fakes'. A quick glance at the students' submissions was all he needed to gauge their authenticity. In an interview with the *New York Times*, Hill explained that the secret of his approach was very simple: the moment he saw a result not having a run of either six or more heads or six or more tails, he deemed it a fake. The probability of such a run occurring, namely, has the surprisingly high value of 96.53%. To underline this, we give in the table below the probability of a run of either r or more heads or r or more tails in n coin tosses for several values of r and n. It is very instructive to take a good look at this table.

Table 12.1 Probabilities for success runs in coin-tossing

n/r	3	4	5	6	7	8	9	10
10	0.826	0.465	0.217	0.094	0.039	0.016	0.006	0.002
25	0.993	0.848	0.550	0.300	0.151	0.073	0.035	0.017
50	1.000	0.981	0.821	0.544	0.309	0.162	0.082	0.041
75	1.000	0.998	0.929	0.703	0.438	0.242	0.126	0.064
100	1.000	1.000	0.972	0.807	0.542	0.315	0.169	0.087
150	1.000	1.000	0.996	0.918	0.697	0.440	0.247	0.131
200	1.000	1.000	0.999	0.965	0.799	0.542	0.318	0.172

How do we calculate the probabilities in the table? The most insightful way is to use a Markov chain with an absorbing state. As seen before in Chapter 6, the absorbing Markov chain model is a very useful tool. It has surprisingly many applications. For fixed r, let state 0 correspond to the start of the coin-toss process, and state i to an uninterrupted sequence of either i heads or i tails for $i = 1, \ldots, r$. State r is taken as an absorbing state. The process describing the evolution of the state is a Markov chain whose probabilities p_{ij} of going from state i to state j in one step are

$$p_{01} = 1,\ p_{i,i+1} = p_{i1} = \frac{1}{2} \text{ for } i = 1, \ldots, r-1, \quad \text{and } p_{rr} = 1.$$

The other p_{ij} are zero. Let $\mathbf{P}$ be the matrix whose elements are the one-step transition probabilities p_{ij}. The element $p_{ij}^{(n)}$ of the n-fold matrix product $\mathbf{P}^n$ is the probability of being in state j after n steps when the starting state is i. In particular, $p_{0r}^{(n)}$ is the probability of getting a run of either r or more heads or r or more tails in n tosses of the coin.

A useful rule of thumb states that the probability distribution of the longest run of either heads or tails in n coin tosses is strongly concentrated around $\log_2(n)$ for n large enough.[1] For 200 tosses of

[1] This rule of thumb comes from Mark Schilling, "*The surprising predictability of long runs*," Mathematics Magazine, Vol. 85 (2012), 141-149.

a coin, 7 is the most probable value for the length of the longest run.

Misconceptions over the way that truly random sequences behave in coin tossing can be grouped together under the heading 'gambler's fallacy'. This refers to the gambler who believes that, if a certain event occurs less often than average within a given period of time, it will occur more often than average during the next period. The tangle of events that occurred on August 18, 1913, at the casino in Monte Carlo when the roulette ball fell on black 26 times in a row, illustrates this concept nicely. Players began to stake larger and larger sums on red after the ball had fallen on black a few times in a row, and continued to do so. How unusual is such an event over a longer period? We can answer this question using the rule of thumb that tells us that the probability mass of the longest run of either red or black in n spins of the roulette wheel is strongly concentrated around the value $\log_{1/p}(n(1-p)) + 1$ for n large, where $p = \frac{18}{37}$ in European roulette. This means that in 100 million spins of the roulette wheel the probability mass of the length of the longest run is concentrated around the value 26, precisely the value that occurred in 1913 in the Monte Carlo Casino. Considering the number of years that roulette has been played in countless casinos around the world, a color-run of 26 is not inconceivably long. But back then, it was.

If the consequences of the gambler's fallacy turned out to be something less than disastrous for the roulette players at the casino in Monte Carlo, this was certainly not the case in Italy during the last weeks of 2004 and the first weeks of 2005, when 'Venice-53' frenzy broke out in the national Italian lottery. In this lottery, there is a bi-weekly draw in each of ten Italian cities, including Venice. Each draw, in each of the ten cities, consists of five numbers picked from the numbers 1 through 90. Participants bet on 1, 2, 3, 4 or 5 numbers, with a payoff of about 10 times the amount staked for a correct 1-number pick, and a payoff of 1 million times the amount staked for a correct 5-number pick. While the number 53 had fallen repeatedly in other cities, it did not come up at all in Venice, in any of the 182 draws occurring in the period from May, 2004, to February, 2005. During this period, more than 3.5 billion euro was bet on the number 53, entire family fortunes risked. In the month of January, 2005, alone, 672 million euro were staked on the number 53. Several professors of probability theory made appearances on Italian television to alert people to the fact that lottery balls have no memory, an attempt to stave off irresponsible betting behavior. All in vain. Many Italians held firmly to the belief that the number 53 was about to fall, and they continued to bet large sums. Some tragic events occurred during that January, 2005, direct consequences of the enormous amounts of money bet and lost on the number 53 in the Venice lottery. One Tuscan housewife drowned herself in the Tyrrhenian Sea, an insurance salesman from Florence shot his wife and son dead before turning the gun on himself, and a man in Sicily was arrested for assault and battery, distress over his mounting debt getting the better of him. Many other incidents occurred: people fell into the hands of ruthless loan sharks, others lost their homes to pay off their debts. After 182 successive draws resulting in no number 53, that number finally made its appearance in the February 9, 2005 draw, thus bringing an end to the gambling frenzy that had held sway over Italy for such a long time. It is estimated that the lottery paid out about 600 million euro to those that had placed bets on the number 53 on that day. This is a lot of money, but it is nothing compared to the amount taken in by the lottery during the Venice-53 gambling craze. This was the worst example of collective gambling madness in Italy since 1941, when the number 8 didn't turn up in a Rome based lottery for 201 successive draws. Common belief is that, in that case, Italian dictator Mussolini manipulated the draw in order to bring in the cash to finance his war effort. It is only a question of time before Venice-53 history repeats itself. There is a large probability that this will happen within 10 years, or 25 years of the event. The probability that in a period of 10 years there is some number that will not appear in some of the ten Italian city lotteries during 182 or more consecutive draws is about 50%, while this probability is about 91% for a period of 25 years.

How do we arrive at these probabilities? To do so, Markov chain analysis and a heuristic argument are combined. Take a specific city (say, Venice) and a particular number (say, 53). Define Q_n as the probability there will be a window of 182 consecutive draws in which number 53 does not appear in the next n draws of the lottery in Venice. The exact value of this probability can be calculated using an absorbing Markov chain. Consider a Markov chain with states $i = 0, 1, \ldots, 182$, where state i means that the particular number 53 is not drawn in the last i draws of the lottery in Venice. State 182 is taken as an absorbing state. Since the probability of getting number 53 in a given draw of the lottery is $\frac{5}{90}$, the one-step transition probabilities p_{ij} of the Markov chain are

$$p_{i0} = \frac{5}{90},\ p_{i,i+1} = \frac{85}{90} \text{ for } i = 0, 1, \ldots, 181, \quad \text{and } p_{182,182} = 1.$$

The other p_{ij} are zero. Let $\mathbf{P}$ be the matrix with the p_{ij} as elements. Some reflections show that the element $p^{(n)}_{0,182}$ of the n-fold matrix product $\mathbf{P}^{(n)}$ is equal to the probability that within the next n draws of the lottery in Venice, there will be some window of 182 consecutive draws in which the number 53 does not appear. This probability has the value $Q_n = 0.00077541$ for $n = 1040$ and $Q_n = 0.0027013$ for $n = 2600$. Next we use a heuristic argument to approximate the probability that in the next n draws of the lottery in the city of Venice there will *not* be some number that remains absent during 182 consecutive draws. The five winning numbers in a draw of the lottery are not independent of each other, but the dependence is weak enough to justify the approximation $(1-Q_n)^{90}$ for this probability. The lottery takes place in 10 cities. Thus, the sought probability that there will be some number not appearing in some of the 10 Italian city-lotteries during 182 or more consecutive draws within the next n draws is approximately equal to

$$1 - (1 - Q_n)^{900}.$$

This probability has the values 0.5025 for $n = 1040$ (period of 10 years) and 0.9124 for $n = 2600$ (period of 25 years). Monte Carlo simulations confirm these probabilities. Alas, these hard facts will have little power to prevent future outbursts of lottery madness.

CHAPTER 13

Euler's Number e is Everywhere

ONLY A FEW NUMBERS are so famous that they have been assigned their own letter, in perpetuity. The magical Euler's number $e = 2.718281\ldots$ is one such number. It is the limit of the sequence $(1+\frac{1}{n})^n$ as n gets larger and larger. In 2004, Google referenced Euler's number in a most original way, when they took the company public and sold stock options. They didn't do this in the ordinary way, through investment banks, as one might expect. Google announced an online auction, open directly to the public at large, at the end of which the target goal of 2 718 281 828 dollars was to have been reached. The powers-that-be in the investment world were stunned by the apparent absurdity and peculiarity of such a precise target amount, but many a mathematician let out an appreciative guffaw. Google was right to be fascinated by the number e. It crops up everywhere in the field of sciences. It is always present, for example, in descriptions of population growth and radio-active waste. The number e appeared superficially for the first time in 1608 in the work of Scottish mathematician John Napier (1550–1617), the discoverer of logarithms. In 1683, Swiss mathematician Jacob Bernoulli (1654–1705) rediscovered the number e in a study of bank account growth when interest is added year after year. But it was the work of Swiss genius Leonhard Euler (1707–1783), the most productive mathematician of all times, that put the number e on the map definitively.[1]

The field of probability is also rife with examples in which the number e plays a main role. Let's go back to the year 1713, when French mathematician Pierre Rémond de Montmort published his book "*Essay d'analyse sur les jeux de hazard*." In this book, De Montmort introduced the following card game, better known today as the Las Vegas card game. Using a well-shuffled deck of 52 cards, the dealer turns over the cards one by one, all the while, for each card turned, calling out 'ace, two, three, . . ., queen, king, ace, two, three, . . ., queen, king', and so forth until each of the 13 ranks of the cards has been called out four times in the process of turning over the 52 cards. A match occurs when the rank of the card named is the same as that of the card being turned over. What is the probability of no match in rank occurring? De Montmort could not solve this problem for 52 cards, but he was able to solve a simpler version of this problem using only 13 cards: ace, two, three, . . ., queen and king. He found an exact solution for the probability of no match occurring, showing this probability to be about

$$e^{-1} \approx 0.3679,$$

which agrees with the exact value in all four decimals. A variant of the simplified problem with 13 cards is the Santa Claus problem: at a Christmas party, each one of a group of children brings a present, after which the children draw lots randomly to determine who gets which present. In the Santa Claus problem, the probability distribution of the number of children winding up with their own presents can be very accurately approximated by a Poisson

[1] A historic overview of the number e can be found in the article Brian J. McCartin "*e: The Master of All*," The Mathematical Intelligencer, Vol. 28 (2006), 10-21.

distribution with expected value 1, regardless of the number of children involved.

A beautiful application of the Santa Claus problem is related to the one-million-dollar challenge of the sceptic and magician James Randi. This warrior in the fight against what he calls pseudo-scientific 'woo-woo,' and charlatanism offered a prize, through the James Randi Educational Foundation, of one million dollars to anyone who could demonstrate paranormal powers. Naturally, such powers would have to be demonstrated in a controlled testing environment, meaning that spoon-bending psychics who bent spoons seemingly without applying physical force were not invited to bring their own spoons to the experiment. A number of individuals took the challenge; none succeeded. One medium claiming to have remote powers of extrasensory perception was subjected to the following test: the medium could single out a child with whom he believed himself to have a telepathic connection. Then the medium was shown ten different toys that would be given to the child one after the other, in random order, out of sight of the medium. The child was taken to an isolated chamber, and each time the child received a toy, the medium was asked to say what toy it was. If the medium was right six or more times, he would win one million dollars. The medium took the challenge, but guessing correctly only once, he lost. James Randi was in very little danger of having to cough up the loot. The probability of six or more correct answers is practically equal to the Poisson probability $1 - \sum_{k=0}^{5} \frac{e^{-1}}{k!}$ and is about 0.06%. The medium perhaps thought, beforehand, that five correct guesses was the most likely outcome, and a sixth correct guess on top of that wasn't that improbable, so, why not go for it.

In contrast to the Santa Claus problem, it is significantly more difficult to get an exact solution to the Las Vegas card game when playing with 52 cards. The exact solution requires in-depth combinatorics. An excellent approximation to the probability of no match occurring in the Las Vegas card game can be obtained with the so-called Poisson heuristic (this heuristic was applied earlier in Chapter 3). This heuristic leads to the approximate value

$$e^{-4} \approx 0.0183,$$

which is quite close to the exact value 0.0162. Lifting a corner of the veil on the Poisson heuristic can be very instructive. This heuristic is generally applicable and, in fact, we used it to get the approximate solution of the Santa Claus problem. The Poisson heuristic applies to the situation of a large number of trials each having a small probability of success. If the trials are independent of each other or show only a 'weak' dependence, then the total number of successes can be approximated by a Poisson distribution, see also Chapter 3. A pleasant feature of the Poisson distribution is that it is fully determined by its expected value: you have only to find the expected value of the number of successes. The Las Vegas card game consists, in fact, of 52 successive trials, each of which is comprised of a new card being turned over. The experiment can be said to be successful if the rank of the card turned over matches the rank of the card being named. Each card has the same success probability as that of the first card to be turned over. After all, the cards are randomly ordered given that the deck has been thoroughly shuffled. So, each experiment has the same probability of success. The success probability of each trial is $\frac{4}{52}$, because each rank includes 4 cards in a deck of 52 cards. This means that the expected number of successes in the 52 trials is equal to $52 \times \frac{4}{52} = 4$. Thus, the probability of no match occurring in the Las Vegas game can be approximated by $e^{-4} \approx 0.0183$.

In the same way, the Santa Claus problem with n children can be seen as a sequence of n trials each having a probability $\frac{1}{n}$ of success (= picking own present), and so the number of children who pick their own presents is approximately Poisson distributed with expected value 1.

The Poisson heuristic also offers a good approach to the solution of the following problem, similar to that of the Las Vegas card problem. Let's say you are a member of a family made up of five brothers and sisters and their partners; you all get together each year for Christmas, and you exchange gifts. Each of the 10 family members purchases a present. The presents are numbered 1 through 10, and each family member knows the number of his or her own present and the number of the present purchased by his or her partner. Ten cards, each displaying a number from 1 through 10, are placed into a hat. Each family member then picks a card at random out of the hat. If anyone picks the number of their own present, or that of their partner, all of the cards go back into the hat and they start over again. What is the probability that the game will progress without anyone picking their own number or that of their partner, such that a new draw will not be necessary?

A good approximation of this probability is

$$e^{-2} \approx 0.1353,$$

or rather, a probability of about 13.53%. This approximation can be found with a comparable argument to the one laid out in the Las Vegas card game, and allies itself to all sufficiently large family sizes. For a family of five brothers and sisters, the approximate value of 13.53% is fairly close to the exact value of 12.31%. If the family consists of seven brothers and sisters and their partners, then the approximate probability of 13.53% is even closer to the exact probability of 12.54%. It has been empirically verified that, for the case of a family of n brothers and sisters and their partners, the approximation e^{-2} can be improved to $e^{-2}(1 - \frac{1}{2n})$.

The number e is not limited to this type of situation in probability theory. Just about everyone has heard of the dating problem, also known as the sultan's dowry problem. This problem can also be described in a more gender-neutral version, thus: a gang of thieves has come together for a yearly gathering at a secret location. Outside, a beat-cop on his rounds has stumbled onto the secret gathering, and his only goal thereafter is to arrest the gang's ringleader. The officer knows that, for safety's sake, the crooks will leave the gathering one by one, in a random order, and he also knows that as soon as he nabs one of them, the rest will be warned of his presence and escape out the back. For this reason, he is determined to act only when he has a strong suspicion that the exiting crook is the ringleader. The officer knows that the ringleader is the group's tallest member; he also knows how many crooks there are in the gang. How can the agent maximize the probability of nabbing the right guy? Let's say that the gang has n members. By approximation, the optimal thing to do is to let the first $\frac{n}{e}$ gang members exit unhindered, arresting the first gang member thereafter, who is taller than all of those who have gone before. The probability of nabbing the ringleader using this method is approximately equal to $e^{-1} = 0.3679$. The approximation is useful for $n \geq 10$. For $n = 10$, the optimal strategy is to allow the first four crooks to get away unhindered; the exact value of the probability of collaring the right guy in this scenario is equal to 0.3987.

Finally, let's look at the role of the number e in a probability problem known as the Oberwolfach dinner problem. Think back to our Christmas family of m persons, where $m = 10$. Let's say this group has reserved a table at a restaurant for both Christmas Day and Boxing Day. They have requested a round table, and asked the restaurateur to make a random seating arrangement for each meal. What is the probability that no two or more of the family members will be seated next to one another at both meals? An excellent approximation for this probability is

$$e^{-2}\left(1 - \frac{4}{m} + \frac{20}{3m^3}\right).$$

For $m = 10$, the approximate probability is equal to 0.0825. For all practical purposes, the correction factor to e^{-2} can be omitted if m is at least 80 (say). That said, round tables with space to seat 80 are not to be found on every street corner. That brings us to the Oberwolfach dinner problem with multiple tables. The question is then: what is the probability that, at two successive dinners, no two or more persons will be seated next to each other when, at both meals, rs persons have been randomly placed around r round tables, each with space for s persons? This is a challenging problem, and as far as I know, it remains unsolved!

CHAPTER 14

The 10 Most Beautiful Formulas in Probability

WHAT ARE the 10 most beautiful formulas in probability? This question is not easily answered. Now, if you wanted to know what the most beautiful mathematical formula is, well, that's another matter. Almost all mathematicians agree that this honor goes to Euler's identity:

$$e^{i\pi} + 1 = 0,$$

where $e = 2.71828\ldots$ is the Euler number and i is the complex number $\sqrt{-1}$. This mind-blowing equation is sometimes called

'Gods equation': it unites the five most important numbers from mathematics ('the dream team of numbers').

Getting back to the 10 most beautiful formulas in probability, here is a list of my favorites:

1. The formula of the Gauss curve. Aristotle believed that symmetry was one of the most important elements of the universal idea of beauty. The concept of symmetry crops up in fundamental laws of physics and in many other areas of the natural sciences. In probability, symmetry is best expressed by the formula for the Gauss curve (or normal curve):

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(x-\mu)^2/\sigma^2}.$$

The numbers μ and σ with $\sigma > 0$ are constants, giving the expected value and standard deviation of the normal density function $f(x)$. The bell-shaped graph of $f(x)$ is symmetric around the point $x = \mu$. Regardless of the values of μ and σ, the total area under the curve is equal to 1, where about 68% of the surface area lies between $\mu - \sigma$ and $\mu + \sigma$, about 95% lies between $\mu - 2\sigma$ and $\mu + 2\sigma$, and about 99.7% lies between $\mu - 3\sigma$ and $\mu + 3\sigma$. The Gauss curve is named after the famous mathematician, Carl Friedrich Gauss (1777–1855), who discovered it in his analysis of errors in astronomical measurements. In pre-euro Germany, Gauss' image appeared on the German 10-mark note, with, next to him, an image of the normal curve and its mathematical formula. Many stochastic phenomena in daily life, such as adult height, annual regional rainfall, pregnancy duration, etc., can be modeled by a normal density function with appropriate values for μ and σ.

The particular function $\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ is called the standard normal density function. It shows up in what is considered the most important theorem in probability and statistics (the central limit theorem). This theorem is expressed by the famous formula:

$$\lim_{n\to\infty} P\Big(\frac{X_1 + X_2 + \cdots X_n - n\mu}{\sigma\sqrt{n}} \leq x\Big) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{1}{2}u^2}\, du$$

for any x when $X_1, X_2, \ldots, X_n$ are independent random variables each having the same probability distribution with expected value μ and standard deviation σ. The integral $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{1}{2}u^2}\, du$ is closely related to the Gauss error function, which is used in many scientific fields.

2. The formula of Bayes. Thomas Bayes (1702−1761) was an English cleric with a great interest in the fields of logic and probability. The formula that would later be named after him was published after his death. Only in the 19th century would the importance of this formula be recognized in the work of Pierre Simon Laplace (1749−1827), who is considered to be one of the most influential of French scientists ('the French Newton'). The basic form of the formula of Bayes is given by

$$P(A \mid B) = \frac{P(A)P(B \mid A)}{P(B)}.$$

This formula updates a prior probability for event A to a conditional posterior probability for event A after having observed the occurrence of a relevant event B, see also Chapter 4. In fact, Bayes' rule enables us to reason back from effect to cause in terms of probabilities.

The beauty of the Bayes formula is that it records rational thinking in a simple mathematical formula. The Bayes formula has countless practical applications: in medical research, in legal argumentation, in the research of genetic disorders, in spam filters, in cryptography, in Google's search engine, etc. During World War II, the famous mathematician Alan Turing used the Bayes formula to crack encrypted Nazi codes. The significance of this feat, in the context of ending the war, cannot be overestimated. Bayes formula is also fundamental to procedures created for the finding of missing objects. A nuclear bomb went missing at sea when a KC-135 tanker aircraft and a B-52 bomber collided near the Spanish fishing village of Palomares, and the black box of a 228-passenger Air France flight from Rio de Janeiro to Paris went missing when the plane crashed into the Atlantic Ocean on June 1, 2009. In both cases, Bayesian analysis was crucial to the process of locating the missing objects.

Bayes' rule is the foundation of Bayesian statistics. In Bayesian statistics, uncertainty regarding the 'state of nature' is modeled through a probability distribution, whereas classical statistics would treat an unknown parameter as a constant. In many situations, the approach of modeling knowledge about the 'state of nature' using a probability distribution is a natural approach. For example, one can meaningfully treat the photon flux of a nonvariable star as a random variable and talk about the probability distribution of the true value of the flux. This probability distribution expresses the state of the investigator's knowledge about the true value and is revised through Bayes' rule as additional data arise. Astronomers have been using the 'Bayesian thinking model' since the 19th century, and many others, including pharmaceutical researchers, are currently using it more and more.

3. The gambler's ruin formula. The foundations of the field of probability theory were laid during an exchange of letters dating from 1654 between French mathematicians Blaise Pascal (1623−1662) and Pierre de Fermat (1601−1665), on the subject of gambling. This exchange of letters spurred Christiaan Huygens (1629−1695) to immerse himself in probability theory, and this led to the famous gambler's ruin formula:

$$P(a, b) = \frac{1 - (q/p)^a}{1 - (q/p)^{a+b}}.$$

The formula should be read as $a/(a + b)$ if $p = q = 0.5$. For two players A and B with initial bankrolls of a euro and b euro, $P(a, b)$ gives the probability that player A will ultimately win all of the money if each of the players bets 1 euro on every game move, and any bet is won by player A with probability p and by player B with probability $q = 1-p$. Through the years, the gambler's ruin formula has had applications in all kinds of probability games with varying levels of complexity. The gambler's ruin formula was relevant in the 17th century, and it is still relevant, today. Nowadays, the classic gambler's ruin formula is also useful for determining capital buffers in banking. An interesting application of the gambler's ruin formula in a notorious legal case revolving around the debts of a gambling addict is described in Chapter 1. The gambler's ruin formula shows that if you want to achieve the highest probability of winning a pre-determined target amount at the casino, it is better to bet large rather than small sums. For example, let's say you go to the casino with 100 euro, and your goal is to double it. You opt to play European roulette, betting each time on red. You will double your stake with probability $p = \frac{18}{37}$ and you will lose with probability $q = \frac{19}{37}$. So, if you stake 5 euro each time ($a = b = 20$), or 25 euro ($a = b = 4$), or 50 euro ($a = b = 2$), the probability of reaching your goal will have the respective values of 0.2533, 0.4461 and 0.4730.

Intuition tells us that larger stakes will increase your probability of reaching your goal: using this strategy, your money is exposed to the casino's house edge for a shorter period of time. In April of 2012, the wisdom of this thinking was quite literally tested by

Englishman Ashley Revell, in a Las Vegas casino. Revell had sold off all of his possessions, down to the clothes on his back, traveling to Las Vegas with a neat 135,300 dollars in the pocket of his only remaining pair of trousers. And just like that, he staked the entire amount on red, watching calmly as the wheel spun. Lo and behold, it fell on red! In one go, Revell had reached his target goal, and doubled his money. Few people would have the nerve to perform such an all-or-nothing act. Before Revell, there was William Lee Bergstrom. This Texas horse trader was a professional gambler who became known as 'The Suitcase Man' after he walked into a Las Vegas casino in 1980 with two suit cases – one empty, the other filled with 777,000 dollars. At the craps table, he bet the whole sum all at once, and a short while later, he departed the casino, both suitcases crammed full with cash.

4. The square-root formula. Abraham de Moivre (1667–1754) was born in France but settled permanently in England in 1686 as a result of the persecution of protestants in his native country. De Moivre is one of the most famous probabilists of the 18th century. In addition to his discovery of what would later be called the normal distribution, he also came up with the square-root law for the standard deviation:

$$\sigma(X_1 + \ldots + X_n) = \sigma\sqrt{n}.$$

This formula says that the standard deviation of the sum of n independent stochastic variables $X_1, X_2, \ldots, X_n$, each having the same standard deviation σ, is not equal to $n\sigma$ but is, rather, equal to $\sigma\sqrt{n}$. The square root law is sometimes called De Moivre's equation. This formula had an immediate impact on methods used to inspect gold coins struck at the London Mint. The standard gold weight, per coin, was 128 grains (this was equal to 0.0648 gram), and the allowable deviation from this standard was $\frac{1}{400}$ of that amount, or 0.32 grains. A test case of 100 coins was periodically performed on coins struck, their total weight then being compared with the standard weight of 100 coins. The gold used in the striking of coins was the property of the king, who sent inspectors to discourage minting mischief. The royal watch dogs had traditionally allowed a deviation of $100 \times 0.32 = 32$ grains in the weight of 100 inspected coins. Directly after De Moivre's publication of the square root formula, the allowable deviation in the weight of 100 coins was changed to $\sqrt{100} \times 0.32 = 3.2$ grains; alas for the English monarchy, previous ignorance of the square root formula had cost them a fortune in gold. The square root law has many applications, providing explanation, for example, for why city or hospital size is important for measuring crime statistics or death rates after surgery. Small hospitals, for example, are more likely than large ones to appear at the top or bottom of ranking lists! This makes sense if you consider that, when tossing a fair coin, the probability that more than 70%, or less than 30%, of the tosses will turn up heads is much larger for 10 coin tosses, than for 100. The smaller the numbers, the larger the chance fluctuations!

5. The Kelly betting formula. This formula has come to be accepted as one of the most useful betting/investment formulas. Suppose you are offered a series of betting opportunities in which you have an edge. How should you bet such that you will be managing your money in the best possible way? The idea is not to bet all of your money at once, but to bet a certain fraction of your current bankroll on each move. The Kelly betting formula says you should bet the same fixed fraction

$$f^* = \frac{pr-1}{r-1},$$

of your current bankroll each time, when each bet pays out r times your stake with probability p and is a losing proposition with probability $1-p$. This formula assumes that $pr > 1$, that is, that the game is advantageous to you. The Kelly fraction $(pr-1)/(r-1)$ can be interpreted as the ratio of the expected net payoff for a one-dollar bet and the payoff odds. It can be shown that the long-run growth factor of your bankroll is maximized when using the Kelly strategy. We come back to this matter in Chapter 16. The idea of the Kelly strategy is not only useful for betting on sports events such as horse racing and soccer matches, but also for investment decisions. A variety of famous investors, including Warren Buffett, have successfully implemented investment strategies based on the Kelly system. The Kelly system became known primarily through the work of Edward Thorp. Thorp was the first one to use the Kelly betting system in casinos; he used it to develop a successful method of card counting in the game of blackjack, a method that gives the player an advantage over the casino. Who says you can't get rich with mathematics?

6. The asymptotic law of distribution of prime numbers. The simple formula

$$\frac{1}{\ln(n)-1}$$

provides an approximation to the probability that a randomly chosen number from the integers 1 to n is prime when n is large. For example, for $n = 10^3$, 10^6 and 10^{12}, the probability has the approximate values of 16.93%, 7.80% and 3.76%; the probability has the exact values 16.80%, 7.85% and 3.76%. Another interesting result is that the probability of two randomly chosen numbers from $1, 2, \ldots, n$ being relatively prime (largest common divisor is 1) is about 60.8% for large n.

7. A square root formula for the drunkard's walk. In the symmetrical linear drunkard's walk, the drunkard will take consecutive steps of length 1 either to the left with probability $\frac{1}{2}$, or to the right with probability $\frac{1}{2}$, each step being taken independently of previous steps. For large n, the formula

$$\sqrt{\frac{2n}{\pi}}$$

gives an approximation for the expected distance between the drunkard's position after n steps and his starting point. It is also an approximation for the expected value of the maximal distance between the drunkard's position and his starting point during the first n steps. On top of that, the expected value of number of returns of the drunkard to his starting position during the first n steps is approximately equal to $\sqrt{2n/\pi}$ for n large.

8. The newsboy formula. The newsboy formula is one of the most useful formulas in inventory control. It applies to a scenario in which a retailer must make a one-time decision on the stocking quantity of a particular item for which the demand is uncertain. The newsboy framework is appropriate for many situations, including perishable goods and seasonal products such as Christmas trees and articles of high-fashion clothing. The probability distribution of the demand for the item in the coming period is assumed to be known. Let p_j be the probability that the total demand for the item will be j for $j = 0, 1, \ldots$. A shortage cost of c_u is incurred for each unit of unsatisfied demand and a leftover cost of c_o is incurred for each item left over. How much of the item to stock? If you want to minimize the expected value of the total cost, the order quantity should be taken equal to the smallest value of Q satisfying the newsboy formula

$$\sum_{j=0}^{Q} p_j \geq \frac{c_u}{c_o + c_u}.$$

This formula balances the risk of shortages with the risk of leftovers. An intuitive argument for understanding the formula is provided by marginal analysis. Suppose that the order quantity is raised from Q to $Q+1$. Then, the expected value of the leftover cost increases with $c_o \sum_{j=0}^{Q} p_j$ and the expected value of the shortage cost decreases with $c_u \sum_{j=Q+1}^{\infty} p_j$. Next, taking the smallest value of Q satisfying $c_o \sum_{j=0}^{Q} p_j \geq c_u \sum_{j=Q+1}^{\infty} p_j$ and rearranging terms, we get the newsboy formula.

9. The Pollaczek-Khintchine formula. This is the most famous formula for queuing systems and it reads as follows:

$$L_q = \frac{1}{2}\left(1 + \frac{\sigma^2}{\mu^2}\right)\frac{\rho^2}{1-\rho}.$$

Here L_q is the long-run average number of customers awaiting service in a single-server queuing system when the service time of a customer has mean μ and standard deviation σ and the arrivals of customers occur completely randomly in time (that is, the probability of an arrival in a very short time interval is proportional to the length of the interval and does not depend on the amount of time elapsed since the last arrival). The quantity ρ must satisfy $\rho < 1$ and is defined as $\rho = \lambda\mu$ with λ being the arrival rate of customers. That is, ρ is defined as the expected number of new customers arriving during the service time of a customer and thus ρ represents the load on the server. The Pollaczek-Khintchine formula shows how the average queue length depends on the variability of the service time.[1] Most importantly, the formula demonstrates that a small increase in the arrival rate causes a disproportionally large increase in the average queue size L_q when the load ρ is close to 1. In stochastic service systems, one should never try to balance the arrival rate with the service capacity of the system! This is an important lesson from the Pollaczek-Khintchine formula.

10. The formula for the waiting-time paradox. You are in Manhattan for the first time. Having no prior knowledge of the bus schedules, you happen upon a bus stop located on Fifth Avenue. According to the timetable posted, buses are scheduled to run at ten-minute intervals. So, having reckoned on a waiting period of five minutes, you are dismayed to find that after waiting for more than twenty, there is still no bus in sight. The following

[1] An amusing example of this is found in Robert Matthews, "*Ladies in waiting*,"' New Scientist 167 (July 29, 2000), 2249.

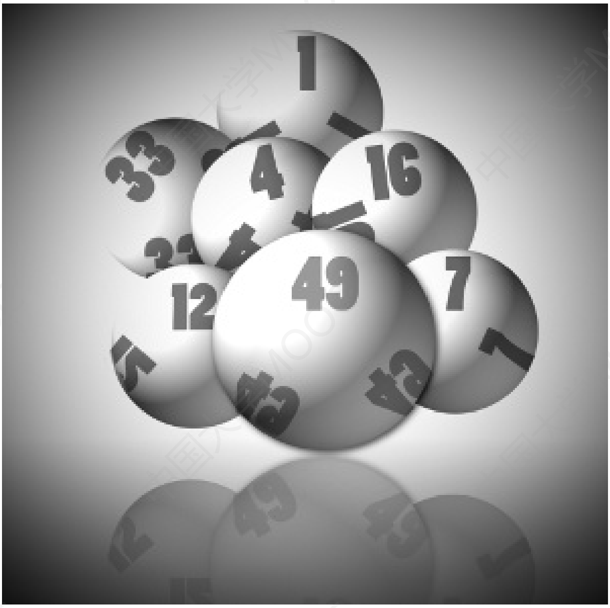
day you encounter a similar problem at another busy spot in the city. How is this possible? Is it just bad luck? No, you have merely encountered the waiting-time paradox. This apparent paradox can be demystified by the mathematical formula

$$\frac{1}{2}\Big(1+\frac{\sigma^2}{\mu^2}\Big)\mu$$

for your expected waiting time when the inter-departure times of the buses are irregular with mean μ and standard deviation σ and when you arrive at a random moment at the bus stop. It is only when buses run precisely at ten-minute intervals ($\sigma = 0$) that your average wait will be equal to the expected five-minute period. For highly irregular inter-departure times ($\sigma > 1$), your average waiting time is even larger than the average inter-departure time of the buses. The paradox can be explained by the fact that you have a higher probability of arriving at the bus stop during a long inter-departure time than during a short one.

CHAPTER 15

Beating the Odds on the Lottery



HOW TO GET rich playing the lottery? Faithfully filling in a lottery ticket each week is a futile undertaking. The odds of winning a jackpot in the 6/49 Lotto with a single ticket are inconceivably small. To get an idea of just how small the odds are, consider that the odds of matching the six numbers drawn are the same as those for getting a specially marked 1 euro coin when picking one coin at random out of a twenty-and-one-quarter mile-high stack of 1 euro coins, all having a thickness of 2.33 mm. All the same, as the amazing stories below prove, people will always seek, find or create loopholes to help them beat the odds of winning the lottery.

The only certain way to win a jackpot is to purchase a sufficient number of tickets to cover every single combination of numbers possible. If the jackpot has grown such that the expected payoff for one ticket is greater than the cost of the ticket, then buying every single combination of numbers possible can pay off. This is what some large syndicates have successfully achieved in lotteries that lend themselves to this strategy. The most famous case is that of the syndicate set up by mathematician Stefan Mandel. In 1992, he pulled off every lottery player's dream. With a firm reputation as a 'lotto-breaker', he set up a consortium, in Australia, of 2,500 investors, each of whom invested 4,000 dollars. In 1992, the syndicate targeted the USA's Virginia lotto, where game regulations allowed for mass ticket purchases. The lottery consists of a draw of six numbers, from 1 through 44, and if you get all six numbers right, you win the jackpot. In the Virginia lotto game, there were about 7.1 million combinations of six numbers possible. In February, 1992, the jackpot had increased to 27 million dollars. Mandel had been following this lotto game for some time while making meticulous plans to set up a win. The operation took about three months: They printed out all 7.1 million tickets in Australia, paid $60,000 to ship them to the US, and negotiated bulk buys with grocery stores all around Virginia about how they could send cashier's checks to buy tens of thousands of lottery tickets. Stefan Mandel had developed a smart algorithm such that each possible combination of numbers was listed once, avoiding duplications. The consortium was not able to turn in all 7.1 million combinations, but they did manage to turn in about 5 million of them. This gives a probability of about $\frac{5}{7.1}$ of the consortium hitting the jackpot on one of its ticket numbers. Lady luck was with them insofar as, as the deadline for the draw approached, no run on tickets for their draw had occurred among ordinary players; this meant that they had a pretty high probability of not having to share the jackpot prize money with others, in the case of a win. How high a probability? We can only estimate the magnitude of this. Assume that there were two million tickets purchased by other participants. What is the probability, then, that the consortium will have to share the jackpot with others? The probability that none of the other participants will have picked the six winning numbers can be estimated by

$$\left(1 - \frac{1}{7.1 \times 10^6}\right)^{2\times 10^6} \approx e^{-2/7.1} = 0.7545,$$

assuming that the vast majority of the tickets of the other participants were filled in randomly. This estimate uses the approximation $1 - x \approx e^{-x}$ for x close to 0. As a side remark, the complementary probability $1 - 0.7545 = 0.2455$ is entirely in agreement with the fact that, among 2 million randomly filled-in tickets for lotto 6/44, the expected number of different combinations of six numbers is about 1.743 million.[1] Thus, we can conclude that there is a probability of about $\frac{5}{7.1} \times 0.7545$, or about 53%, that the consortium would be the only winner of the jackpot. And so it was: the consortium won the jackpot, along with several second and third prizes, and thousands of minor prizes. They weren't able to collect their winnings without a struggle, but in the end, the consortium made a hefty profit on a considerably risky investment. This was Stefan Mandel's final gambit. Nowadays, the Romanian born mathematician is to be found communing with the crabs on a tropical island in the South Pacific.

In 2004, the Cash WinFall lottery was introduced in the US Commonwealth of Massachusetts. This lottery was launched because the jackpot of its forerunner was so seldom won that the rate of participation had decreased dramatically. The lottery organization decided that WinFall would avoid this awkward situation by limiting the jackpot. If the jackpot rose to \$2 million without a winner, the jackpot would 'roll-down' and instead be split among the players who had matched three, four, or five numbers. Lower-tier prizes were \$4000, \$150, or \$5 for matching five, four, or three numbers respectively, and those prizes were increased by a factor of five to ten if the jackpot reached \$2 million and was not won. In the lottery six different numbers were drawn from the numbers 1 to 46. Each week, the lottery published the estimated amount of the jackpot for that week's draw. Each time a roll-down draw approached, several syndicates bought a very large number of tickets. This was not

[1] Mathematically, if b times a ball will be randomly put into one of c cells, then the expected number of empty cells is $c\left(\frac{c-1}{c}\right)^b \approx ce^{-b/c}$ for c large. For $b = 2 \times 10^6$ and $c = 7.1 \times 10^6$, the expected number of empty cells is about 5.357×10^6.

too risky for them since the jackpot was seldom hit, and ordinary players barely bought more tickets as a roll-down draw approached. What can be said about the cash winnings of the syndicates? Let's say that one syndicate invested \$400,000 in 200 thousand lottery tickets of \$2 per ticket when a roll-down was expected. Under the assumption that those tickets are Quick Pick tickets whose ticket numbers are randomly generated by the lottery's computers, let's make some rough calculations for the case that the jackpot reached \$2 million and was not won. We take the conservative estimates $a_1 = \$25{,}000$, $a_2 = \$925$, and $a_3 = \$27.50$ for the payoff a_k on any ticket that matches exactly k of the six winning Winfall numbers (of course, the actual amounts depend on the number of winning tickets). The probability of a single ticket matching exactly k of the six winning numbers given that it didn't hit the jackpot is practically equal to

$$p_k = \frac{\binom{6}{k}\binom{40}{6-k}}{\binom{46}{6}}.$$

This probability has the numerical values $p_5 = 2.56224 \times 10^{-5}$, $p_4 = 1.24909 \times 10^{-3}$, and $p_3 = 2.10957 \times 10^{-2}$. Let's define the random variable X_k as the number of syndicate tickets that match exactly k of the six winning numbers. In view of the physical background of the Poisson distribution – the distribution of the number of successes in a large number of experiments each having a small probability of success – it is reasonable to approximate the distribution of X_k by a Poisson distribution with expected value $\lambda_k = 200{,}000 \times p_k$. The numerical values of the λ_k are $\lambda_5 = 5.12447$, $\lambda_4 = 249.8180$ and $\lambda_3 = 4219.149$. This leads to the estimate

$$\lambda_1 a_1 + \lambda_2 a_2 + \lambda_3 a_3 = 475{,}220 \text{ dollars}$$

for the expected cash winnings of the syndicate. An expected profit of more than \$75,000, a healthy return of about 19%. What is an estimate for the probability that the syndicate does not see a return of its \$400,000 investment? In order to find this, we need the standard deviation of the cash winnings of the syndicate. The random variables X_1, X_2 and X_3 are nearly independent of each other. Therefore, letting $\sigma(X_k)$ denote the standard deviation of X_k, the standard deviation of cash winnings of the syndicate is approximately equal to

$$\sqrt{a_1^2\sigma^2(X_1) + a_2^2\sigma^2(X_2) + a_3^2\sigma^2(X_3)}.$$

Using the basic fact that the standard deviation of a Poisson distributed random variable is the square root of the expected value of the random variable, we have $\sigma^2(X_k) = \lambda_k$. This leads to the estimate \$58,479 for the standard deviation of the cash winnings of the syndicate. The Poisson distribution can be accurately approximated by the normal distribution when the expected value of the distribution is large enough, say 25 or more. Using this fact together with the fact that a linear combination of independent normally distributed random variables is again normally distributed, we have that the distribution of the cash winnings $a_1X_1 + a_2X_2 + a_3X_3$ can be approximated by a normal distribution with expected value \$475,220 and standard deviation \$58,479. Thus, denoting by $\Phi(x)$ the cumulative probability distribution function of the standard normal distribution, the probability that the syndicate will not earn back its investment of \$400,000 can be estimated as

$$\Phi\Big(\frac{400{,}000 - 475{,}220}{58{,}479}\Big) = 0.099.$$

Three syndicates, one of which was a group of MIT students, won millions of dollars by making clever use of the 'roll-down' character of the lottery, and in fact, profiting from a jackpot that had been amassed by other participants. By 2011, syndicate activity was getting negative publicity, which prompted lottery officials to adjust the rules, and ultimately to abandon the game altogether.

Lotteries and duplicity seem to go hand-in-hand. An exceptional form of lottery fraud occurred in Canada. Employees of certain establishments where lottery tickets were sold adopted the practice of stealing winning tickets from their mainly elderly clientele, who unsuspectingly tendered their tickets to be 'checked' against the winning numbers. When one savvy customer registered a complaint, a Canadian television station mounted an investigation, calling in Jeffrey Rosenthal, well-known professor of probability, to help with the statistical analysis. It appeared that, in the period of 1999−2006, a total of 5713 big prizes (\$50,000 or more) had been hit in the province of Ontario, 200 of which prizes had definitely gone to individuals employed as lottery ticket vendors. The question was: was the total of 200 winners among vendors more than could be attributed to coincidence? To answer this question, some additional information was necessary. Research done by the television station revealed an estimate of 60,000 vendors of lottery tickets. The vendors also purchased lottery tickets, and it was estimated that the average outlay on lottery tickets among all ticket vendors was about 1.5 times greater than the average outlay on lottery tickets among all adult inhabitants of Ontario. At the time, the adult population of Ontario was estimated to be about 8,900,000. Using this data, the expected number of big winners among vendors can be estimated as:

$$5713 \times \frac{60\,000 \times 1.5}{8\,900\,000} = 57.$$

Due to the physical nature of the Poisson distribution, a Poisson model for the number of winners is justified. The standard deviation of the Poisson distribution is equal to the square root of the expected value of the distribution, and the Poisson distribution has nearly all of its probability mass within three standard deviations from the expected value when the expected value is more than 25. Thus, two hundred or more winners comes in at

$$\frac{200 - 57}{\sqrt{57}} \approx 19$$

standard deviations above the expected value. The probability of this occurring is inconceivably small (on the order of 10^{-49}). The lottery organizers objected to the calculations and came with new figures. Therefore the calculations were redone for 101,000 vendors using a correction factor of 1.9 for the average outlay instead of the correction factor 1.5. These figures give us a value of 123 for the expected number of winners among vendors. Two hundred winners still comes in at

$$\frac{200 - 123}{\sqrt{123}} \approx 7$$

standard deviations above the expected value, and again, the probability of this actually occurring is extremely small (on the order of 10^{-7}). This shows quite convincingly that large scale lottery fraud was occurring at shops selling lottery tickets.[2] The investigation led to quite a bit of upheaval, and security procedures were adjusted in order to better protect customers. The stores' ticket checking machines must now be viewable by customers, and make loud noises to indicate wins. Customers are now required to sign their names on their lottery tickets before redeeming them, to prevent switches.

[2]More in the striking publication of Jeffrey Rosenthal,"*Statistics and the Ontario lottery retailer scandal*," https://protect-us.mimecast.com/s/eIG-CERZP5f3k9934UpvwFg?domain=probability.ca

Another notorious case of lottery fraud known as the 'Triple Six Fix' occurred in 1980 in the US state of Pennsylvania. The lottery in this case went like this: during a TV show, three air-blown containers of balls numbered 0 through 9 were activated until three balls were drawn into a vacuum tube and ejected into a display. The winning combination of three numbers was determined by the order in which the balls entered the display. A group of defrauders hatched a plan to rig the game, and their leader was none other than Nick Perry, the TV show's presenter. Perry had white latex paint injected into all of the balls except those numbering 4 and 6. This meant that only balls bearing 4's and 6's would be light enough to attain the height of the vacuum tube. Perry's group had purchased a great number of tickets for winning numbers in eight combinations: 666, 664, 646, 644, 466, 464, 446 and 444. When the lottery took place, the winning combination turned out to be 666, which explains the moniker 'Triple Six Fix'. The ring of defrauders didn't have much time to enjoy their approximately 1.8 million dollars in spoils, however. A few of them had purchased a large number of tickets, all consisting exclusively of combinations of fours and sixes, at a pub, and made a telephone call, in Greek, while finalizing the purchase. These things did not go unremarked by one observant pub employee, who, upon hearing reports of local bookmakers suspicious of a draw preceded by unusually high ticket-sales for number combinations of fours and sixes, called in to the TV station to offer a tip. The telephone conversation placed from the pub was quickly traced to the TV studio, and suspicion quickly fell onto Nick Perry, who was known to speak Greek. The gang were arrested and convicted, each receiving a stiff sentence, including 7 years for Perry. The film Lucky Numbers, based on this event and starring John Travolta, came out in 2000. It is not known whether Perry ever went to a screening of the film after his release.

CHAPTER 16

Investing and Gambling with Kelly

IN HIS BOOK *A Mathematician Plays the Stock Market*, John Allen Paulos describes a scenario that occurred during the wild times when dotcom companies were going public on a daily basis. A certain investor is offered the following opportunity: Every Monday for a period of 52 weeks the investor may invest funds in the stock of one dotcom company. On the ensuing Friday, the investor sells. The following Monday, he purchases new stock in another dotcom company. Each week, the value of the stock purchased has a probability of $\frac{1}{2}$ of increasing by 80%, and a probability of $\frac{1}{2}$ of decreasing by 60%, depending on market conditions in the previous weeks. This means that on average, the increase in value of the purchased stock is equal to $0.8 \times \frac{1}{2} - 0.6 \times \frac{1}{2} = 0.1$, giving a return of 10% per week. The investor, who has a starting bankroll of ten thousand dollars to invest over a period of the coming 52 weeks, doesn't hesitate for a moment; he decides to invest the full amount, every week, in the stock of a dotcom company. After 52 weeks, it appears that our investor only has 2 dollars left of his initial ten-thousand-dollar bankroll. He is, quite literally, at a loss to figure it all out. But in fact, this investment result is not very surprising when you consider how dangerous it is to rely on averages in situations involving uncertainty. A person can drown, after all, in a lake that has an average depth of 25 cm. For situations involving uncertainty factors, you should never work with averages, but rather with probabilities! It is easily explained that the probability of nearly depleting the bankroll is large if the investor invests his whole bankroll in each transaction. The most likely path to develop over the course of 52 weeks is one in which the stock increases in value 50% of the time, and decreases in value 50% of the time. This path results in a bankroll of

$$1.8^{26} \times 0.4^{26} \times \$10{,}000 = \$1.95$$

after 52 weeks. Running one hundred thousand simulations of these investments over 52 weeks renders a probability of about 50% that the investor's final bankroll will not exceed \$1.95, and a meagre probability of 5.8% that the investor's final bankroll will be greater than his starting bankroll of ten thousand dollars.

Misled by seemingly favorable averages, our foolhardy investor stakes the full amount of his bankroll every week. Apparently, he is unacquainted with the Kelly strategy. According to this strategy, rather than investing the full amount of his current bankroll for every transaction, he would do better to invest a same fixed fraction of his current bankroll each time. In this specific case of the investor, the Kelly strategy requires him to invest a fraction $\frac{5}{24}$ of his current bankroll for each transaction. In practical terms, this renders a practically zero probability of his ending with \$1.95 or less after 52 weeks. In fact, applying the Kelly strategy would give the investor about a 70% probability of ending with more than ten thousand dollars after 52 weeks, and about a 44% probability of ending with more than twenty thousand dollars. The Kelly fraction

$f^* = \frac{5}{24}$ is what we get by inserting the values $p = 0.5$, $r_1 = 1.8$ and $r_2 = 0.4$ into the famous Kelly betting formula:

$$f^* = \frac{pr_1 + (1-p)r_2 - 1}{(r_1 - 1)(1 - r_2)}.$$

This formula applies to the following general situation. Imagine that you can repeatedly make bets in a particular game. The game is assumed to be 'favorable' for you, where favorable means that the expected value of the net payoff of the game is positive. For every dollar staked on a repetition of the game, you receive r_1 dollars back with probability p, and r_2 dollars with probability $1 - p$, where $0 < p < 1$, $r_1 > 1$ and $0 \leq r_2 < 1$. We then assume that $pr_1 + (1-p)r_2 > 1$, that is, the game is favorable for you in terms of expected value. You start with a certain bankroll, and it is assumed that you may stake any amount up to the maximum of your current bankroll. If you want to maximize the growth factor of your bankroll over the long run, the Kelly formula advises you to stake a same fixed fraction f^* of your current bankroll each time. This fraction is called the *Kelly fraction*. In the special case when $r_2 = 0$, the Kelly fraction reduces to

$$f^* = \frac{pr_1 - 1}{r_1 - 1},$$

which can be interpreted as the ratio of the expected net gain per staked dollar and the payoff odds.

A sketch of the derivation of the Kelly formula should not be omitted here. Let's say your initial bankroll is W_0 and that you will stake a same fixed fraction f of your current bankroll on each bet. By staking that fraction f, your bankroll will increase by factor $1 - f + fr_1$ with probability p, and it will decrease by factor $1 - f + fr_2$ with probability $1 - p$. Denoting your bankroll after n bets with the random variable W_n, we get

$$W_n = (1 - f + fr_1)^W \times (1 - f + fr_2)^L \times W_0,$$

where the random variable W represents the number of times that you win the bet, and the random variable L represents the number of times you lose the bet ($W + L = n$). Now define G_n through

$$e^{nG_n} W_0 = W_n,$$

or $G_n = \frac{1}{n} \ln(W_n/W_0)$. The random variable G_n represents the growth factor of your bankroll over the first n bets. If we take the logarithms of both sides of the formula for W_n, then we find that

$$G_n = \frac{W}{n} \ln\left(1 - f + fr_1\right) + \frac{L}{n} \ln\left(1 - f + fr_2\right).$$

According to the strong law of large numbers, $\frac{W}{n}$ will converge to p and $\frac{L}{n}$ to $1-p$ if n gets very large. So, the long-run growth factor of your bankroll can be expressed as

$$g(f) = p \ln(1 - f + fr_1) + (1 - p) \ln(1 - f + fr_2).$$

Now it is just a question of simple algebra to find the Kelly formula. Putting the derivative of $g(f)$ equal to 0 shows that $g(f)$ is maximal for $f = f^*$.

It is interesting to note the following: if $u(w) = \ln(w)$ is the utility of your wealth w, then investing the Kelly fraction of your wealth w maximizes the expected value of the utility of the wealth resulting from this investment. The utility function $u(w) = \ln(w)$ has as derivative $u'(w) = 1/w$. This means that for this utility function the marginal value of one extra monetary unit is inversely proportional to the value of your present wealth. The utility function $\ln(w)$ was introduced in 1738 by Daniel Bernoulli in his approach to the St. Petersburg paradox. This coin-toss problem poses the question: what is a fair price for entering a game in which you repeatedly toss a fair coin until heads appears and you win 2^k ducats if your coin lands heads for the first time on the kth toss? A linear utility function would give that a 'fair' price for playing this game is infinitely large, which is of course absurd. No one in their right mind would pay no more than a small fee to enter the game. That's why Daniel Bernoulli introduced the utility function $\ln(w)$.

What is the long-run rate of return of your bankroll when you invest a same fixed fraction f of your current bankroll each time? This long-run rate of return is given by

$$\gamma(f) = (1 - f + fr_1)^p (1 - f + fr_2)^{1-p} - 1.$$

To explain this, we define the random variable γ_n through $W_n = (1 + \gamma_n)^n W_0$. Then γ_n is the rate of return over the first n investments. Using the earlier definition $e^{nG_n} W_0 = W_n$, it then follows that

$$\gamma_n = e^{G_n} - 1.$$

As shown before, the random variable G_n converges with probability 1 to $g(f)$ if n gets very large, and that means that γ_n converges with probability 1 to

$$e^{g(f)} - 1 = (1 - f + fr_1)^p (1 - f + fr_2)^{1-p} - 1$$

When using the Kelly strategy, in some cases the stakes can be very large. This means that big swings in the time path of your bankroll may occur, which can wreak havoc with a player's ability to get a good night's rest, to say the least. For example, if $p = 0.95$, $r_1 = 2$ and $r_2 = 0$, then, under the Kelly strategy, 90% of your bankroll must be staked each time. If you lose a Kelly bet, your bankroll then decreases by 90%. Your bankroll undergoes a roller coast ride in this situation! This is why in practice, a fractional Kelly strategy is employed.[1] Under the fractional Kelly strategy, you would stake a fraction $f = cf^*$ of your bankroll each time, where $0 < c < 1$. In practice c will mostly be chosen between 0.3 and 0.5. Empirical evidence shows that

$$\frac{\gamma(cf^*)}{\gamma(f^*)} \approx c(2 - c).$$

That is, using the fractional Kelly strategy, the long-run rate of return of your bankroll will be reduced to about $c(2 - c)$ times the long-run rate of return you would have if you used the full Kelly strategy. The fractional Kelly strategy $f = cf^*$ is less risky. This is also expressed by the approximation

$$\frac{1 - b^{2/c-1}}{1 - (b/a)^{2/c-1}}$$

for the probability of reaching a bankroll of aW_0 without falling down first to a bankroll of bW_0, where a and b are any constants with $0 < b < 1 < a$. For example, by staking half of the Kelly fraction instead of the (full) Kelly fraction, you sacrifice a quarter of your maximum long-run rate of return, but you increase the probability of doubling your bankroll without having it halved first from 0.67 to 0.89.

The Kelly formula cited above only works for games with single-bet-only opportunities. Multiple bets may be placed on sport

[1] See also Leonard MacLean, Edward Thorp, Yonggan Zhao, and William Ziemba, "*How does the fortune's formula Kelly capital growth model perform?*," The Journal of Portfolio Management, Vol. 37 (2011), 96-111.

events such as soccer matches and horse races. Let's say that the soccer club Manchester United is hosting a match against Liverpool, and that a bookmaker is paying out 4.5 times the stake if Liverpool win, 4.5 times the stake if the match ends in a draw, and 1.75 times the stake if Manchester United win. You estimate Liverpool's chance of winning at 25%, the chance of the game ending in a draw at 25%, and the chance of Manchester winning at 50%. If you are prepared to bet 100 euro, how should you divide your stake on this match? The Kelly formula given above is not the appropriate formula to decide how to place your bets because in this situation, you are able to place multiple bets at the same time. The Kelly betting strategy can, however, be extended to cover this more general situation. Although the soccer match is just a one-time event, you could stake bets using the strategy you would follow in the hypothetical situation of a soccer match being played a very large number of times under identical circumstances. In practice, this appears to be a good rule of thumb. Let's say our sporting event gives us the opportunity of placing multiple bets at the same time on s possible outcomes. If the event ends in outcome i, the bookmaker's pay-out is equal to β_i times your stake on outcome i. Suppose our estimate of the probability of outcome i occurring is p_i, where $\sum_{i=1}^{s} p_i = 1$. We assume that $p_i\beta_i > 1$ for at least one i and that not all $p_i\beta_i$ are greater than 1. In the example of the soccer match,

$$s = 3,\ \beta_1 = \beta_2 = 4.5,\ \beta_3 = 1.75 \ \text{ and }\ p_1 = p_2 = 0.25,\ p_3 = 0.5.$$

In the hypothetical case of a sporting event that can be repeated very often under identical circumstances, and in which every time you use the same fixed fraction f_i of your bankroll to stake a bet on outcome i for any i, the long-run growth factor of your bankroll is given by

$$\sum_{i=1}^{s} p_i \ln\big[1 - (f_1 + \cdots + f_s) + f_i\beta_i)\big].$$

The details of the derivation of this formula are skipped. The optimal value of the betting fractions $f_1, \ldots, f_s$ are found by maximizing the growth factor under the conditions $\sum_{i=1}^{s} f_i \leq 1$ and $f_i \geq 0$ for all i. A closed-form expression for the f_i cannot be given, but advanced optimization theory allows us to design the following algorithm for calculating the f_i:

Step 0. Renumber the indices such that $p_1\beta_1 \geq p_2\beta_2 \geq \cdots \geq p_s\beta_s$.

Step 1. Determine the largest index r for which $\sum_{i=1}^{r} \frac{1}{\beta_i} < 1$. Calculate

$$B(k) = \frac{1 - (p_1 + \ldots + p_k)}{1 - \big(1/\beta_1 + \ldots + 1/\beta_k\big)} \quad \text{for } k = 1, \ldots, r.$$

Let q represent index k for which $B(k)$ is minimal (in the case where $B(k)$ is minimal for multiple indices, use the smallest index for q).
Step 2. Calculate the betting fractions $f_i = p_i - \frac{B(r)}{\beta_i}$ for $i = 1, \ldots, q$ and $f_i = 0$ for $i > q$.

For the soccer match between Manchester United and Liverpool, the algorithm is as follows:

Step 0. For outcomes 1 (a win for Liverpool), 2 (draw) and 3 (a win for Manchester United), we have $p_1\beta_1 \geq p_2\beta_2 \geq p_3\beta_3$.
Step 1. We have $\frac{1}{\beta_1} = \frac{10}{45}$, $\frac{1}{\beta_1} + \frac{1}{\beta_2} = \frac{20}{45}$ and $\frac{1}{\beta_1} + \frac{1}{\beta_2} + \frac{1}{\beta_3} > 1$. So, $r = 2$. Further, $B(1) = \frac{27}{28}$ en $B(2) = \frac{9}{10}$. This results in $q = 2$.
Step 2. $f_1 = f_2 = 0.25 - \frac{0.9}{4.5} = \frac{1}{20}$ and $f_3 = 0$.

To conclude, the Kelly strategy proposes that you stake 5% of your bankroll of 100 euro on a win for Liverpool, 5% on a draw, and 0% on a win for Manchester United. For this strategy, the subjective expected value of your bankroll after the match is equal to $90 + 0.25 \times 22.5 + 0.25 \times 22.5 = 101.25$ euro. It is interesting to note that the two concurrent bets on the soccer match act as a partial hedge for each other, reducing the overall level of risk.

The Kelly strategy was first used in casinos by mathematician Edward Thorp, in order to try out his winning blackjack system. Later, Thorp and a host of famous investors including Warren Buffett, successfully applied the Kelly strategy to guide their stock market decisions. Nowadays, the Kelly strategy is also frequently used for sporting events such as horse racing and soccer matches. This was not what John Kelly Jr. had in mind when he developed his famous formula in 1956 at the Bell Labs research institute. As is often the case, fundamental research leads to unforeseen applications.

CHAPTER 17

To Stop or Not to Stop? That is the Question

EVERY DECISION involves the weighing of risk factors. When the weather suddenly worsens, meteorologists must decide when to stop taking measurements and when to start warning the public. When Apple has a new version of the iPhone in the pipeline, they must decide when to stop testing and when to start marketing. These are examples of decisions made in uncertain environments, and they can result in significant, negative consequences if made at the wrong moment in time. But large companies are not alone in having to make such decisions; during the course of everyday life, everyone encounters situations that prompt time related decision-making quandaries, such as determining the right moment to purchase a house, or to accept an offer for a new job.

In the field of probability, decisions that require consideration of whether to 'stop or continue on' are known as optimal stopping problems.[1] The most famous of these is the dating problem. Newspaper and magazine editors love this subject and give it wide, popular coverage from time to time. The problem goes like this: let's say you are swiping your way, one-by-one, through a broadly assembled collection of profiles that describe potential partners. The profiles appear in random order. You may arrange a date with just one of the individuals profiled. After looking at a profile, you must immediately decide whether to select or reject that candidate. You may not return to any candidate once you have opted to reject. Reviewing a candidate's profile, your only option is to judge whether the candidate is 'better' or 'worse than those seen previously (and rejected). In other words, you can only ascribe relative values to the candidates. Under these circumstances, what strategy gives you the highest probability of choosing the best candidate when the number of candidates is known? As so often is the case, it's mathematics to the rescue. When the number of candidates is sufficiently large, say 10 or more, then the optimal decision rule is to pass through about 37% of the candidate profiles in order to obtain a baseline to work with, and select the first candidate thereafter who appears to be better than those that came before. In this way, you have a probability of about 37% of finding the best candidate (the maximum probability of getting the best candidate converges toward $1/e \approx 0.3679$ as the number of candidates increases). Many people are surprised to learn that the probability of success in such an endeavor is so high, and that it is not related to the total number of candidates when that number is sufficiently large. But that surprise will be tempered if they consider that you already have a greater than 25% probability of finding the best candidate if you swipe through half of the profiles and select the first candidate thereafter who is better than the previous ones. In

[1] A nice survey article about optimal stopping problems is Ted Hill, "*Knowing when to stop*," American Scientist, Vol. 97 (2009), 126-133.

any case, this simple strategy will always help you to locate the best candidate, provided that the second-best candidate is profiled in the first half of the list of candidates, and the best candidate is profiled in the second half of that list. Assuming $n = 20$ candidates, the probability of this occurring is $\frac{10}{20} \times \frac{10}{19} \approx 0.263$. This can be seen as follows. The probability that the second-best candidate will be grouped among the first 10 candidates is $\frac{10}{20}$, while the probability that the best candidate will be grouped among the last 10 candidates, given that the second-best candidate is grouped among the first 10, is equal to $\frac{10}{19}$. An analogous argument shows that the lower bound of 25% applies whatever the number of candidates is, even if you have one million candidates!

A nice variant of the dating problem is the Googol problem for two players. Let one person take as many slips of paper as he likes, writing down a number on each one. The number may be as great or small as he likes, selected from between a miniscule number and a very large number, the size of a 'googol' (a 1 followed by one-hundred 0's), or even larger if he likes. The slips of paper are then shuffled thoroughly and placed number-side down on a table. One by one, you turn the slips over, stopping when you have turned over the number you believe to be the largest. A great many people will estimate your probability of finding the largest number to be lower than is actually the case. If you were a betting man or a gambling gal, you could profit from this fact.

Several variations of the dating problem have also come under scrutiny, such as one that determines a strategy for achieving the maximal probability of selecting one of the two best profiles. For this case, the maximal probability of success increases to about 57%. The optimal strategy is now characterized by two change-over points c_1 and c_2: observe the first c_1 candidates without picking one, next observe the candidates $c_1 + 1, \ldots, c_2$ and stop if you see a candidate with the highest rank seen so far, and, if no record occurs in this stretch of candidates, you continue to the next stage in which you stop as soon as you see a candidate with one of the highest two ranks seen so far. The optimal change-over points satisfy $c_1 \approx 0.3470\,n$ and $c_2 \approx 0.6667\,n$ when the number n of candidates is sufficiently large. If the problem is to focus on the selection of one of the best three profiles, the maximal probability of success increases further to about 71% and the optimal change-over points are $c_1 \approx 0.3367\,n$, $c_2 \approx 0.5868\,n$, and $c_3 \approx 0.7746\,n$.

The dating problem is a rewarding subject of research for heuristics that are easy to use. The problem has also gained the attention of cognitive psychologists. In analyzing this problem, German psychologist Peter Todd identified the 'magical' number 12: when your goal is to maximize your probability of selecting a partner from the top 10% of a group of one-hundred to several hundred candidates, you reject the first 12, and select the first one after that who is better than the previous ones. In order for this strategy to work, it is not necessary to know the precise number of candidates.

An entirely different situation arises when you can rate each candidate on a scale of 0 to 1, say. Let's look at this situation in the following context: imagine that someone has made n independent draws $a_1, \ldots, a_n$ from the uniform distribution on $(0, 1)$. The outcomes of these draws are unknown to you, but you do know that they come from the uniform distribution on $(0, 1)$. These numbers are shown to you one by one, in a random order. You want to maximize your probability of identifying the largest number. Each time you are shown a number, you must immediately decide whether you think it is the largest one, and stop the game, or to reject that number as the largest and move on to the next number. You may not go back and change your mind about numbers previously seen. What should your strategy be?

The optimal strategy is hard to determine, but there is simple heuristic that performs quite well. It works as follows. Suppose that you are shown a number a that is the largest number you have seen so far, and that there are still k numbers to come. The heuristic rule says that you should select this number a only if $a^k \geq 0.5$, that is, if there is a probability of 50% or more that each of the k remaining numbers is less than or equal to a. Under this heuristic rule, the probability of selecting the largest number is about 60.5% for $n = 10$ and about 58.9% for $n = 25$, as can be verified by computer simulation. The probability stabilizes at a value of about 58% if n increases. If the game involves betting and the monetary payout is equal to the number at which you stop the game, the heuristic rule also provides good results if you want to maximize the average payout per game. Simulation reveals that under the heuristic rule the average payout per game is about 0.814 for $n = 10$ and about 0.853 for $n = 25$ when the game is played very often. In this gaming problem, another good heuristic is the square-root heuristic that disregards the first $\sqrt{n} - 1$ numbers and then picks the first number that is larger than all previous numbers shown. In the popular press this heuristic was wrongly interpreted as a good heuristic for the classical dating problem. In the foregoing discussion, the numbers $a_1, \ldots, a_n$ were independent draws from the uniform distribution on $(0, 1)$. What should you do when these numbers are independent draws from a continuous random variable X with cumulative probability distribution function $F(x)$? This case can be reduced to the previous case of the uniform distribution by replacing a_i by $a_i' = F(a_i)$ for all i. The rationale of this transformation is the fact that the random variable $F(X)$ is uniformly distributed on $(0, 1)$ when $F(x)$ is strictly increasing.

In a discussion on optimal stopping problems, the *one-stage-look-ahead rule* may not be omitted. This is an intuitively appealing rule that looks just one step ahead, as the name suggests. The rule prescribes stopping in states in which it would be at least as good to stop now, as it would be to continue for one more step, and then stop. In many cases, this approach leads to an optimal or nearly-optimal stopping rule.

Let's look at an example in which the one-stage-look-ahead rule yields a nearly-optimal stopping rule. This example takes the famous Chow-Robbins game as case-in-point. In this game, you repeatedly toss a fair coin. You can stop whenever you want. Your payoff is the proportion of heads recorded at the time you stop. When to stop if your goal is to maximize your expected payoff? This is a problem that, mathematically speaking, has not yet been fully solved, and that, some years ago, became a hype among the 'quants' running Wall Street and other stock exchanges. The optimal stopping rule has a rather complex form: it is characterized by integers $\beta_1, \beta_2, \ldots$ such that you stop after the nth toss when the difference between the number of heads and the number of tails after n tosses is greater than or equal to β_n. It is very difficult to compute the β_n. Also, the precise value of the maximal payoff is still unknown; numerical investigations have only shown that this value must be between $0.7929530\ldots$ and $0.7929556\ldots$. Using the principle of looking one step ahead, however, you can obtain a very simple stopping rule for which the expected payoff is very close to the maximal expected payoff. In order to find this rule, suppose that you have executed n tosses. Let f_n be the proportion of heads, or, equivalently, let nf_n be the total number of heads after n tosses. If you decide to go for one more toss, then the expected value of the proportion of heads after $n + 1$ tosses is

$$\frac{1}{2} \times \frac{nf_n + 1}{n + 1} + \frac{1}{2} \times \frac{nf_n}{n + 1}.$$

This proportion is smaller than f_n only if $f_n > \frac{1}{2}$. This suggests a strategy of stopping as soon as $f_n > \frac{1}{2}$, or, stated differently, you stop as soon as the number of heads exceeds the number of tails. The expected payoff under this simple and appealing stopping rule can be shown to be $\frac{\pi}{4} = 0.785398\ldots$. The expected payoff in this case is quite close to the maximal expected payoff. The Chow-Robbins game is filled with mathematical surprises. Just try making a simulation to determine the expected value of the game's duration under the sub-optimal stopping rule!

An appealing problem that has intrigued psychologists studying high-risk behaviors is the devil's penny game. In this game of chance a number of identical, closed boxes are placed in front of you. Every box contains a sum of money, except for one box. This 'devil's box' contains the devil's penny. The object of the game is to open the boxes one by one, and collect the money inside. You may continue to open boxes and keep the money inside as long as you don't encounter the box containing the devil's penny. As soon as you open that box, the game is over and you forfeit the entire sum of money you have amassed up to that point. The total value of the individual sums contained in the boxes is made known to you when the game starts. When should you stop if you want to maximize the expected value of the sum you will end up with?

The answer to this is quite simple. You stop as soon as the amount of money collected is greater than or equal to half of the total amount placed in the boxes. This is the one-stage-look-ahead rule, which is optimal for the devil's penny game. The break-down goes like this: let's say you are playing the game with $n+1$ boxes, where the n boxes, not including the one containing the devil's penny, contain sums $a_1, \ldots a_n$. Let $A = a_1 + \cdots + a_n$ denote the total sum available in the game. You know the value of the total amount A. Suppose that $k+1$ boxes are still waiting to be opened at a given moment, the box containing the devil's penny being among them. Let a be the sum of money collected up to that point. To calculate the expected value of the amount by which the capital a you have collected so far will change if you decide to open one more box, you do need not to know how the remaining amount of $A - a$ dollars is distributed over the k remaining closed boxes not containing the devil's penny. To see this, imagine that the dollar amounts $b_1, \ldots b_k$ (say) are in these boxes. Then, by $b_1 + \cdots + b_k = A - a$, the expected value of the amount by which your current capital will change is

$$\frac{1}{k+1}\,b_1 + \cdots + \frac{1}{k+1}\,b_k - \frac{1}{k+1}\,a = \frac{1}{k+1}(A-a) - \frac{1}{k+1}\,a.$$

The expected change in your current capital is positive only if $a < \frac{1}{2}A$, regardless of the value of k. This suggests a strategy of stopping as soon you have collected $\frac{1}{2}A$ dollars or more and to continue otherwise. This strategy is optimal.[2] It is remarkable that the optimal strategy does not depend on the number of boxes and how the total amount of A dollars is distributed over the boxes. If there is another box with a second devil's penny, then, by the same arguments as above, it is optimal to stop when you have collected $\frac{1}{3}A$ or more dollars.

The devil's penny game can also be played with eleven cards: ace, two, three, ..., nine, ten, and joker, where the joker takes on the role of the devil's penny, and any other card counts for its face value, with ace $= 1$. The total number of points in the game is $1 + 2 + \ldots + 10 = 55$. The cards are turned over one by one and the game is over with zero points when the joker is turned over. It is optimal to stop as soon as the cumulative number of points of the cards turned up is greater than or equal to 28. If you play this game very often, you will get about 15.45 points per game on average. The number of points you get in a single execution of the card game is rather erratic with a standard deviation close to the expected value. The card game is an amusing game to play with a group of friends to throw light on which of you are the risk takers, and which are the risk averse.

[2] In general, the one-stage-look-ahead rule is optimal if another unfavorable state occurs when you do not stop in an unfavorable state. In other words, the set of unfavorable states must be closed.

Acknowledgement of Illustrations

Image Chapter 9: A quack doctor selling remedies from his caravan; satirizing. Credit: Wellcome Collection, London. This work is licensed under CC BY.
Image Chapter 13: the 10 Swiss Franc Banknote of Leonhard Euler. Permission of SNB Archives (ASNB): Archival reference.
Image Chapter 14: the 10 German Mark Banknote of Carl Friedrich Gauss. Courtesy of the Deutsche Bundesbank.

Index